

SECTION 6C

BQ (WORKMEN) HOUSING

DULIAJAN, ASSAM

DESIGN BASIS REPORT

PREPARED FOR



PREPARED BY



RITES LTD.

(A Government of India Enterprise)

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CONTENTS

1	GROUP HOUSING (BQ-WORKMEN)	1
1.1	PROJECT INTRODUCTION	1
1.2	EXISTING SITE	1
1.2.1	LOCATION AND CONTEXT	1
1.2.2	CONNECTIVITY & CIRCULATION	1
1.2.3	SITE ANALYSIS:	2
1.3	OIL'S REQUIREMENTS	3
1.4	APPLICABLE CODES & STANDARDS	3
1.4.1	CLIMATE AND ENVIRONMENTAL CONSIDERATIONS	6
1.5	DESIGN PROPOSAL	7
1.5.1	MASTER PLANNING CONCEPT	7
1.5.2	BUILDING PLAN (BQ HOUSING - WORKMEN):	9
1.5.3	BQ INFRASTRUCTURE DETAILS:	11
1.6	CONCEPT ELEVATION– BQ HOUSING	13
1.7	AREA DETAILS	15
2	GREEN BUILDING CONSIDERATION	16
2.1	GRIHA- RATING SYSTEM AND CERTIFICATION	16
2.2	TARGET GRIHA RATING & VERSION:	16
2.3	IMPORTANT SUSTAINABLE FEATURES AND PARAMETERS	16
3	LANDSCAPE DESIGN CONSIDERATIONS	21
3.1	VISION FOR THE LANDSCAPE	21
3.2	KEY OBJECTIVES	21
3.3	CLIMATIC AND SITE CONTEXT	21
3.4	LANDSCAPE ZONING:	21
3.5	PLANTING STRATEGY	22
3.6	SHRUBS & GROUND COVERS:	22
3.7	GREEN DESIGN ELEMENTS	23

3.8	IRRIGATION MANAGEMENT	23
3.9	OUTDOOR INFRASTRUCTURE & ACCESSIBILITY.....	23
3.10	MAINTENANCE & SUSTAINABILITY	23
4	STRUCTURE CONSIDERATIONS.....	25
4.1	INTRODUCTION.....	25
4.2	DESIGN AND DETAILING METHODOLOGY	25
4.3	STRUCTURAL SYSTEM.....	25
4.3.1	SUPERSTRUCTURE.....	26
4.3.2	SUBSTRUCTURE	26
4.3.3	RETAINING WALLS.....	26
4.3.4	WATER RETAINING STRUCTURES	26
4.3.5	CRACK WIDTH CONTROL/LIMITATIONS	27
4.3.6	EXPOSURE CONDITION.....	27
4.3.7	NOMINAL COVER	27
4.3.8	CONCRETE SPECIFICATION	27
4.3.9	REINFORCEMENT AND STRUCTURAL STEEL SPECIFICATION.....	28
4.3.10	EXCAVATION.....	28
4.4	STRUCTURAL MODELING AND SOFTWARE.....	28
4.5	LOADING PARAMETERS.....	29
4.5.1	MATERIAL PROPERTIES	29
4.5.2	DEAD LOADS (DL)	29
4.5.3	IMPOSED LOAD/LIVE LOAD (IL/LL)	30
4.5.4	EARTHQUAKE LOADS (EQL).....	30
4.5.5	WIND LOADS (WL)	30
4.5.6	TEMPERATURE LOAD (TL)	31
4.6	LOAD COMBINATIONS	31
4.7	DETAILING OF STRUCTURES.....	42
4.8	CONTROL OF DEFLECTIONS	42
4.9	GEOTECHNICAL DATA.....	44

4.10	INDIAN CODE OF PRACTICE USED	44
5	ELECTRICAL, HVAC, ELV & FIRE FIGHTING SYSTEM DBR	47
5.1	PROJECT DESCRIPTION	47
5.2	GENERAL	47
5.3	SCOPE OF WORK	47
5.4	ELECTRICAL POWER REQUIREMENT	48
5.5	SOURCE OF ELECTRICAL POWER SUPPLY	48
5.6	SCOPE OF EPC CONTRACTOR	49
5.7	ELECTRIC SUB-STATION, 11/ 0.433 KV- 1 NO.....	50
5.8	DG SET FOR BACKUP SUPPLY	52
5.9	EMERGENCY & LIFE SAFETY SYSTEMS – POWER SUPPLY REQUIREMENTS	53
5.10	CONSTRUCTION STANDARDS FOR SUBSTATION AND DG SET.....	54
5.11	PIPING WORK	54
5.12	SMART METERING	54
5.13	SMART METERING AND MONITORING SYSTEM	55
5.14	ELECTRICAL POWER DISTRIBUTION	57
5.15	GAS SUPPERSSION SYSTEM	58
5.16	EARTHING SYSTEM	59
5.17	LIGHTNING PROTECTION SYSTEM.....	60
5.18	INTERNAL ELECTRIFICATION, LV & ALLIED WORKS	61
5.19	INTERNAL LIGHTING DESIGN & LIGHTING FIXTURE	64
5.20	EXTERNAL/ STREET LIGHTING	66
5.21	MASTER ANTENNA TELEVISION SYSTEM (MATV)	68
5.22	UPS (UNINTERRUPTIBLE POWER SUPPLY)	69
5.23	SOLAR PHOTOVOLTAIC POWER GENERATION SYSTEM.....	70
5.24	CLOSED CIRCUIT TELEVISION SURVEILLANCE SYSTEM (CCTV)	71
5.25	IP BASED PUBLIC ADDRESS SYSTEM (PA SYSTEM):.....	73
5.26	IP BASED EPABX TELEPHONY SYSTEM.....	74

5.27	FIRE DETECTION AND ALARM SYSTEM.....	76
5.28	BOOM BARRIERS.....	78
5.29	ESTIMATED ELECTRICAL LOAD SHEET:	79
5.30	ELECTRICAL SLD	83
5.31	LIFTS.....	88
5.32	SAFETY REQUIREMENTS	90
5.33	ELECTRIC VEHICLE CHARGING STATION:	91
6	DESIGN BASIS REPORT FOR HVAC SYSTEM	93
6.1	INTRODUCTION.....	93
6.2	SCOPE OF WORK	93
6.3	HVAC SYSTEM	94
6.3.1	OBJECTIVE	94
6.3.2	SCOPE OF WORK.....	94
6.3.3	TERMINOLOGY	95
6.3.4	REFERENCE STANDARDS	99
6.3.5	ABBREVIATIONS.....	99
6.3.6	BASIS OF DESIGN	100
6.3.7	AIR CONDITIONING LOAD AND TYPE OF SYSTEM.....	104
6.3.8	SYSTEM PROPOSED	107
6.3.9	VENTILATION	111
6.3.10	DESIGN PARAMETERS FOR MAJOR EQUIPMENTS	112
6.3.11	BRIEF OUTLINE SPECIFICATIONS FOR MAJOR EQUIPMENT	113
6.3.12	VENTILATION CALCULATIONS WITH EQUIPMENT SELECTION.....	116
6.3.13	EMERGENCY SMOKE VENTILATION CALCULATIONS	120
6.3.14	LIFT WELL PRESSURIZATION TYPICAL TOWER	121
6.3.15	EXCLUSIONS FROM HVAC SCOPE OF WORK.....	121
7	PLUMBING & FIREFIGHTING SYSTEMS.....	123
7.1	PREAMBLE.....	123
7.1.1	SCOPE OF WORK.....	123

7.1.2	PROJECT OBJECTIVES	123
7.1.3	BROAD CONCEPT OF SERVICES:.....	123
7.1.4	ENERGY CONSERVATION FEATURES:.....	123
7.1.5	DESIGN PHILOSOPHY AND DESIGN CONSIDERATION	124
7.1.6	SPACE REQUIREMENT	125
7.1.7	INTEGRATED DESIGN APPROACH WITH CONSULTANT TEAM	125
7.1.8	HIGH-END DEVELOPMENT	125
7.2	PLUMBING SYSTEM	125
7.2.1	REFERENCE STANDARDS	125
7.2.2	APPROACH TO PLANNING.....	125
7.2.3	SYSTEM REQUIREMENTS	126
7.2.4	DESIGN CALCULATIONS.....	126
7.2.5	WATER STORAGE	131
7.2.6	CAPACITY OF UNDERGROUND & OVERHEAD WATER STORAGE TANK	131
7.2.7	SOURCE OF WATER	132
7.2.8	WATER TYPE & TREATMENT	133
7.2.9	WATER DISTRIBUTION	133
7.2.10	APPURTENANT	134
7.2.11	SEWAGE AND STORM WATER DRAINAGE	136
7.2.12	DESIGN CRITERIA	137
7.2.13	SANITARY FIXTURES & FITTINGS	138
7.2.14	SEWAGE TREATMENT PLANT	138
7.2.15	STORM WATER DRAINAGE.....	143
7.3	FIRE FIGHTING	160
7.3.1	INTRODUCTION	160
7.3.2	NBC REQUIREMENT AND PROPOSALS.....	160
7.3.3	MANUAL FIRE EXTINGUISHER AND GAS FLOODING SYSTEM	169

LIST OF FIGURES

FIGURE 1: GOOGLE IMAGE LOCATION OF PROPOSED BQ HOUSING	1
FIGURE 2: BQ HOUSING SITE PHOTOS	2
FIGURE 3: BYELAWS FOR F.A.R. (FLOOR AREA RATIO).....	4
FIGURE 4: BYE LAWS FOR EXTERIOR OPEN SPACE	4
FIGURE 5: BYELAWS FOR PARKING	5
FIGURE 6: NBC 2016 BYLAWS FOR GROUP HOUSING	5
FIGURE 7: AVERAGE ANNUAL TEMPERATURE IN DULIAJAN	6
FIGURE 8: CLIMATE DATA OF DULIAJAN.....	7
FIGURE 9: PROPOSED SITE PLAN FOR BQ HOUSING	10
FIGURE 10: GOOGLE EARTH IMAGERY SUPERIMPOSED	10
FIGURE 11: TYPICAL UNIT PLAN FOR BQ HOUSING	11
FIGURE 12: BQ HOUSING CONCEPT ELEVATION.....	14
FIGURE 13: DIAGRAM GENERAL CCTV SYSTEM.....	71
FIGURE 14: IP BASED PA SYSTEM	74
FIGURE 15: IP BASED EPABX SYSTEM.....	76
FIGURE 16: BOOM BARRIER SCHEMATIC	78
FIGURE 17: ELECTRICAL SLD	83
FIGURE 18: WATER BALANCE DIAGRAM IN NON-RAINY DAYS	130
FIGURE 19: WATER BALANCING IN RAINY DAYS.....	130
FIGURE 20: RAINFALL DATA, IMD.....	147
FIGURE 21: IGBC CRITERIA FOR 4 POINTS	148
FIGURE 22: PROCESS FLOW DIAGRAM FOR WATER TREATMENT PLANT.....	155
FIGURE 23: REFERENCE FOR OCCUPANCY/POPULATION CALCULATION.....	156
FIGURE 24: REFERENCE FOR WATER REQUIREMENT CALCULATION	157
FIGURE 25: REFERENCE FOR UG & OHT STORAGE TANK CAPACITY.....	158
FIGURE 26: REFERENCE FOR FIRE FIGHTING SYSTEM	171

LIST OF TABLES

TABLE 1: PROPOSED BUILDING PARAMETERS	13
TABLE 2: LANDSCAPE ZONING	22
TABLE 3: PLANTING STRATEGY	22
TABLE 4: IS CODE USED FOR STRUCTURE ELEMENTS	44
TABLE 5: SOURCE METERING REQUIREMENTS AS PER GRIHA MANUAL	55
TABLE 6: LIGHTING FIXTURES PARAMETERS.....	65
TABLE 7: LED LIGHTING FIXTURES PARAMETERS.....	66
TABLE 8: ELECTRICAL LOAD SHEET	79
TABLE 9: LIST OF APPLICABLE IS CODES.....	83
TABLE 10: LIFT DETAILS	88
TABLE 11: INDOOR DESIGN CONDITIONS FOR HVAC.....	101
TABLE 12: MECHANICAL VENTILATION	102
TABLE 13: ASSEMBLY U VALUE	102
TABLE 14: OCCUPANT DENSITY/LIGHTING LOA/EQUIPMENT LOAD.....	103
TABLE 15: HEAT LOAD SUMMARY OF COMMUNITY CENTRE.....	104
TABLE 16: HEAT LOAD SUMMARY OF GUEST HOUSE BLOCK	105
TABLE 17: COMPARISON OF DIFFERENT AC SYSTEMS	107
TABLE 18: ELECTRICAL POWER REQUIREMENT (AC SYSTEM).....	111
TABLE 19: LIFT WELL PRESSURIZATION	121
TABLE 20: DESIGN CONSIDERATIONS FOR FIRE FIGHTING & PLUMBING SYSTEM.....	124
TABLE 21: WATER DEMAND CALCULATIONS.....	128
TABLE 22: WATER STORAGE CALCULATIONS	131
TABLE 23: CAPACITY OF UNDERGROUND & OVERHEAD WATER STORAGE TANK	131
TABLE 24: FLOW RESISTORS DETAILS	135
TABLE 25: DESIGN PARAMETERS FOR STP	139
TABLE 26: CALCULATIONS FOR STORM WATER LOAD.....	144

TABLE 27: VALVES DETAILS.....	150
TABLE 28:DRINKING WATER CHARACTERISTICS (IS: 10500).....	151
TABLE 29: LIST OF MATERIALS	152
TABLE 30: COMPONENTS OF STP	154
TABLE 31: FIRE FIGHTING REQUIREMENTS FOR HOUSING BLOCK.....	161
TABLE 32: FIRE FIGHTING REQUIREMENTS FOR GUEST HOUSE	163
TABLE 33: FIRE FIGHTING REQUIREMENT FOR COMMUNITY CENTRE	166

ARCHITECTURE DBR

1 GROUP HOUSING (BQ-WORKMEN)

1.1 PROJECT INTRODUCTION

The upcoming group housing project in Oil Township, Duliajan, Assam, is designed to provide well-structured living spaces for workmen. The development includes BQ housing, thoughtfully planned to enhance resident's quality of life while fostering a strong community. It features residential towers, community centre, guest house, children's park, on-street parking facility, ancillary services ensuring a well-rounded living environment with essential amenities for comfort and convenience.

1.2 EXISTING SITE

1.2.1 LOCATION AND CONTEXT

The existing site of BQ housing covers 29,400 sqm (7.26 acres) area (approx.) and is located in the South-West of DX housing at a distance of 300m approx. in Duliajan (Assam). The design approach intends to make an environmentally friendly habitable space.

1.2.2 CONNECTIVITY & CIRCULATION

Proposed BQ Housing site is located at the co-ordinates (27°22'05.0" N 95°19'06.0" E) and the connectivity to the site is augmented via Duliajan – Dibrugarh Highway through the Buddha Mandir Road and further through the Moitry Nagar Road after St. Xavier's Higher Secondary School but before DX housing location.

FIGURE 1: GOOGLE IMAGE LOCATION OF PROPOSED BQ HOUSING



1.2.3 SITE ANALYSIS:

The site is located in a residential area, with its southern boundary parallel to Assam Public School. On the eastern side, it connects to DX Housing via a public road, providing direct access.

The terrain is predominantly flat, with existing utilities like Gas supply lines, electrical overhead and underground lines, street light poles, CCTV Cables, Water Supply lines, etc. are passing through the proposed site. These shall be located and shifted by M/S OIL. Site is densely vegetated with significant number of trees are spread across the site. A road which is presently passing through the site is to be diverted by M/S OIL to the southern boundary as per their requirement to enhance privacy of gated housing.

Infrastructure passing near the site include water lines running along the western boundary and drainage channels situated on both the northern and western sides. Additionally, an existing electrical substation is positioned on the northeastern side of the proposed site.

FIGURE 2: BQ HOUSING SITE PHOTOS





1.3 OIL'S REQUIREMENTS

- a) 192 Units of Flats
- b) Guest house with 20 nos. of rooms with attached toilets and 04 Nos. Dormitory.
- c) Community centre
- d) Boundary wall, Entry-Exit Gate
- e) Children Park and Green Area
- f) Security and CCTV Surveillance Room

1.4 APPLICABLE CODES & STANDARDS

The site falls under the jurisdiction of Duliagan, Assam. Applicable building bye-laws/ regulations considered is Assam Unified Building Construction (Regulation) Bye Laws 2022. Further, other building bye-laws also considered for the designing of residence which are mentioned below:

- NBC 2016
- MoHUA Model Building Bylaws, 2016

Extracts of some important parameters are shown below from Assam Unified Building Construction (Regulation) Bye Laws 2022 and NBC 2016:

FIGURE 3: BYELAWS FOR F.A.R. (FLOOR AREA RATIO)

Base FLO OR ARE A RATIO (FAR)	Existing Road Width (m)	Plot Size (Sq m/ Bigha /Katha)								Plot Size above 13380 sq m (10 B)	
		Plot Size up to 670 sq m (2.5 K)		Plot Size above 670 sq m up to 1338 sq m (2.5 K- 1 B)		Plot Size above 1338 sq m up to 6690 sq m (1B- 5 B)		Plot Size above 6690 sq m (5 B) up to 13380 sq m (10 B)			
			FLOOR AREA RATIO (FAR)		FLOOR AREA RATIO (FAR)		FLOOR AREA RATIO (FAR)		FLOOR AREA RATIO (FAR)		FLOOR AREA RATIO (FAR)
100	Above 3.6 upto 4.5		125		125		125		125		125
125	Above 4.5 upto 6.6		125		125		125		150		150
150	Above 6.6 upto 8.0		150		160		175		175		175
150	Above 8.0 upto 15		150		175		225		225		275
160	Above 15		175		200		250		275		300

Buildings comprising of Residential and Commercial use.

FIGURE 4: BYE LAWS FOR EXTERIOR OPEN SPACE

(b) SETBACK REGULATIONS

Minimum setback of the building or the structure from the prescribed street line-

(i) FRONT SETBACK:

Every building fronting a street shall have a front space from the prescribed street line forming an integral part of the site as below:—

Existing width of Street fronting the plot	Minimum Front Open Space		
	Upto 9.6m	Up to 15.6 m*	Above 15.6 m*
Up to 6.6 Mts.	3.6 m	4.5 Mts.	6.0 Mts.
More than 6.6 to 15.0 Mts.	4.5 m	6.0 Mts.	7.5 Mts.

(ii) SIDE AND REAR SETBACK:

Sl.no	Height of the building *	Side and Rear Open space to be left around the Building
1	9.6 m	1.8 m
2	12.6 m	2.4 m
3	15.6 m	3.6 m
4	18.6 m	4.2 m
5	21.6 m	5.0 m
6	24.6m	5.5 m
7	27.6m	6.0 m
8	30.6m	7.0 m
9	36.6m	9.0 m
10	45.6 m	10.0 m
11	54.6 m and above	12.0 m

FIGURE 5: BYELAWS FOR PARKING

"MINIMUM NO. OF OFFSTREET PARKING SPACE"				
Sl No	Building Types	Parking Requirements		Visitors' Car
		Car	Two Wheeler	
1	Residential Buildings/ Group Housing / Apartment Building	1 per Dwelling Unit measuring 66 Sqm and above in carpet area; 2 Per Dwelling Unit measuring 120 sq m and above in carpet area including units constructed under PMAY.	1 per Dwelling unit of 66 sq.m. of carpet area or below.	(i) For EWS and LIG Category of houses up to 66 sqmt of carpet area shall provide 1 car per 10 dwelling unit. (ii) For dwelling units above 66sqm 1 car per 4 dwelling units shall be provided.

FIGURE 6: NBC 2016 BYELAWS FOR GROUP HOUSING

Table 23 Minimum Requirements for Fire Fighting Installations

(Clauses 4.18.2, 6.1.2, 6.2.3, 6.3.2, 6.4.3, 6.5.2, 6.5.2.1, 6.5.2.2, 6.5.2.3, 6.5.2.4, 6.5.2.5, 6.6.2, 6.7.2, 6.8.2 and 6.9.2)

Sl No.	Type of Building Occupancy	Type of Installation									Water Supply (in l)		Pump Capacity (in l/min)	
		Fire Extinguisher	Hose Reel	Dry Riser (see Note 6)	Wet Riser	Down-Corner	Yard Hydrant	Automatic Sprinkler System	Manually Operated Electric Fire Alarm Systems	Automatic Detection and Alarm System	Underground Static Water Storage Tank	Terrace Tank	Pump Near Underground Static Water Storage Tank (Fire Pump) with Minimum Pressure of 3.5 kg/cm ² at Terrace Level	At the Terrace Tank Level with Minimum Pressure of 2.0 kg/cm ²
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
RESIDENTIAL BUILDINGS (A)														
1)	Less than 15 m in height	R	R	NR	NR	NR	NR	R (see Note 2)	NR	NR	NR	5 000 (5 000) (see Note 4)	NR	450 (450) (see Note 4)
2)	15 m and above but not exceeding 35 m in height	R	R	NR	NR	R	NR	R (see Note 2)	R (see Note 7)	NR	NR	25 000	NR	900

For Community Centre

ASSEMBLY BUILDINGS (D) (see Note 16)													
a)	Buildings (D-1 to D-5)												
1)	Less than 10 m in height												
	i) Up to 300 persons	R	R	NR	R	NR	R (see Note 4)	R	NR	NR	20 000 (5 000) (see Note 6)	NR	450 (450) (see Note 6)

For Guest House

d)	Hotels (A-5)												
1)	Less than 15 m in height												
	i) Floor area not exceeding 300 m ² on any of the floor	R	R	NR	NR	NR	R (see Note 4)	R	NR	NR	5 000 (5000) (see Note 6)	NR	450 (450) (see Note 6)
	ii) Floor area exceeding 300 m ² but not more 1 000 m ² on any of the floor	R	R	R (see Note 7)	NR	NR	R (see Note 4)	R	R	10 000 for every 500 m ² floor area subject to minimum of 50 000 (see Note 7)	10 000 (5000) (see Note 6)	(see Note 14 in case of provision of wet riser)	450 (450) (see Note 6)

1.4.1 CLIMATE AND ENVIRONMENTAL CONSIDERATIONS

Duliajan experiences a humid subtropical climate marked by high humidity, heavy monsoon rainfall, and mild winters. The annual average temperature is around 25.6°C, with peak summer temperatures reaching up to 32.7°C in April and lows of approximately 13.3°C in January. Rainfall peaks during the monsoon, particularly in July, which records around 389-489 mm. Predominant wind flows come from the east-northeast to southeast directions, with typical wind speeds of 3.7–9.3 km/h. Solar analysis shows high sun angles in summer with sunrise around 4:15 AM and sunset by 6:07 PM, peaking at a solar altitude of ~86° in June. Hence, optimal building orientation would align longer facades along the North–South axis to minimize direct heat gain, especially from low-angle East and West sun.

FIGURE 7: AVERAGE ANNUAL TEMPERATURE IN DULIAJAN

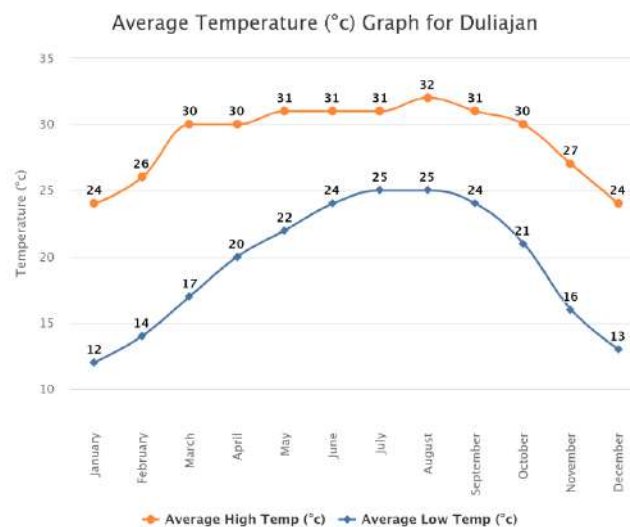


FIGURE 8: CLIMATE DATA OF DULIAJAN

Climate Duliajan Oil Town: Weather By Month

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Nov	Oct	Dec	Year
Record high °C (°F)	30.0 (86.0)	31.0 (87.8)	37.0 (98.6)	41.0 (105.8)	39.0 (102.2)	38.0 (100.4)	39.0 (102.2)	38.0 (100.4)	36.0 (96.8)	33.0 (91.4)	30.0 (86.0)	28.0 (82.4)	41.0 (105.8)
Average high °C (°F)	24.48 (76.06)	26.0 (78.8)	29.7 (85.46)	32.67 (90.81)	31.19 (88.14)	31.46 (88.63)	29.14 (84.45)	32.37 (90.27)	29.93 (85.87)	29.37 (84.87)	26.71 (80.08)	24.53 (76.15)	28.96 (84.13)
Daily mean °C (°F)	20.31 (68.56)	22.06 (71.71)	25.29 (77.52)	28.19 (82.74)	27.96 (82.33)	29.14 (84.45)	27.42 (81.36)	29.8 (85.64)	27.66 (81.79)	26.27 (79.29)	23.1 (73.58)	20.45 (68.81)	25.64 (78.15)
Average low °C (°F)	13.28 (55.9)	15.64 (60.15)	18.04 (64.47)	20.71 (69.28)	22.16 (71.89)	24.62 (76.32)	24.61 (76.3)	25.49 (77.88)	23.79 (74.82)	20.88 (69.58)	17.16 (62.89)	13.82 (56.88)	20.02 (68.04)
Record low °C (°F)	10.0 (50.0)	13.0 (55.4)	13.0 (55.4)	16.0 (60.8)	15.0 (59.0)	19.0 (66.2)	19.0 (66.2)	23.0 (73.4)	17.0 (62.6)	15.0 (59.0)	13.0 (55.4)	11.0 (51.8)	10.0 (50.0)
Average precipitation mm (inches)	9.55 (0.38)	34.08 (1.34)	34.82 (1.37)	59.87 (2.36)	124.62 (4.91)	215.94 (8.5)	389.36 (15.33)	231.55 (9.12)	193.8 (7.63)	76.31 (3.0)	19.71 (0.78)	6.39 (0.25)	116.33 (4.58)
Average precipitation days (≥ 1.0 mm)	1.73	3.0	4.0	5.82	7.27	8.0	8.45	8.09	7.55	5.91	2.82	0.73	5.28
Average relative humidity (%)	61.92	63.96	60.85	62.84	76.81	85.24	91.74	86.08	89.37	84.83	80.02	68.37	76.0
Mean monthly sunshine hours	10.29	10.65	11.1	10.8	10.3	11.69	10.65	9.85	9.15	9.91	10.23	10.51	10.43

1.5 DESIGN PROPOSAL

1.5.1 MASTER PLANNING CONCEPT

The design philosophy of this project focuses on creating a comfortable and sustainable living environment. It incorporates modern design, ensuring it aligns with both existing and evolving habitats. The plan includes rainfall protection, green building features, and sustainable infrastructure to enhance durability and environmental responsibility. With community spaces and green landscapes, the project supports a practical and balanced lifestyle for its residents.

The building is oriented such that its longer façades run along the North–South axis.

RESIDENTIAL UNITS:

The residential units in the housing area are designed to offer comfort, and functionality. Each unit features:

Spacious Layouts: Well-ventilated and spacious areas are designed to maximize natural light and airflow, creating a comfortable and refreshing indoor environment. Thoughtful spatial planning ensures seamless movement, while large openings and strategically placed windows enhance cross-ventilation and promote a sense of openness. These features contribute to a healthier, more energy-efficient living space

Inclusive Design: The development prioritizes inclusivity, integrating thoughtful provisions for ease of movement and a supportive living environment.

COMMUNITY CENTRE

The community centre is providing a versatile space for social gatherings, cultural events, and recreational activities. It is planned to include:

- Multipurpose Hall: Adaptable spaces that can be used for meetings, celebrations, and other events is planned at first floor.
- Fitness Centre/Activity Area: An activity area may be proposed to promote health and wellness among residents.
- Indoor Sports Game room: A room has been planned for indoor sports such as table tennis, Carrom board, chess.
- Kids Sports Room: Kids sports room is also planned for indoor kids' activity
- ATM and Utility Rooms: A space dedicated for ATM facility and 2 utility rooms for various purposes has been planned in the community centre.

PARKING SOLUTION:

The parking allocation is structured to ensure efficient use of space while maintaining accessibility. Each of the 192 residential units is allocated 1 ECS, with visitor parking provisions calculated based on a ratio of 1 ECS per 4 units, resulting in 48 ECS reserved for visitor use. Additionally, the guest house parking follows a 1 ECS per 3 rooms ratio, providing a total of 8 ECS. This arrangement supports both residents and visitors, optimizing convenience and functionality within the development

In compliance with relevant bylaws, the project includes designated parking for two-wheelers at stilt level and on-site, with an allocation of one space per residential unit. With 192 residential units, the total provision amounts to approximately 192 two-wheeler parking spaces.

As per the client's requirements, on-site car parking has been planned perpendicular to the road to ensure efficient manoeuvring and smooth vehicle movement. All 192 resident parking bays and 8 guest house parking slots are proposed to be covered, while the remaining 48 visitor parking spaces will remain open.

CHILDREN'S PARK AND GREEN AREAS:

The project incorporates dedicated spaces for children and green areas, designed to promote outdoor engagement and strengthen community interaction. Key features include a Central Park with a dedicated children's play area, ensuring a safe and vibrant recreational space. Additionally, green zones are integrated throughout the development, including a green lane along the boundary wall, landscaped areas around the community centre and guest house, and several tot-lot spaces. Well-planned walking paths encourage relaxation, fostering a welcoming and connected environment for residents.

EXISTING INFRASTRUCTURE

Any existing infrastructure infringing the new proposal at BQ site is proposed for dismantling strategically before development of new housing.

All the utilities running through the site shall be shifted by M/s OIL.

PUBLIC ROAD DIVERSION:

To ensure privacy and gated community concept for the project, the existing public road passing through the BQ housing area is proposed to be diverted towards the southern side of the site, running alongside Assam Public School. This adjustment aims to enhance traffic flow and ensure a safer, more organized circulation pattern within the community

This diversion will:

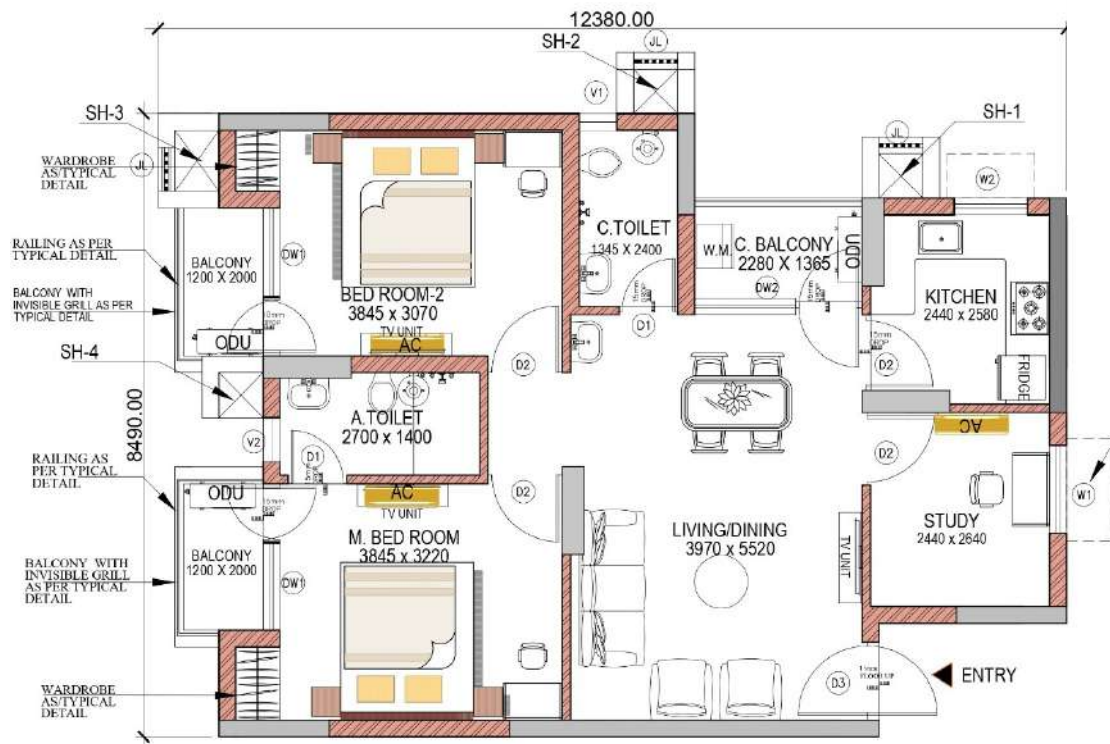
- *Improve Safety:* Reduce the risk of accidents and ensure a safer environment for residents.
- *Enhance Privacy:* Create a more private and serene living space.

1.5.2 BUILDING PLAN (BQ HOUSING - WORKMEN):

The BQ housing site area covers approx. 29,400 sqm and includes 192 units as per OIL's requirements. The housing society is complemented by community centre and guest house. This housing type is designed to provide comfortable and functional living spaces for workmen, fostering a sense of community and belonging. Infrastructure planned for development is as under:

- 08 Residential Towers with Stilt+6 Floor Configuration for 192 nos., 2.5BHK Units (2 Bedrooms + 1 Study Room)
- One G+3 Guest House having 23 nos. of Single rooms and 4 (six bedded) dormitories.
- Community Centre - G+1 with community hall
- Children Park and Green Area
- Two Wheelers Stilt Parking and Onsite Parking for Four Wheelers.
- Ancillary Infrastructure (UGT, STP, ESS, Guard Room with CCTV Surveillance, Security Hut)
- Typical unit plinth area is 90 sqm excluding service shafts
- Typical Cluster (Floor) area is 484 sqm

FIGURE 11: TYPICAL UNIT PLAN FOR BQ HOUSING



1.5.3 BQ INFRASTRUCTURE DETAILS:

RESIDENTIAL TOWERS: The BQ Workmen Housing comprises eight residential towers, each with a stilt + six-floor configuration. The towers have two-wheeler parking at the stilt level, providing dedicated space for resident's vehicles and multipurpose space for functional use.

A 1.8-meter-wide pathway runs alongside each tower, enhancing pedestrian movement, while on-site car parking is strategically placed nearby for added convenience.

BQ TYPICAL UNIT:

The BQ housing units are designed to provide a comfortable, well-ventilated, and spacious living environment. Each 2.5 BHK unit consists of two bedrooms, a study room, a combined living and dining area, along with a kitchen and Two toilets. The layout features large windows to enhance natural light and airflow, creating a bright and airy atmosphere. Additionally, each unit includes three balconies (two attached to the bedrooms and one to the living area) offering outdoor access and improved ventilation with invisible grills for safety without obstructing the view. To support efficient utility management, dedicated shafts are planned for essential services, accommodating rainwater drainage, wastewater disposal, and gas pipeline systems, ensuring seamless integration of utilities within the

development. Doors are carefully planned for movement of regular size residential appliances.

CLUSTER AND LOBBY CORE:

Each typical cluster consists of four units, strategically arranged to ensure privacy for residents. The lobby core is designed with two staircases and two lifts—a 15-person capacity passenger lift and a stretcher lift (15 Passenger)—to facilitate efficient vertical circulation. Additionally, the lobby integrates dedicated areas & shafts for essential services, including electrical systems and fire protection, ensuring seamless functionality and compliance with safety standards.

GUEST HOUSE – BQ

The guest house has been thoughtfully planned to accommodate 23 rooms with attached toilets, along with 4 nos. 6 bedded dormitories each with an attached toilet, a common commercial kitchen, and a dining area along with a multipurpose room for various activities. The proposed G+3 structure, has been designed to provide a functional and well-equipped living environment. Additionally, utility spaces for washing, storage, and other essential functions ensure practicality and convenience.

Each guest room is spacious, featuring an attached balcony to enhance natural light. To ensure smooth and safe vertical circulation, the design incorporates a staircase and a stretcher lift (15 passenger), facilitating efficient movement between floors while prioritizing accessibility and convenience.

ENTRANCE GATES AND SECURITY:

To ensure security and controlled movement, two entrance gates have been planned (both equipped with surveillance facilities)—one as the main entrance from the road connecting to DX Housing and the other providing secondary access from adjacent colony road of TD+ housing. Additionally, a wicket gate has been designated for pedestrian movement.

For enhanced safety and monitoring, one security hut is strategically positioned at the southern side of the site, ensuring continuous oversight.

Following are the broad features of overall conceptual plan of BQ housing:

- Maximum Stilt+6 Storey is proposed for BQ Housing
- Proposed Site Area: 29,400 sqm (7.26 Acres)
- Proposed Built-up area – 30,824 Sqm
- Proposed Ground Coverage: 5140 sqm (18%) (Excluding STP and UGT Tank)
- Floor to Floor height –Dwelling Units- 3.15m; Stilt– 3.00m; Community Centre: 4.2m, Guest House-3.15m
- Built-up Area at typical floor of housing block: 484 sqm
- Built-up Area per housing unit: 90 sqm (excluding service shafts)
- Carpet Area per Unit: 69 sqm

- Common Area/Core Area: 124 sqm (Including Service Shafts)
- Car Parking Provided: 192 ECS (Housing Units) + 48 ECS (Visitors) + 8 ECS (Guest House)-On Street Covered Parking Sheds for 208 Nos.
- Two Wheelers Parking: 192 Nos. in Stilt floor

TABLE 1: PROPOSED BUILDING PARAMETERS

S. NO.	Particular	Minimum / Permissible *	Proposed
1	Plot Area	1337.8 Sqm (Residential)	29,400 sqm (7.26 Acres)
2	Front Setback	7.5 m	9.0 m
3	Side Setback	6.0 m	6.0m
4	F.A.R.	1.5 (max)	0.90
5	Minimum Approach Road Width	9.0 (min.)	12
6	Height	1.5xRoad width + Front Open Space	22.35 m (Up to Terrace Level)
7	Ground Coverage	-	18%
8	Car Parking	1(ECS/Unit)	192 Units + 48 Visitors Parking+8 Guest House Parking and 192 Two Wheelers Parking
9	GREEN AREA	15% of total Plot Area (NBC) 20% Assam Unified Building Bylaws	6,000 sqm Approx. (20%)

*Assam Unified Building By-Laws 2022

1.6 CONCEPT ELEVATION– BQ HOUSING

The concept elevation for the proposed BQ housing has been developed based on inputs from OIL, incorporating a flat roof at the Mumty level and the removal of flats at the stilt level.

The front facade features include covered balconies for enhanced privacy and weather protection, invisible grills for safety without obstructing the view, and projections at

windows and parapet levels to shield against rain. Additionally, aesthetic projections have been incorporated to enhance the visual appeal of the design.



FIGURE 12: BQ HOUSING CONCEPT ELEVATION

1.7 AREA DETAILS

BQ HOUSING BUILDING WISE AREA CHART DETAIL

SN	DESCRIPTION	AREA (SQM)/ NOS.	PERCENTAGE
01	TOTAL PLOT AREA (APPROX.)	29,400	100.00%
02	GROUND COVERAGE Excluding UGT, STP	5140	(18%)
03	TOTAL BUILT-UP AREA	30,823.40	
04	TOTAL NO. OF HOUSING BLOCKS	08	
05	TOTAL NO. OF HOUSING UNITS	192	
(A) HOUSING(G+6)		FOR EACH TOWER	TOTAL
I	STILT FLOOR AREA	467.70	3741.60
II	FIRST FLOOR AREA	483.60	3868.80
III	SECOND FLOOR AREA	483.60	3868.80
IV	THIRD FLOOR AREA	483.60	3868.80
V	FOURTH FLOOR AREA	483.60	3868.80
VI	FIFTH FLOOR AREA	483.60	3868.80
VII	SIXTH FLOOR AREA	483.60	3868.80
VIII	MUMTY AREA	50	400
(B) GUEST HOUSE (G+3)			
I	GROUND FLOOR AREA	465	465
II	FIRST FLOOR AREA	495	495
III	SECOND FLOOR AREA	500	500
IV	THIRD FLOOR AREA	500	500
V	MUMTY AREA	30	30
(C) COMMUNITY CENTRE (G+1)			
I	GROUND FLOOR AREA	490	490
II	FIRST FLOOR AREA	490	490
III	MUMTY AREA	55	55
(D) SERVICES: ESS, PUMP ROOM, STP ROOM & TWO GUARD ROOM, SECURITY HUT			
I	ESS BUILDING BLOCK	368	
II	GUARD ROOMS & SECURITY HUT	76	
III	PUMP ROOM & UGT (TENTATIVE)	205	
IV	STP (TENTATIVE)	270	
E	TOTAL BUILT-UP AREA EXCLUDING PUMP ROOM & STP		30,823.40
F	TOTAL BUILT-UP AREA EXCLUDING STILT & MUMTY FOR FAR		26,596.80
06	F.A.R (Excluding Stilt & Mumty Area)		0.90
07	Covered Parking (Housing Blocks)	192	248 ECS
	Covered Parking (Guest House)	8	
	Open Parking (Visitors)	48	
	Two Wheelers Parking	192	

2 GREEN BUILDING CONSIDERATION

2.1 GRIHA- RATING SYSTEM AND CERTIFICATION

The GRIHA V 2019 rating system consists of 30 criteria categorized under various sections such as Site Planning, Construction Management, Occupant Comfort and Wellbeing, Sustainable Building Materials, Performance Monitoring and Validation, and Innovation.

GRIHA is a performance-oriented system where points are earned for meeting the design and performance intent of the criteria. Each criterion has certain points assigned to it. It means that a project demonstrating compliance with a criterion would achieve the associated points. GRIHA is a 100-point system consisting of some core points, which are mandatory, while the rest are optional. Different levels of certification (one star to five stars) are awarded based on the number of points earned. The minimum points required for certification are 25.

The thresholds for achieving GRIHA rating are mentioned below:

Percentile Threshold	Achievable Star Rating
25–40	★
41–55	★★
56–70	★★★
71–85	★★★★
86 and more	★★★★★

2.2 TARGET GRIHA RATING & VERSION:

- The project aims to achieve 3-Star GRIHA rating under the latest GRIHA version (including any subsequent amendments).
- GRIHA compliance will be ensured across all project phases including Planning & Design, Construction, Post-Construction, and Operations.
- The contractor shall ensure compliance with the latest GRIHA checklist to achieve the targeted rating.
- The contractor shall try to get maximum points under Energy Optimization & Water Management.

2.3 IMPORTANT SUSTAINABLE FEATURES AND PARAMETERS

- Preserve top soil during construction, maintain its fertility (during construction phase) and use that preserved top soil for landscape post-construction.
- Increase total number of trees on site by 25% above the pre-construction phase OR Plant 3 trees for every 1 tree cut of the same native/naturalized species

- Reduce landscape water demand by at least 40% from the GRIHA base case.
- Water and solid waste management with waste water recycling, water conservation and rain water harvesting.
- Rainwater harvesting for managing the rainwater runoff from roof and ground surfaces.
- The STP installed on site meets the CPCB norms.
- Potable water quality, treated waste water and captured rainwater quality test reports from various sources before and after treatment shall be as per standard.
- Development of surroundings with indigenous vegetation and plantation.
- Use of energy efficient glass, cladding materials, wall appropriate to the use of the building. The ratio of glass to wall shall vary depending the direction and as required for THREE STAR GRIHA certification.
- All the fenestration shall meet the SHGC requirement of ECBC – 2017/ weighted façade average SHGC (for each orientation) meets SHGC requirements of ECBC 2017.
- Window to Wall Ratio (WWR) and Skylight Roof Ratio (SRR) shall not exceed 60% and 5% respectively.
- VLT (Visible Light Transmission) shall meet the requirement of ECBC 2017.
- The contractor shall use only those materials which conform to the GRIHA Rating System criteria & procured within 500 kms from project site.
- All sanitary and plumbing fixtures shall be of low flow category and shall reduce water demand through selection of low-flow fixtures by 70% below the GRIHA base case. Supportive documents, calculations and photographs shall be submitted by the contractor.

- Use of low emitting materials, adhesives and sealants to reduce/avoid use of materials, which are irritating and naturally cause health problems to the construction workmen and occupants.
- Limits of Low VOC content (grams of VOC per litre) for (Interior/Exterior coatings, wood polish, Adhesives etc.) should be within the prescribed limit of GRIHA v2019 and supportive documents shall have to be produced
- All the insulation used in building shall be CFC/HCFC free.
- All the fire suppression system and fire extinguisher installed in the building shall be Halon free.
- Artificial lighting design to fall within limits (lower and higher range limits) as recommended space/task specific lighting levels as per NBC and to meet a minimum uniformity ratio of 0.4.
- The indoor noise levels should be within the acceptable limits as specified in NBC and key noise source on site (like DG sets etc.) should have sufficient acoustic insulation as per NBC 2005 norms.
- Concrete blocks (hollow, solid, or AAC) with minimum 40% fly ash content by weight. Documents supporting the statement shall have to be produce.
- Ensure that at least 70% of all roads and vehicular pathways within site premises are constructed with one, or any combination, of the following:
 - Bituminous road with minimum 6% plastic waste content by weight of bitumen
 - Cast in-situ cement concrete road with minimum 30% fly ash content by weight of total cementitious material
 - Concrete blocks with minimum 40% fly ash content by weight of block

- Paver blocks containing minimum 10% C&D waste content by weight of block
- Stones from India
- Any other product with minimum 10% recycled content by weight.
- Minimum 25% replacement of Ordinary Portland Cement (OPC) with fly ash in plaster/masonry mortar or Use of Portland Pozzolana Cement (PPC) .
- Project demonstrates that at least 50% of all materials (calculated by surface area) used for building interiors meets the GRIHA criterion low-impact material requirements. Documents in support shall be required.
- All ceiling fan shall have minimum sweep size 900mm and BEE Star rated and shall conform GRIHA standard.
- For the regular monitoring of project's energy consumption, Mandatory digital meters shall be installed at the following point of source:

Energy

- Utility Grid
- On-Site renewable energy system
- Diesel Genset/ Gas – Genset (If any)
- Each building level (at each apartment level for residential and at each tenant level for retail and office buildings)

Water

- Municipal supply
- Treated water outlet from STP
- Captured Rain Water

Energy

Residential

- For basement parking lighting, community/recreation centre, water pumping, outdoor

lighting

- Lifts and common areas

Hospitalities:

- Distributed units (split/ window ACs)]
- Lighting (indoor and outdoor)

Ensure the installation of smart metering system (one-way or two-way communicable system) which is capable of tracking energy and water consumption through a web-hosted portal for all installed meters as mentioned in Appraisal 28.1.1 GRIHA Manual Vol-1.

Air & Water pollution control measures including but not limited to:

- Provision of at least 3-meter-high barricading around the construction area.
- Wheel washing facility at the vehicular entrance of the site
- Covering of fine aggregate and excavated earth on site with plastic/geotextile sheets
- Water sprinkling on fine aggregate (sand) and excavated earth
- Diesel gensets on site must have proper chimneys with their outlet facing away from the site.

Adopting strategies to reduce water use during construction like:

- Using gunny bags for curing and using ponding for curing
- Monitoring to avoid leaks and water wastage.
- Use of additives to reduce water requirements during curing iv. Use of treated waste water/captured storm water.

Implementation waste management practices during construction including but not limited to:

- Allocation of separate waste storage spaces
- Safe disposal of waste
- Hiring of waste haulers
- Provide infrastructure (multi-coloured dustbins/different garbage chutes) to building occupants to ensure segregation of waste at source
- Provide dedicated, segregated and hygienic storage spaces in the project site to store different wastes before treatment /recycling.
- Provide contractual tie-ups with waste recyclers for safe recycling for recyclable wastes like metal, paper, plastic, glass etc. Submit document highlighting tie-up with recyclers for ensuring safe recycling of recyclable wastes.

3 LANDSCAPE DESIGN CONSIDERATIONS

3.1 VISION FOR THE LANDSCAPE

The landscape design aims to create a peaceful, attractive, and eco-friendly outdoor space for residents. Set within a region known for its high rainfall and rich biodiversity, the strategy emphasizes environmental harmony, passive comfort, and functional aesthetics. Key priorities include ample green areas, integrated stormwater management, climate-adaptive planting, and user-centric open spaces that enhance the quality of life while fostering a strong connection with nature.

3.2 KEY OBJECTIVES

- Promote comfort and well-being through green, shaded environments.
- Use native species for low-maintenance, water-efficient landscaping.
- Proper drainage to manage rainwater naturally.
- Create inclusive, accessible, and engaging outdoor spaces for all age groups.
- Foster community interaction through shared green areas.

3.3 CLIMATIC AND SITE CONTEXT

Annual Rainfall: Around 2,500 mm – necessitates a robust drainage strategy and use of flood-resilient planting.

Temperature Range: Varies from 8°C in winters to 38°C in summers, indicating the need for temperature-moderating landscape elements.

Soil Composition: Primarily alluvial and loamy soils, suitable for planting due to good permeability and moderate fertility.

Vegetation Profile: A mix of deciduous and semi-evergreen species naturally thrive here, forming a base for regional planting palettes.

3.4 LANDSCAPE ZONING:

The site is divided into the following landscape zones:

TABLE 2: LANDSCAPE ZONING

Zone	Purpose
Entry & Drop-offs	Formal entry, shade, visual appeal
Internal Roads	Avenue planting, guidance, cooling
Central Park	Community use, play, recreation
Courtyards/Lawns	Microclimate near homes
Play Areas	Safe play, soft shade
Parking Zones	Shade, heat reduction
Utility Areas	Screening, low maintenance

3.5 PLANTING STRATEGY

TABLE 3: PLANTING STRATEGY

Botanical Name	Common Name	Purpose / Ideal Use
Melusi (Bakul)	Mimusops Elengi	Medium-sized evergreen tree with dense foliage
Lagerstroemia speciosa	Queen's Crape Myrtle	Flowering tree – avenues, visual nodes, parks
Polyalthia longifolia	Ashoka	Narrow columnar – roadsides, parking islands, boundaries
Arjuna	Terminalia Arjuna	Deciduous nature and broad canopy offer seasonal shading and cooling.
Azadirachta Indica	Neem	Hardy, medicinal – boundary buffers, roads, utility edges

3.6 SHRUBS & GROUND COVERS:

- IXORA COCCINEA
- HIBISCUS ROSA-SINENSIS

- DURANTA ERECTA
- FOXTAIL FERN (ASPARAGUS DENSIFLORUS)

3.7 GREEN DESIGN ELEMENTS

- Ornamental trees: At entries and drop-offs for shade and aesthetics.
- Native flowering and fruit trees: In parks and along walkways.
- Shrub borders: around parking, services, and boundaries for screening.
- Layered planting: near buildings for cooling and privacy.
- Perimeter tree belts: for dust and noise buffering.

3.8 IRRIGATION MANAGEMENT

- Provision of water connection for shrub, hedge and landscape areas.
- Rainwater reuse from STP treated water for landscape irrigation

3.9 OUTDOOR INFRASTRUCTURE & ACCESSIBILITY

- Accessible walkways with ramps.
- Shaded seating near parks and play areas.
- Jogging/walking tracks within green zones.
- Well-lit paths for safety.
- Open lawns.

3.10 MAINTENANCE & SUSTAINABILITY

- STP treated water reuse.
- Native plants for low maintenance.
- Informative signboards for residents.

STRUCTURAL DBR

4 STRUCTURE CONSIDERATIONS

4.1 INTRODUCTION

Location: Duliajan , Assam

Earthquake Zone: Zone V

The scope of work for this report includes aspects of design methodology considered for structure design/drawings of buildings and other structure in this project. This report generally covers the basis on which structural analysis, design and details will be done. The report covers:

- Structural systems
- Material
- Loading
- Design limit state combinations
- Applicable codes and standards and reference documents.

Data referred/required for design

- Approved layout Plan of the area,
- Approved Architecture drawings,
- Survey Plan of the concerned area,
- GT Report- GT Report should be borehole wise specific to the location where structure has to be constructed.

4.2 DESIGN AND DETAILING METHODOLOGY

Limit State design methods is adopted for the design of the structure. In this method, each member of the structure is designed to satisfy Serviceability criteria, and ultimate collapse criteria. This ensures satisfactory behavior and low probability of failure. The structure system consists of frame structure with conventional beam slab system. The reinforced concrete elements will be designed as per limit state method given in IS 456-2000, Handbook of reinforced concrete detailing- SP34, the steel elements will be designed as per the limit state method given in IS 800:2007.

4.3 STRUCTURAL SYSTEM

The structural system is as following:

- G+6 Storey RCC Building - Ductile RC structural walls with RC SMRFs
- Other RCC buildings-Special Moment Resisting Frame

The project is located in a seismic zone classified as V according to IS1893 (Part1): 2016, the design must consider minimizing the structural damages resulting from an earthquake & basic wind speed classified as 50m/s according to IS 875 part III.

The main considerations followed for the design of structure are:

- Structural safety and stability
- To meet the demands of aesthetics conceived by the architect
- Availability of material, equipment and expertise.
- Constructability and ease of maintenance.
- Durability
- Economy

4.3.1 SUPERSTRUCTURE

The structural system for RCC floors will comprise of RCC column and shear walls framed with RCC Beams & Slab. The stability of building against lateral forces will be entirely provided by the ductile RC structural walls with RC SMRFs for G+6 storey building and by special moment resisting frame (SMRF) for all other buildings in the scheme.

4.3.2 SUBSTRUCTURE

The substructure i.e. the foundation system will consist of pile foundation with pile cap for G+6 storey building and Pile/raft/isolated/combined footings for all other buildings on the recommendations to be made by geotechnical consultant in the GT Report.

4.3.3 RETAINING WALLS

Retaining walls shall be designed for stability (overturning and sliding) and checked for bending & shear. Active earth pressure coefficient shall be applied based on the retained soil fill at the back of walls. To avoid shrinkage cracks on later stage, Shrinkage strips/control joint shall be provided at regular intervals. Foundation of retaining wall structures shall be provided as per soil report. Retaining walls shall be analysed and designed as Beam element. All possible loading of earth pressure, surcharge, fire tender and water pressure shall be applied to retaining wall.

4.3.4 WATER RETAINING STRUCTURES

Water retaining structures shall be designed as per IS: 456-2000 and IS: 3370-2009. Since water retaining structures are subjected to cyclic loading (full or empty conditions) thus as structural

concept all water retaining structures shall be designed based on working stress method, and stress limitations in concrete and steel shall be as per IS:3370–2009.

4.3.5 CRACK WIDTH CONTROL/LIMITATIONS

Cracking of concrete will occur whenever the tensile strength of concrete is exceeded. As concrete has relatively low tensile strength as well as low failure strain in tension, cracking is usually in inevitable in normal reinforced concrete members. However, the degree of cracking (in terms of width and spacing of cracks) can be controlled through proper design. In order to avoid excessive cracking in the flexural members, maximum diameter and spacing of the reinforcements is restricted as per the building rules indicated in IS: 456: 2000. To control cracking maximum spacing of bars in all structural elements shall not be more than 200mm center to center.

As per clause 35.3 of IS-456-2000 the permissible crack width is as following.

- For Slabs and beams not greater than 0.3mm.
- For water retaining structures and elements in direct contact with soil not greater than 0.2mm.

4.3.6 EXPOSURE CONDITION

The exposure condition considered in design is as stated below as per IS 456:

For surfaces exposed to external weather: Severe

For other surfaces : Moderate

4.3.7 NOMINAL COVER

Clear cover to reinforcement shall be as follows:

Foundation	:	75 mm
Column	:	50 mm
Beams:	:	40 mm
Slab	:	25 mm
RCC Wall	:	30mm

4.3.8 CONCRETE SPECIFICATION

Concrete design grades used shall be as follows:

Elements	:	Minimum Concrete Grade
Beams & Slab	:	M30
Columns	:	M30
Retaining Wall	:	M30
Foundation	:	M30
Piles and Pile cap	:	M30
Lean Concrete (PCC)	:	1:4:8/1:3:6 (M15/M10)

4.3.9 REINFORCEMENT AND STRUCTURAL STEEL SPECIFICATION

Grade of reinforcement	:	Fe550D for all RCC components
Structural Steel (built-up sections)	:	350MPa
Base plates/connection plates	:	350MPa
Hot-Rolled Standard Sections	:	350MPa
Tubular Sections	:	240MPa / 310MPa
Anchor Bolts	:	4.6, 4.8 & 8.8

4.3.10 EXCAVATION

Excavation shall be done as per structural drawing for excavation. It is contractor's responsibility to design earth retaining system and submit design along with calculation for review and approval by the site in charge. In case of water-logged site, water table shall be brought down by installing discharge point. Contractor shall prepare scheme for discharging water volume from site.

4.4 STRUCTURAL MODELING AND SOFTWARE

(ETABS / STAAD Pro, MS EXCEL, AUTOCAD, etc.)

3-D Analysis of all the buildings Structures will be carried out using ETABS / STAAD Pro Software package and analysis/design results shall be used for designing/detailing the various elements. All the beams and columns will be idealized as Beam Elements. The structure shall be analyzed and designed for all possible Combinations of gravity loads (dead and live load), and lateral loads (Earthquake/Wind loads).

Apart from ETABS/STAAD, design of Foundations, Slab, etc shall be done as per the Standard methods (In-house MS Excel sheet) and relevant codal provisions. Drawings shall be prepared in Auto-cad.

4.5 LOADING PARAMETERS

4.5.1 MATERIAL PROPERTIES

Unit Weight of Reinforced Concrete	=	25.0 kN/m ³
Unit Weight of Plain Concrete	=	24.0 kN/ m ³
Unit Weight of Steel	=	78.5 kN/ m ³
Unit Weight of AAC block	=	8.5 kN/ m ³
Modulus of elasticity of steel (Es)	=	2.1 x 10 ⁵ N/mm ²
Modulus of elasticity of concrete (Ec)	=	5000√f _{ck} N/mm ²
Dry density of Soil	=	18.0 KN/m ³
Density of saturated Soil	=	18.0 KN/m ³
Density of Water	=	10.0 KN/m ³

4.5.2 DEAD LOADS (DL)

Self-Weight

Self-weight of RCC/steel column and beams are automatically considered by the analysis software i.e. (ETABS / STAAD Pro.)

Dead load at RCC floor level shall be as follows:

200mm Slab (0.200 x 25)	=	5.00 KN/m ²
150mm Slab (0.150 x 25)	=	3.75 KN/m ²
125mm Slab (0.125 x 25)	=	3.125 KN/m ²
Wall Brick Load		
100mm brick	=	0.85 KN/m/meter height
225mm brick	=	1.9125 KN/m/meter height

4.5.3 IMPOSED LOAD/LIVE LOAD (IL/LL)

All Rooms	:	3.00 kN/m ²
Toilet & Bathrooms & Kitchen	:	2.00 kN/m ²
Balconies	:	3.00 kN/m ²
Steel/RCC Roof	:	0.75 kN/m ² without access

Steel/RCC Roof	:	1.5 kN/m ² with access
Corridors and Staircase	:	3.00 kN/m ²

Average finishing on terrace = 150 mm

Solar panel load = 0.5 kN/m²

4.5.4 EARTHQUAKE LOADS (EQL)

The structure is to be designed for the minimum state seismic bases hear set out by IS-1893(Part-I) using the parameters given below:

Location of Building	:	Duliajan, Assam
Soil Type	:	As per G.T. Report
Damping	:	5%
Zone Factor (Z)	:	0.36 for Zone V
Importance Factor (I)	:	1.2
Response Reduction Factor (R)	:	5

4.5.5 WIND LOADS (WL)

The structure will be designed for the wind load as per the IS 875(Part3)

Basic Wind Velocity	:	50.0m/sec
Probability factor (k ₁)	:	1
Topography factor (K ₂)	:	1.07 (may vary depending on height of building)
Topography factor (K ₃)	:	1.0
Importance Factor for the cyclonic region (K ₄)	:	1.0

Design wind speed (V_z) : 53.5/s

Design wind pressure (P_z) : 1717.35 N/m²

4.5.6 TEMPERATURE LOAD (TL)

Analysis may be carried out for ambient temperature variation. The temperature variation shall be considered as average maximum annual variation in temperature. The average maximum annual variation in temperature for this purpose shall be taken as the difference between the mean of the

daily minimum temperature during the coldest month of the year and mean of daily maximum temperature during the hottest month of the year.

Temperature Differential (ΔT)

$T_{\max} = 45^\circ\text{C}$ & $T_{\min} = 7.5^\circ\text{C}$

Temperature load shall be $2/3$ of average of maximum & min temperature i.e. $= 17.5^\circ\text{C}$

Coefficient of thermal expansion for concrete and steel is written above.

4.6 LOAD COMBINATIONS

For Calculating Seismic Weight

Load Combination as per IS-1893 (Part 1) and IS: 875

- For imposed UDL up to 3 kN/m² : (DL + 0.25 LL)
- For imposed UDL above 3 kN/m² : (DL + 0.50 LL)

Note: Live Load on roof not considered while calculating the seismic weight of building.

Primary Loads

- Earthquake X direction ($\text{EQ} \pm \text{X}$)
- Earthquake Y direction ($\text{EQ} \pm \text{Y}$)
- Earthquake Z direction ($\text{EQ} \pm \text{Z}$)
- Dead load (DL)
- Live load (LL)
- Wind load X direction ($\text{WL} \pm \text{X}$)
- Wind load Z direction ($\text{WL} \pm \text{Z}$)

Note: As per clause B-2.3 of IS 456:2000, wind and seismic forces need not to be considered as acting simultaneously.

****ULS LOAD COMBINATIONS**

LOAD COMB 100 1.5DL+1.5LL

22 1.5 23 1.5 20 1.5 21 1.5

LOAD COMB 101 1.2 DL+1.2 LL+1.2 EQX

22 1.2 23 1.2 20 1.2 25 1.2 21 1.2

LOAD COMB 102 1.2 DL +1.2 LL - 1.2 EQX

22 1.2 23 1.2 20 1.2 25 -1.2 21 1.2

LOAD COMB 103 1.2 DL+1.2 LL+1.2 EQZ

22 1.2 23 1.2 20 1.2 26 1.2 21 1.2

LOAD COMB 104 1.2 DL+1.2 LL-1.2 EQZ

22 1.2 23 1.2 20 1.2 26 -1.2 21 1.2

LOAD COMB 105 1.2 DL+1.2 LL+1.2 WX

22 1.2 23 1.2 20 1.2 14 1.2 21 1.2

LOAD COMB 106 1.2 DL +1.2 LL - 1.2 WX

22 1.2 23 1.2 20 1.2 14 -1.2 21 1.2

LOAD COMB 107 1.2 DL+1.2 LL+1.2 WZ

22 1.2 23 1.2 20 1.2 15 1.2 21 1.2

LOAD COMB 108 1.2 DL+1.2 LL-1.2 WZ

22 1.2 23 1.2 20 1.2 15 -1.2 21 1.2

LOAD COMB 109 1.5 DL+1.5 EQX

22 1.5 23 1.5 25 1.5 21 1.5

LOAD COMB 110 1.5 DL-1.5 EQX

22 1.5 23 1.5 25 -1.5 21 1.5

LOAD COMB 111 1.5 DL+1.5 EQZ

22 1.5 23 1.5 26 1.5 21 1.5

LOAD COMB 112 1.5 DL-1.5 EQZ

22 1.5 23 1.5 26 -1.5 21 1.5

LOAD COMB 113 1.5 DL+1.5 WX

22 1.5 23 1.5 14 1.5 21 1.5

LOAD COMB 114 1.5 DL-1.5 WX

22 1.5 23 1.5 14 -1.5 21 1.5

LOAD COMB 115 1.5 DL+1.5 WZ

22 1.5 23 1.5 15 1.5 21 1.5

LOAD COMB 116 1.5 DL-1.5 WZ

22 1.5 23 1.5 15 -1.5 21 1.5

LOAD COMB 117 0.9 DL+1.5 EQX

22 0.9 23 0.9 25 1.5 21 0.9

LOAD COMB 118 0.9 DL-1.5 EQX

22 0.9 23 0.9 25 -1.5 21 0.9

LOAD COMB 119 0.9 DL+1.5 EQZ

22 0.9 23 0.9 26 1.5 21 0.9

LOAD COMB 120 0.9 DL-1.5 EQZ

22 0.9 23 0.9 26 -1.5 21 0.9

LOAD COMB 121 0.9 DL+1.5 WX

22 0.9 23 0.9 14 1.5 21 0.9

LOAD COMB 122 0.9 DL-1.5 WX

22 0.9 23 0.9 14 -1.5 21 0.9

LOAD COMB 123 0.9 DL+1.5 WZ

22 0.9 23 0.9 15 1.5 21 0.9

LOAD COMB 124 0.9 DL-1.5 WZ

22 0.9 23 0.9 15 -1.5 21 0.9

LOAD COMB 125 1.2 DL+1.2 LL+1.2 EQX+ 0.36 EQZ

22 1.2 23 1.2 20 1.2 25 1.2 26 0.36 21 1.2

LOAD COMB 126 1.2 DL +1.2 LL - 1.2 EQX + 0.36 EQZ

22 1.2 23 1.2 20 1.2 25 -1.2 26 0.36 21 1.2

LOAD COMB 127 1.2 DL +1.2 LL + 1.2 EQX - 0.36 EQZ

22 1.2 23 1.2 20 1.2 25 1.2 26 -0.36 21 1.2

LOAD COMB 128 1.2 DL +1.2 LL - 1.2 EQX - 0.36 EQZ

22 1.2 23 1.2 20 1.2 25 -1.2 26 -0.36 21 1.2

LOAD COMB 129 1.2 DL+1.2 LL+0.36 EQX+1.2 EQZ

22 1.2 23 1.2 20 1.2 25 0.36 26 1.2 21 1.2

LOAD COMB 130 1.2 DL +1.2 LL -0.36 EQX + 1.2 EQZ

22 1.2 23 1.2 20 1.2 25 -0.36 26 1.2 21 1.2

LOAD COMB 131 1.2 DL +1.2 LL +0.36 EQX - 1.2 EQZ

22 1.2 23 1.2 20 1.2 25 0.36 26 -1.2 21 1.2

LOAD COMB 132 1.2 DL +1.2 LL - 0.36 EQX - 1.2 EQZ

22 1.2 23 1.2 20 1.2 25 -0.36 26 -1.2 21 1.2

LOAD COMB 133 0.9 DL+1.5 EQX + 0.45 EQZ

22 0.9 23 0.9 25 1.5 26 0.45 21 0.9

LOAD COMB 134 0.9 DL+1.5 EQX - 0.45 EQZ

22 0.9 23 0.9 25 1.5 26 -0.45 21 0.9

LOAD COMB 135 0.9 DL-1.5 EQX + 0.45 EQZ

22 0.9 23 0.9 25 -1.5 26 0.45 21 0.9

LOAD COMB 136 0.9 DL-1.5 EQX - 0.45 EQZ

22 0.9 23 0.9 25 -1.5 26 -0.45 21 0.9

LOAD COMB 137 0.9 DL+0.45 EQX + 1.5 EQZ

22 0.9 23 0.9 25 0.45 26 1.5 21 0.9

LOAD COMB 138 0.9 DL+0.45 EQX - 1.5 EQZ

22 0.9 23 0.9 25 0.45 26 -1.5 21 0.9

LOAD COMB 139 0.9 DL-0.45 EQX + 1.5 EQZ

22 0.9 23 0.9 25 -0.45 26 1.5 21 0.9

LOAD COMB 140 0.9 DL-0.45 EQX - 1.5 EQZ

22 0.9 23 0.9 25 -0.45 26 -1.5 21 0.9

FOUNDATION DESIGN COMBINATIONS

LOAD COMB 200 1 DL+1 LL

22 1.0 23 1.0 20 1.0 21 1.0

LOAD COMB 201 1 DL+1 LL+1 EQX

22 1.0 23 1.0 20 1.0 25 1.0 21 1.0

LOAD COMB 202 1 DL +1 LL - 1 EQX

22 1.0 23 1.0 20 1.0 25 -1.0 21 1.0

LOAD COMB 203 1 DL+1 LL+1 EQZ

22 1.0 23 1.0 20 1.0 26 1.0 21 1.0

LOAD COMB 204 1 DL+1 LL-1 EQZ

22 1.0 23 1.0 20 1.0 26 -1.0 21 1.0

LOAD COMB 205 1 DL+1 LL+1 WX

22 1.0 23 1.0 20 1.0 14 1.0 21 1.0

LOAD COMB 206 1 DL +1 LL - 1 WX

22 1.0 23 1.0 20 1.0 14 -1.0 21 1.0

LOAD COMB 207 1 DL+1 LL+1 WZ

22 1.0 23 1.0 20 1.0 15 1.0 21 1.0

LOAD COMB 208 1 DL+1 LL-1 WZ

22 1.0 23 1.0 20 1.0 15 -1.0 21 1.0

LOAD COMB 209 1 DL+1 EQX

22 1.0 23 1.0 25 1.0 21 1.0

LOAD COMB 210 1 DL-1 EQX

22 1.0 23 1.0 25 -1.0 21 1.0

LOAD COMB 211 1 DL+1 EQZ

22 1.0 23 1.0 26 1.0 21 1.0

LOAD COMB 212 1 DL-1 EQZ

22 1.0 23 1.0 26 -1.0 21 1.0

LOAD COMB 213 1 DL+1 WX

22 1.0 23 1.0 14 1.0 21 1.0

LOAD COMB 214 1 DL-1 WX

22 1.0 23 1.0 14 -1.0 21 1.0

LOAD COMB 215 1 DL+1 WZ

22 1.0 23 1.0 15 1.0 21 1.0

LOAD COMB 216 1 DL-1 WZ

22 1.0 23 1.0 15 -1.0 21 1.0

LOAD COMB 217 1 DL+1 EQX

22 1.0 23 1.0 25 1.0 21 1.0

LOAD COMB 218 1 DL-1 EQX

22 1.0 23 1.0 25 -1.0 21 1.0

LOAD COMB 219 1 DL+1 EQZ

22 1.0 23 1.0 26 1.0 21 1.0

LOAD COMB 220 1 DL-1 EQZ

22 1.0 23 1.0 26 -1.0 21 1.0

LOAD COMB 221 1 DL+1 WX

22 1.0 23 1.0 14 1.0 21 1.0

LOAD COMB 222 1 DL-1 WX

22 1.0 23 1.0 14 -1.0 21 1.0

LOAD COMB 223 1 DL+1 WZ

22 1.0 23 1.0 15 1.0 21 1.0

LOAD COMB 224 1 DL-1 WZ

22 1.0 23 1.0 15 -1.0 21 1.0

LOAD COMB 225 1 DL+1 LL+1 EQX+ 0.3 EQZ

22 1.0 23 1.0 20 1.0 25 1.0 26 0.3 21 1.0

LOAD COMB 226 1 DL +1 LL - 1 EQX + 0.3 EQZ

22 1.0 23 1.0 20 1.0 25 -1.0 26 0.3 21 1.0

LOAD COMB 227 1 DL +1 LL + 1 EQX - 0.3 EQZ

22 1.0 23 1.0 20 1.0 25 1.0 26 -0.3 21 1.0

LOAD COMB 228 1 DL +1 LL - 1 EQX - 0.3 EQZ

22 1.0 23 1.0 20 1.0 25 -1.0 26 -0.3 21 1.0

LOAD COMB 229 1 DL+1 LL+0.3 EQX+1 EQZ

22 1.0 23 1.0 20 1.0 25 0.3 26 1.0 21 1.0

LOAD COMB 230 1 DL +1 LL -0.3 EQX + 1 EQZ

22 1.0 23 1.0 20 1.0 25 -0.3 26 1.0 21 1.0

LOAD COMB 231 1 DL +1 LL +0.3 EQX - 1 EQZ

22 1.0 23 1.0 20 1.0 25 0.3 26 -1.0 21 1.0

LOAD COMB 232 1 DL +1 LL - 0.3 EQX - 1 EQZ

22 1.0 23 1.0 20 1.0 25 -0.3 26 -1.0 21 1.0

LOAD COMB 233 1 DL+1 EQX + 0.3 EQZ

22 1.0 23 1.0 25 1.0 26 0.3 21 1.0

LOAD COMB 234 1 DL+1 EQX - 0.3 EQZ

22 1.0 23 1.0 25 1.0 26 -0.3 21 1.0

LOAD COMB 235 1 DL-1 EQX + 0.3 EQZ

22 1.0 23 1.0 25 -1.0 26 0.3 21 1.0

LOAD COMB 236 1 DL-1 EQX - 0.3 EQZ

22 1.0 23 1.0 25 -1.0 26 -0.3 21 1.0

LOAD COMB 237 1 DL+0.3 EQX + 1 EQZ

22 1.0 23 1.0 25 0.3 26 1.0 21 1.0

LOAD COMB 238 1 DL+0.3 EQX - 1 EQZ

22 1.0 23 1.0 25 0.3 26 -1.0 21 1.0

LOAD COMB 239 1 DL-0.3 EQX + 1 EQZ

22 1.0 23 1.0 25 -0.3 26 1.0 21 1.0

LOAD COMB 240 1 DL-0.3 EQX - 1 EQZ

22 1.0 23 1.0 25 -0.3 26 -1.0 21 1.0

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SLS LOAD COMBINATIONS

LOAD COMB 300 1 DL+1 LL

22 1.0 23 1.0 20 1.0 21 1.0

LOAD COMB 301 1 DL+1 LL+0.8 EQX

22 1.0 23 1.0 20 1.0 25 0.8 21 1.0

LOAD COMB 302 1 DL +1 LL - 0.8 EQX

22 1.0 23 1.0 20 1.0 25 -0.8 21 1.0

LOAD COMB 303 1 DL+1 LL+0.8 EQZ

22 1.0 23 1.0 20 1.0 26 0.8 21 1.0

LOAD COMB 304 1 DL+1 LL-0.8 EQZ

22 1.0 23 1.0 20 1.0 26 -0.8 21 1.0

LOAD COMB 305 1 DL+1 LL+0.8 WX

22 1.0 23 1.0 20 1.0 14 0.8 21 1.0

LOAD COMB 306 1 DL +1 LL - 0.8 WX

22 1.0 23 1.0 20 1.0 14 -0.8 21 1.0

LOAD COMB 307 1 DL+1 LL+0.8 WZ

22 1.0 23 1.0 20 1.0 15 0.8 21 1.0

LOAD COMB 308 1 DL+1 LL-0.8 WZ

22 1.0 23 1.0 20 1.0 15 -0.8 21 1.0

LOAD COMB 309 1 DL+1 EQX

22 1.0 23 1.0 25 1.0 21 1.0

LOAD COMB 310 1 DL-1 EQX

22 1.0 23 1.0 25 -1.0 21 1.0

LOAD COMB 311 1 DL+1 EQZ

22 1.0 23 1.0 26 1.0 21 1.0

LOAD COMB 312 1 DL-1 EQZ

22 1.0 23 1.0 26 -1.0 21 1.0

LOAD COMB 313 1 DL+1 WX

22 1.0 23 1.0 14 1.0 21 1.0

LOAD COMB 314 1 DL-1 WX

22 1.0 23 1.0 14 -1.0 21 1.0

LOAD COMB 315 1 DL+1 WZ

22 1.0 23 1.0 15 1.0 21 1.0

LOAD COMB 316 1 DL-1 WZ

22 1.0 23 1.0 15 -1.0 21 1.0

LOAD COMB 317 1 DL+1 EQX

22 1.0 23 1.0 25 1.0 21 1.0

LOAD COMB 318 1 DL-1 EQX

22 1.0 23 1.0 25 -1.0 21 1.0

LOAD COMB 319 1 DL+1 EQZ

22 1.0 23 1.0 26 1.0 21 1.0

LOAD COMB 320 1 DL-1 EQZ

22 1.0 23 1.0 26 -1.0 21 1.0

LOAD COMB 321 1 DL+1 WX

22 1.0 23 1.0 14 1.0 21 1.0

LOAD COMB 322 1 DL-1 WX

22 1.0 23 1.0 14 -1.0 21 1.0

LOAD COMB 323 1 DL+1 WZ

22 1.0 23 1.0 15 1.0 21 1.0

LOAD COMB 324 1 DL-1 WZ

22 1.0 23 1.0 15 -1.0 21 1.0

LOAD COMB 325 1 DL+1 LL+0.8 EQX+ 0.24 EQZ

22 1.0 23 1.0 20 1.0 25 0.8 26 0.24 21 1.0

LOAD COMB 326 1 DL +1 LL - 0.8 EQX + 0.0.24 EQZ

22 1.0 23 1.0 20 1.0 25 -0.8 26 0.24 21 1.0

LOAD COMB 327 1 DL +1 LL + 0.8 EQX - 0.0.24 EQZ

22 1.0 23 1.0 20 1.0 25 0.8 26 -0.24 21 1.0

LOAD COMB 328 1 DL +1 LL - 0.8 EQX - 0.24 EQZ

22 1.0 23 1.0 20 1.0 25 -0.8 26 -0.24 21 1.0

LOAD COMB 329 1 DL+1 LL+0.24 EQX+0.8 EQZ

22 1.0 23 1.0 20 1.0 25 0.24 26 0.8 21 1.0

LOAD COMB 330 1 DL +1 LL -0.24 EQX + 0.8 EQZ

22 1.0 23 1.0 20 1.0 25 -0.24 26 0.8 21 1.0

LOAD COMB 331 1 DL +1 LL +0.0.24 EQX - 0.8 EQZ

22 1.0 23 1.0 20 1.0 25 0.24 26 -0.8 21 1.0

LOAD COMB 332 1 DL +1 LL - 0.0.24 EQX - 0.8 EQZ

22 1.0 23 1.0 20 1.0 25 -0.24 26 -0.8 21 1.0

LOAD COMB 333 1 DL+0.8 EQX + 0.0.24 EQZ

22 1.0 23 1.0 25 0.8 26 0.24 21 1.0

LOAD COMB 334 1 DL+0.8 EQX - 0.0.24 EQZ

22 1.0 23 1.0 25 0.8 26 -0.24 21 1.0

LOAD COMB 335 1 DL-0.8 EQX + 0.24 EQZ

22 1.0 23 1.0 25 -0.8 26 0.24 21 1.0

LOAD COMB 336 1 DL-0.8 EQX - 0.24 EQZ

22 1.0 23 1.0 25 -0.8 26 -0.24 21 1.0

LOAD COMB 337 1 DL+0.24 EQX + 0.8 EQZ

22 1.0 23 1.0 25 0.24 26 0.8 21 1.0

LOAD COMB 338 1 DL+0.24 EQX - 0.8 EQZ

22 1.0 23 1.0 25 0.24 26 -0.8 21 1.0

LOAD COMB 339 1 DL-0.24 EQX + 0.8 EQZ

22 1.0 23 1.0 25 -0.24 26 0.8 21 1.0

LOAD COMB 340 1 DL-0.24 EQX - 0.8 EQZ

4.7 DETAILING OF STRUCTURES

Detailing of all structural members shall be done as per IS:456 and IS:13920 and SP:34. Detailing shall be done in such a way that it meets minimum & maximum R/F requirements, minimum and maximum spacing of bars.

For Beam

- Longitudinal R/F-Minimum $0.24\sqrt{f_{ck}/f_y}$ & maximum 0.025 of total cross sectional area at tension face.
- Shear R/F-Minimum as per IS: 456 & IS:13920 using stress of Fe550D steel for design.
- Side Face R/F- When web of beam is more than 750mm. R/F equivalent to 0.1% of web area shall be distributed equally on two faces. Spacing shall not exceed 300mm or web thickness whichever is less.

For Slab

- Minimum R/F- 0.12% of total cross sectional area.
- Maximum Spacing of bars- 3d or 200 mm whichever is less.

For Pedestal/Columns

- Min 0.8% of area of columns section required to resist direct stress and maximum 4% of column section. In extreme cases 6% of area of column section may be provided.

RCC Wall

- Min 0.4% of total cross sectional area in both directions.

Retaining Wall

- Min 0.12% of cross-sectional area as vertical R/F. Min 0.2% of cross sectional area of horizontal R/F equally divided on each face.

4.8 CONTROL OF DEFLECTIONS

The deflections due to service loads should not exceed the following values:

Vertical Deflections

Concrete Structure

As per IS 456, clause no. 23.2, final deflection due to all loads including the effects of temperature, creep and shrinkage and measured from the as-cast level of the supports, floors, roof and all other horizontal members should not exceed $\text{span}/250$.

The deflection including the effects of temperature, creep and shrinkage reoccurring after the erection of partitions and the application of finishes should not exceed $\text{span}/350$ or 20mm, whichever is less.

Steel Structure

As per the clause 5.6.1 of IS800, the deflection under serviceability loads of a building or a building component should not impair the strength of the structure or components or cause damage to the finishing. Deflections are to be checked for the most adverse but realistic combination of service loads and the rearrangement, by elastic analysis, using a load factor of 1.0.

Table 6 gives recommended limits of deflections for certain structural members and systems.

Table 6 Deflection Limits

Type of Building (1)	Deflection (2)	Design Load (3)	Member (4)	Supporting (5)	Maximum Deflection (6)
Industrial Buildings	Vertical	Live load/ Wind load	Purlins and Girts	Elastic cladding Brittle cladding	Span/150 Span/180
		Live load	Simple span	Elastic cladding Brittle cladding	Span/240 Span/300
		Live load	Cantilever span	Elastic cladding Brittle cladding	Span/120 Span/150
		Live load/ Wind load	Rafter supporting	Profiled Metal Sheetting Plastered Sheetting	Span/180 Span/240
		Crane load (Manual operation)	Gantry	Crane	Span/500
		Crane load (Electric operation up to 50 t)	Gantry	Crane	Span/750
		Crane load (Electric operation over 50 t)	Gantry	Crane	Span/1 000
		No cranes	Column	Elastic cladding Masonry/Brittle cladding Crane (absolute)	Height/150 Height/240 Span/400
	Lateral	Crane + wind	Gantry (lateral)	Relative displacement between rails supporting crane	10 mm
		Crane+ wind	Column/frame	Gantry (Elastic cladding; pendent operated) Gantry (Brittle cladding; cab operated)	Height/200 Height/400

Horizontal Deflections

Concrete Structure

- Drift due to EQ: The inter-storey drift should not exceed $H/250$.
- Drift due to WL: The overall drift of the structure under wind load should not exceed $H/500$.

Steel Structure

Lateral	{	No cranes	Column	{ Elastic cladding	Height/150
				Masonry/Brittle cladding	Height/240
	{	Crane + wind	Gantry (lateral)	Crane (absolute)	Span/400
				Relative displacement between rails supporting crane	10 mm
	{	Crane+ wind	Column/frame	Gantry (Elastic cladding; pendent operated)	Height/200
				Gantry (Brittle cladding; cab operated)	Height/400

4.9 GEOTECHNICAL DATA

As per the G.T. report.

4.10 INDIAN CODE OF PRACTICE USED

Specific applicable codes and standards will be identified and adopted in the design philosophies as appropriate to the structural elements. The latest editions of the Codes and standards will be used in designs. All design work shall be based on Indian Standards and Codes with latest revision.

List of main Indian standards codes and publications used for analysis and design is given below:

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TABLE 4: IS CODE USED FOR STRUCTURE ELEMENTS

IS 456-2000	Plain and Reinforced Concrete Code of Practice
IS: 800 – 2007	General construction in steel code of practice.
IS: 875 (Part I) – 1987	Code Of Practice for Design Loads (Other Than Earthquake) For Buildings and Structures. (DEAD LOAD)
IS: 875 (Part 2) – 1987	Code Of Practice for Design Loads (Other Than Earthquake) For Buildings and Structures (LIVE LOAD)
IS: 875 (Part 3) – 2015	Code Of Practice for Design Loads (Other Than Earthquake) For Buildings and Structures (WIND LOAD)

IS: 875 (Part 5) – 1987	Code Of Practice for Design Loads (Other Than Earthquake) For Buildings and Structures (Special loads & Combinations)
IS 1893 (Part 1) :2016	Criteria For Earthquake Resistant Design of Structures: Part 1 General Provisions and Buildings
IS: 1904 – 2021	General Requirements for Design and Construction of Foundations in Soils: code of practice
SP 34:1987	Handbook on Concrete Reinforcement and Detailing
IS 2911:2010	Design and construction of pile foundations — code of practice
IS 1786-2008	Specification for High Strength Deformed Steel Bars and wires for Concrete Reinforcement
IS 13920:2016	Ductile design & detailing of reinforced concrete structures subjected to seismic forces-Code of practice
IS 3764	Code of safety for Excavation work
IS: 8009-1976	Code of practice for calculation of settlements of foundation
IS:2950(Part1)-1981	Indian Standard Code of Practice for Design & Construction of raft foundation - (Part -1)
IS: 3370(Part-1) - 2009	Concrete Structures for Storage of Liquids - code of Practice

ELECTRICAL, ELV & FIRE FIGHTING SYSTEM DBR

5 ELECTRICAL, HVAC, ELV & FIRE FIGHTING SYSTEM DBR

5.1 PROJECT DESCRIPTION

The aim of the project is to Construct BQ Housing complex for OIL India Limited, Duliajan, Assam. It comprises of 8 blocks of (G+6) floor each residential complex including G+3 guest house and G+1 community center building inside the housing complex.

5.2 GENERAL

The EPC Contractor shall be carried out Design, Supply, Installation, Testing & Commissioning of complete Internal & External Electrification works including Low Voltage (LV) and other works as required for smooth functioning of the “Construction of BQ Housing Complex at Duliajan, Assam.” All Electrical & LV works shall be executed as per CPWD Specifications, Govt. Building Bylaws, NBC 2016, ECBC 2017, Central Pollution Control Board, relevant IS Codes, Indian Electricity Rules and Acts, Central Electricity Authority all amended up to date and if there are any special requirements for smooth functioning of the work.

The rating and capacity of equipment's indicated herein below are minimum to be provided. However, during detailed designing, if required and found necessary, the capacity/rating of the equipment's may be upgraded by the EPC Contractor.

5.3 SCOPE OF WORK

Electrical & Allied covers Electric Sub Stations, Silent type DG Set with AMF Panel Installation work, Internal Electrical Installations, HT/ LT Panels, Distribution Boards, External Electrical Installations, 11 kV/ 0.433 KV Substation equipment comprising HT Panel, transformers, HT and LT Cables, bus trunking from transformer to LT Panel, LT Panels, Automatic power factor correction panel if required, active harmonic filters, TVSS (transient voltage suppression system), SPD (surge protection system), Essential panels, floor distribution panels, panels for guest house building, panels for community center building, earthing, required inter connections, substation safety equipment's, Danger Notice Board, safety instruction charts, LT cabling from substation to the buildings, Road/ Compound/ Facade Lighting, Solar Lighting with Poles, UPS system, Internal & External Electrical Distribution work. It shall also include grid interactive roof top solar photo voltaic (PV) power

generation system, IP-based Telephones System for guest house and community center, LAN Networking System, CCTV System, Fire detection & Alarm System, Public Address System, Facade Lighting, Lifts, rising mains for each tower and EV charging Points in parking area as per GRIHA-3 Star rating etc. Suitable size shafts, cutouts, Niche, openings etc. shall be provided to facilitate installation of Rising Mains, Cable Trays, Ducts, etc. in all floor slabs of various buildings for various service areas, as required. All shafts, cutouts, Niche, openings etc. provided on floor slabs shall be suitably closed after laying of services lines as per fire safety norms of NBC 2016. Suitable doors shall be provided for all shafts at all floors as per fire safety norms stipulated in NBC 2016/ CPWD specifications.

All Services, as required like electrical power panels, Light & Fan Points, Power Points, Telephone Points LAN/Data Points, UPS Panels & UPS Power Points, Earth Pits, Earthing Network, HVAC provisions, ducting etc. shall be adequately provided as per OEM requirements.

5.4 ELECTRICAL POWER REQUIREMENT

The electrical load requirement has been calculated based on the covered area of various buildings/blocks considering the lighting load and power load of various buildings. Loads for HVAC, services, lifts, pumps, external street lights, External power, STP etc. are also considered.

The electrical load demand for the BQ Housing Complex including Guest House and Community Centre buildings and amenities being provided under the present work scope has been assessed as 1919 kVA. The electrical load calculation sheet is given in para 28 below.

5.5 SOURCE OF ELECTRICAL POWER SUPPLY

Oil India Limited shall provide 2 Nos. 11 KV HT Power connection through minimum 2 Nos. 11 KV Independent Feeder Lines (1Working+ 1Stand by) as per their technical feasibility for doing 11 kV HT Power connection shall be made available by OIL at a designated point up to the Electrical Substation of Proposed BQ Housing complex at Duliajan to meet total electrical load requirements.

5.6 SCOPE OF EPC CONTRACTOR

EPC Contractor shall commence from the point where 11 KV HT power supply available, where 11KV, 3-phase, 50 Hz, HT power supply is made available by through minimum 2 No. 11 KV Independent Feeder Lines (1W+1S) near to the proposed Substation of BQ Housing Complex. Further Incoming/Outgoing 11 KV VCB (2I/C+1 BC+ 6 O/G) Panels (1 No.) and 11 /0.433 kV Substation along with suitable size rooms/ buildings, common amenities etc. shall be provided by EPC Contractor as per details herein below.

The scope of the EPC Contractor shall include the complete work related to tapping of electrical power from the existing source up to the proposed substation. This includes, but is not limited to:

- Survey, design, and engineering of the power tapping system.
- Procurement and laying of High Tension (HT) XLPE cables of appropriate voltage grade.
- Supply and installation of all associated cable accessories such as terminations, straight-through joints, and clamps.
- Execution of the complete feeder work including trenching, cable laying, cable protection, route markers, and backfilling.
- Restoration of surfaces after cable laying, as per applicable norms and local authority requirements.
- Securing **Right of Way (RoW)**, including obtaining all necessary permissions and clearances from OIL, municipal authorities, and other statutory bodies.
- Handling of all **statutory approvals**, scrutiny and sanction procedures from relevant utility or government departments.
- Payment of all associated **fees**, including application, inspection, scrutiny, and sanction charges.

- Coordination with the OIL/ local power distribution/transmission company or authority to ensure seamless integration and energization.

The entire scope shall be the sole responsibility of the EPC contractor and compliance with applicable standards, regulations, and safety codes.

5.7 ELECTRIC SUB-STATION, 11/ 0.433 KV- 1 NO.

11/ 0.433 kV Electric Substation will be established in Proposed BQ Housing complex by EPC Contractor in the location demarcated in Master Plan. 11 KV VCB Panel (1 No.) along with suitable size rooms shall be provided near boundary wall as per statutory norms/cpwd specifications etc.

Electric Sub Station shall be provided with Indoor Type HT & LT Switchgears, 2 Nos. Dry type transformers will consist of 11 KV VCB Panel Board (Indoor), Dry type transformers, 2 X 2000 KVA – indoor type, 415 V silent type DG Set (indoor) with AMF Panel, indoor sandwich bus- ducts, LT Switch Board, HT/LT power cables, control cables and all other items/accessories as required. DG set shall be of capacity 380 KVA. DG set shall be 415 V, Radiator Cooled silent type housed in weather proof acoustic enclosure. The Transformers shall be connected to Main LT Panel through suitable size indoor AL sandwich bus ducts in the substation. DG Set shall be connected to AMF Panel through suitable size copper armoured cables. Further AMF Panel shall be connected to Essential Panel/Emergency panel through suitable size copper armoured cables.

DG AMF Panel shall have suitable ATS, control relays for auto/manual start stop, auto load management and auto load sharing. Main LT Panel shall have suitable relay for auto changeover. Also, DG Set shall have DG controller of latest version for auto & manual start/ stop, auto changeover, auto load management and DG PLC based load sharing etc. For this purpose, suitable armoured copper control cables shall be provided between Main LT Panel, AMF Panel & DG Set.

The losses of the 11/0.433 kV Dry type Transformer shall be as per latest ECBC Level-2 or above Norms for achieving GRIHA 3-star rating.

11 KV HT Panel shall have 2 Nos. Incomer, 1 No. Bus coupler and 6 Nos. Outgoing Feeders.

11 KV HT Panel shall be provided with suitable VCB mounting trolley (1 No.) for service and maintenance purpose.

11 kV Panels will be provided with tinned Copper material bus bar.

EPC contractor shall conduct fault calculation study for the project and determine the fault rating at different levels, i.e., 11 kV Panel in substation, LT side of transformer, at LT panel incoming & outgoings, at incoming panels of various buildings etc.

All necessary Electrical parameters of Multifunction Meters (i.e. Voltage, current, Frequency, Power factor, KWH, KVAR, KVA and Harmonic component etc.) and status of Numerical Relays provided in 11 KV HT panel, Main LT Panel, Hybrid Capacitor Panels if required, HVAC Panels and LT Panels of all buildings. Necessary control, communication cables for this purpose shall be provided.

HT Panel shall conform to be as per IEC-62271-100/200 as amended up to date.

The connections between Transformers to Main LT Panel through Al Conductor Sandwich Bus Duct. The minimum IP Class of the bus duct will be IP-54.

Main LT Panel in 11/0.433 KV Substation, Hybrid Type Capacitor Panels if required, AMF Panel, HVAC Panel shall be compliant to IEC-61439 Part 1 & Part 2. Trenches with suitable width & depth shall be provided for installation of HT/LT/ Emergency/AMF/ Capacitor Panels etc. and for Laying of HT/LT Power cables & Control Cables. Electrical Substation shall comprise of all ancillary equipment like Battery Chargers with suitable capacity batteries etc. Suitable size MS Chequered Plates, duly painted of minimum thickness 6 mm shall be provided for trenches inside the panel rooms, suitable size MS angles as required. Hot Dip Galvanized Perforated type Cable trays of suitable size shall be used as required.

All armoured HT/LT power cables, control cables, telephone cables, signal cables etc. shall be laid underground/ Directly buried in ground along the roads & pathways at suitable depth as per CPWD specifications. Adequate no. of Hume pipes/ DWC HDPE Pipes having suitable diameter with spare & fittings shall be laid across the roads/pathways etc. Heat sensor cable

for early detection of temperature rise inside electrical panels to be proposed for HT, LT & other electrical panels inside substation which must be connected to fire alarm control panel.

Maximum allowable Dry type transformer losses at 50% & 100% load shall comply with latest ECBC Norms and minimum GRIHA 3-star rating. All Substation/HT/LT Panel Rooms/Floor distribution panel rooms/DG room shall be provided with safety equipment/items like suitable rubber mats (as per relevant IS codes), fire buckets, fire extinguishers, first Aid box, hand gloves, danger plates (HT/LT ratings), safety charts, framed Schematic/SLD, framed shock treatment chart etc. Suitable civil foundation/trenches etc. for all substation equipment shall be provided as per design load of respective equipment. All LT Panels shall have 20% spare outgoing feeders for different rating of feeders. At least one spare outgoing feeder shall be of highest breaker rating of the panel.

Panel Bus Bar current rating shall be provided at 25% higher current rating of the highest current rating of Incomer Breaker/switchgear of the panel. 20% spare terminal blocks shall be provided for motor starter feeders and control wiring purpose in the panel.

Civil foundations and structures for Electrical Panel Room, Battery Room, Control Room, DG Room, HT Panel Room, Transformer Room, Electric Substation, and all electrical equipment's as required will be constructed by EPC Contractor.

To prevent entry of storm water into the ESS buildings and DG set area through soak-pits, the floor level of the Substation area / ESS buildings / DG set area shall be at a suitable elevation above the highest flood water level anticipated in the locality or as directed by the Engineer-in-Charge.

11/ 0.433 KV ESS shall meet the electrical load requirements of entire BQ Housing complex. Dry Type Transformers with On-Load Taps Changer $\pm 5\%$ in steps of 2.5% of capacity 2 X 2000 KVA (1W+1S) shall be provided as per GRIHA 3 Star rating.

5.8 DG SET FOR BACKUP SUPPLY

Diesel Generator sets will be provided in all the buildings and facilities of BQ Housing Complex by the EPC Contractor to meet the backup supply requirements of common area

lighting, outdoor lighting, lifts, LV system, STP, water supply pumps etc. in case of power failure.

The Rating & No. of Transformers, DG Set is summarized as below:

Sr.No.	SUBSTATION	TRANSFORMERS	DG SET
1.	11 /0.433 kV	2X 2000 KVA (1W+1Stand by)	380 KVA

Diesel Generator (DG) Set Installation and Power Supply for Emergency & Life Safety Systems

DG Set Exhaust and Noise Control Measures

- Hospital-grade silencers shall be provided for DG Set.
- *Suitable size* exhaust pipe shall be installed for DG Set.
- The exhaust stack height shall comply with CPCB latest norms/CPWD/Local Bye-Laws standards.
- A self-supporting mild steel (MS) exhaust stack structure of suitable height shall be installed as per latest CPCB norms. It shall be:
 - (i) Duly painted with synthetic enamel paint.
 - (ii) Equipped with expansion bellows as required.
 - (iii) All exhaust pipes and silencers shall be:
 - (a) Properly insulated and clad with aluminium sheets as per CPCB norms, CPWD specifications as per latest amendments.

5.9 EMERGENCY & LIFE SAFETY SYSTEMS – POWER SUPPLY REQUIREMENTS

A reliable and continuous power/control supply shall be ensured for the following emergency and life safety systems:

- (a) Fire Pumps
- (b) Lift well Pressurization and Smoke Venting Systems.
- (c) Lifts
- (d) Exit Signage Lighting

- (e) Emergency Lighting
- (f) Supply water Pumps & STP
- (g) Fire Alarm System
- (h) External Lighting
- (i) Public Address System (for emergency voice evacuation and annunciation).

5.10 CONSTRUCTION STANDARDS FOR SUBSTATION AND DG SET.

- (j) Substation and DG Set works shall strictly adhere to CPWD Specifications.
- (k) DG set should conform to latest revised norms of CPCB.
- (l) DG set shall be provided with day oil tank adjacent to DG set having capacity of 990 litres.

5.11 PIPING WORK

All pipe lines and fittings and accessories requirement inside the room/ enclosure and outside for exhaust piping shall be provided by the EPC contractor. This shall include necessary flexible pieces in the exhaust, fuel, lubricating oil, and water lines as are necessary in view of the vibration isolation requirement in the installation. Piping of adequate size shall be used for lubricating oil of the material as per manufacturer standard. However, only M.S. pipes for the exhaust shall be used. For fuel lines within the acoustic

enclosure, PVC braided pipe as per manufacturer recommendations can be used. However, for fuel lines outside the acoustics enclosure only MS pipe be used. The pipe work shall be inclusive of all fittings, Valves and accessories required such as bends, reducers, elbows, flanges, flexible connections, necessary hardware etc. The installation shall cover clamps, supports, hangers etc. as are necessary for completing the work. However, the work shall be sectionalized with flanged connections as are necessary for easy isolation for purposes for maintenance of unit as approved by Engineer-in-charge.

5.12 SMART METERING

The EPC Contractor shall be provided smart metering for Energy, water meter (for appropriate bore size) as per NBC2016/ECBC-2017 as amended up to date and as per GRIHA-3-star rating. smart energy meter to be provided for each building unit of residential complex. In addition to any other energy consuming equipment and systems as per GRIHA 3-star rating etc.

5.13 SMART METERING AND MONITORING SYSTEM

The EPC contractor shall be provided smart metering and monitoring of energy and water consumption to analyse the project performance.

(A) The source metering requirements as per the GRIHA manual are as follow:

TABLE 5: SOURCE METERING REQUIREMENTS AS PER GRIHA MANUAL

Source	Description
Energy	Ensure regular monitoring of the project's energy consumption by installing digital meters at the following point sources at the project level for: • Utility grid • On-site renewable energy system • Diesel generator set, gas genset, etc. • Each building level (at each apartment level for residential and at each tenant level for retail and office buildings).
Water	Ensure regular monitoring of the project's water consumption by installing digital meters at the following point sources: • Municipal supply • Borewell • Treated water outlet from sewage treatment plant (grey/black) • Captured rainwater for reuse in project

The EPC Contractor must provide the extended metering requirements mentioned below:

Source	Description
Energy	Digital energy meters to be installed at the following points to monitor energy consumption: A. Office/institutional/retail/transit terminals/healthcare/hospitalities: • Heating, ventilation, and air conditioning central plant: Air handling unit, cooling tower, chillers [British thermal unit meters/multi-function metering, and/or distributed units (split/ window ACs)] • Lighting (indoor and outdoor) • Basement parking lighting B. Residential • For basement parking lighting, community/recreation centre, water pumping, outdoor lighting. • Lifts and common areas.
Water	Digital water meters to be installed at the following points to monitor water consumption: • Flushing (at each building for bulk metering) • Domestic (at each building for bulk metering) • Irrigation • Heating, ventilation, and air conditioning

All smart metering and monitoring systems should be capable of the following:

- Hourly data reporting in near real-time (no more than 15-minute delay).
- Energy mix breakdown and consumption patterns.
- Water consumption patterns from various sources.
- Ability to set energy and water consumption targets, alarms, and pricing.
- Ability to compare historical trends and benchmark data.
- Real-time monitoring with user interface that operates even on mobile devices.

Ensure installation of smart metering systems which are capable of tracking energy and water consumption through a web-hosted portal for all meters.

Communicable Systems	Points
Installation of one-way communicable system	1
Installation of two-way communicable system	3

5.14 ELECTRICAL POWER DISTRIBUTION

The Electrical Power Distribution for electric supply shall be as detailed below:

- a. Indoor type Compact Aluminium Sandwich Bus Duct of suitable capacity shall be provided from Transformers to Main LT Panel, DG set to AMF Panel, AMF Panel to Main LT Panel, suitably supported with MS supports duly painted as per directions of Engineer-In- Charge. One vertical aluminium sandwich rising mains with suitable rating tap off box in each floor shall be provided for each tower, guest house and community centre building. Suitable tap off box will be provided at the top of rising main for roof top solar power system of various buildings.
- b. Adequate size and runs of XLPE insulated armoured Aluminium conductor cables shall be laid from Main LT Panel to LT Panels & Rising mains of individual buildings & blocks. Voltage drop in LT cable shall be as per norms.
- c. Separate Aluminium sandwich type rising mains to be provided for each tower, guest house building & community centre for lighting load & power load. Each distribution system shall be with Electrical panels, Floor Panels, Double door type MCB DBs etc.
- d. Separate LT room to receive power from the substation through XLPE AL armoured cable and distribute power to the entire buildings through LT Panel located in the LT room.
- e. Building (above 10 meter) shall have suitable sandwich type rising mains with suitable no. of tap off boxes for catering to loads of Lighting, Power, HVAC equipments etc. as required.
- f. Each rising main shall be Aluminium sandwich type and have tap off box at every floor, feeding the Floor Panels with incoming ACBs/ MCCBs & outgoing MCCBs of required capacities and numbers feeding the double door DBs.
- g. Power distribution from Floor Panels to DBs shall be done with copper armoured cables on surface/cable tray or through Sub- Main wiring in conduits. Earthing system capacity shall be as per CPWD norms.
- h. Hot Dip Galvanized Perforated type Cable trays of suitable size with perforation not more than 17.5% shall be provided as required in all the buildings.
- i. All power cables, up to & including 25 sq. mm. size, shall be 1.1 KV, Copper conductor, armoured, XLPE cables conforming to IS-7098 code as amended up to date. All power cables

above 25 sq. mm. sizes shall be 1.1 KV, Aluminium conductor, armoured, XLPE cables conforming to IS-7098 code as amended up to date. All power cables must be 4 cores.

- j. Voltage drop for feeders shall not exceed 2% at design load. Voltage drop for branch circuit shall not exceed 3% at design load as per ECBC norms.
- k. Surge Protection Devices (SPD) of Type-1 with suitable back-up protection shall be provided at all Incoming Feeders of Main LT Panel, Emergency panel in Substations & HVAC Panel. Likewise, Surge Protection Devices (SPD) of Type-2 shall be provided with suitable back-up protection for Building emergency Panels, Floor Panels, firefighting panel, plumbing panel etc. as required.

5.15 GAS SUPPRESSION SYSTEM

Electrical Panel Protection System: This includes Supply, Installation, Testing and Commissioning of FK-5-1- 12 (Dodecafluoro-2-Methylpentan-3-one)/Novec-1230/IG-541 gas Suppression system in accordance with the Contract Documents. Tube based Gas suppression system and Fire detection system shall be installed in substation panels (HT Panel, LT panel, Emergency panel, AMF Panel panels etc.). The detection tube shall be installed throughout the compartment of panels. Fire suppression systems and fire extinguishers installed in the building should be free of halon.

SCOPE OF WORK:

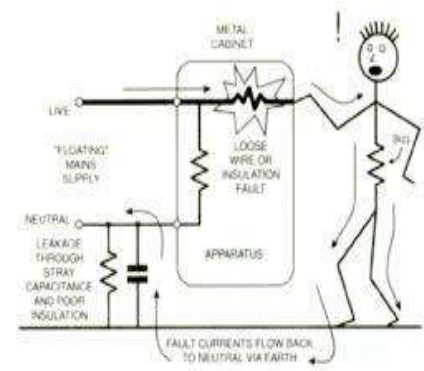
- a. Supply, Installation, Testing and Commissioning of clean Agent (Novec 1230/ FK5-1-12/IG-541) Fire Suppression system designed to provide a uniform concentration within the electrical panels in accordance with NFPA 2017
- b. Provide all engineering design and materials for a complete agent suppression system including FK-5-1- 12/IG-541 storage cylinders with steel bracket, extinguishing agent, detection tube, cylinder valve and associated accessories including but not limit to; adaptors, pressure switch, tube fittings etc, required for complete operation of system.
- c. All necessary safety requirements such as warning signs, discharge alarm shall be part of the system. The cylinders should be seamless & Petroleum and Explosives Safety Organization (PESO) approved. Dates of approval and manufacture should not be more than 6 months old.
- d. The necessary nomenclature such as pressurization level, agent volume, gross/net weight of

cylinder shall be clearly marked on cylinder.

- e. Prior to supply of material at site. Contractor must submit samples of material for approval of Engineer- in-charge.
- f. Drawing & documents clearly showing the panel, routing of tube inside the panel, location, and fixing arrangement of cylinder & system components.
- g. All doors and openings in electrical enclosures or equipment shall be properly closed or sealed to maintain the enclosure's integrity and ensure protection against dust, moisture, and other external elements.

5.16 EARTHING SYSTEM

Earthing shall be carried out using a combination of GI Plate Earthing and Copper Plate Earthing systems, as required, for substations, electrical panel boards, UPS systems, and other electrical installations in each building. The entire earthing system shall comply with the provisions of the Indian Electricity Rules, 1956, IS:3043, and IEEE 80, as amended up to date. Copper or GI earth strips shall be used to connect electrical equipment to earth pits as per relevant norms. Earth



Leakage Circuit Breakers (ELCBs) or RCBOs shall be provided in distribution boards (DBs) for individual units. The sensitivity of ELCBs or RCBOs shall be 30 mA or 300 mA. In the substation area, Copper Earth Strips and Copper earth Electrode (Copper plate earth electrode) Pits shall be installed for both body and neutral earthing of all electrical equipment, in line with CPWD Specifications. To achieve the required earth resistance based on soil conditions, earth conductivity enhancing compounds conforming to IEC 62561- 7 shall be used, with chemical type compounds preferred in hilly areas. Separate and dedicated earth pits with copper electrodes and copper earth strips shall be provided for the following Electrical equipment's:

- HT Panel
- Main LT Panel, Emergency Panel, AMF Panel etc.
- Sandwich Bus ducts.
- UPS system - Body & Neutral.

- EPABX, LAN/DATA Server and other ELV Equipment.
- Transformers – Neutral, Body & surge arrester.
- DG Sets - Neutral & Body.
- AMF Panel-Body
- Firefighting Panel.
- Plumbing Panel.
- Floor Distribution panels.
- Sub emergency panels.
- Any other equipment as required However, Gi strip earthing with GI electrode earth pits shall be provided for Electrical Floor panels, sandwich rising mains, DBs located in various buildings. For Lifts, Copper/GI strip earthing with Copper/GI electrode earth pit shall be provided as per OEM requirements.
- Suitable sized safety rubber mats with suitable thickness as per norms shall be provided for all HT& LT Panels installed in the substations and all buildings, as required.
- All three phase electrical installations shall be provided with double Earth connection and single-phase electrical installations with one Earth connection as per CPWD specifications & NBC 2016. EPC contractor to conduct soil resistivity test and fault calculation and then determine the size & number of earth pits and conductor sizing for the entire campus.

5.17 LIGHTNING PROTECTION SYSTEM

Lightning protection of various buildings and blocks shall be provided as per IS/ IEC-62305-1:2010 (latest as amended), CPWD Specifications and NBC 2016 norms. The main and most effective measure for protection of structures against physical damage is considered to be the lightning protection system (LPS). An external LPS which consists of air-termination system, down-conductor system and earthing system is intended to:

- a) Intercept a lightning flash to the structure (with an air-termination system),
- b) Conduct the lightning current safely towards earth (using a down-conductor system)
- c) Disperse the lightning current into the earth (using an earth-termination system).

Accordingly, a standard lighting protection system will be provided in all the buildings as per NBC 2016 Standards, using single prone finials, horizontal and vertical aluminium conductor/strips of suitable size, terminating in the G.I Plate Earth Pits. Aviation Obstruction Light (AOL) shall be provided in various buildings as per Civil Aviation regulations, NBC norms & CPWD Specifications as applicable. All Aviation Obstruction Lights shall be fed with Emergency power supply.

5.18 INTERNAL ELECTRIFICATION, LV & ALLIED WORKS

Following works shall be carried out in coordination with the civil work with in the buildings complete in all respects as per latest IS Codes and CPWD Specifications.

- i. Wiring & Conduiting (MS/ PVC Conduits) for internal electrification, LV & Allied works.
- ii. LED Light fixtures, Fan (Ceiling & Wall) & Exhaust Fans.
- iii. 6A Light Point /UPS Modular Switch & Socket Outlets.
- iv. 16A/ 20A Power/UPS Modular Switch & Socket Outlets
- v. L T Cables and Sub main wiring, circuit wiring.
- vi. HOT dipped galvanized perforated Cable Trays
- vii. Raceways if required.
- viii. Sandwich Type Rising Mains / Sandwich type Bus Ducts
- ix. Floor Panels, Distribution Boards
- x. Earthing
- xi. Extra Low Voltage system like Telephone/IPBAX, LAN, Fire Detection & Alarm System, CCTV System, Public Address System, Boom Barriers etc.
- xii. PVC conduit to be Heavy duty and minimum size for each service, viz electrical & ELV
- xiii. All buildings Drawings/ Tender documents/ Scope of work mentioning minimum requirements of light points, power points, AC points, geyser points, ceiling/exhaust fan points, TV points, telephone points, network points. It is mandatory as EPC contractor will tend to provide less than required points.
- xiv. Any other system as required.
- xv. Internal areas like rooms, corridors, lobbies, staircases, terraces, washrooms of all buildings and blocks shall be adequately illuminated conforming to provisions stipulated in NBC

2016, ECBC and CPWD technical specifications maintaining the indicated Lux levels and Light Power Density.

- xvi. The internal & External electrification work shall be carried out using recessed or surface-mounted MS or PVC conduits, in accordance with CPWD General Specifications for Electrical Works Part-I (Internal) – 2023 and Part-II (External) – 2023, including all up-to-date amendments.
- xvii. MS/ PVC Conduits shall be surface mounted or laid on GI angle/channels with suitable hanging GI supports in areas wherever there is false ceiling provision. In case there is no provision for false ceiling, MS/PVC Conduits shall be recessed/ concealed in concrete during slab casting. Wiring for lighting/power and ELV services shall be generally done in Heavy duty PVC Conduits unless stated otherwise.
- xviii. FRLS PVC insulated Copper conductor wires will be used for points, circuit & sub-main wiring conforming to IS 694 as amended up to date. Wiring shall be carried out with following sizes of PVC insulated FRLS multiple stranded single core copper conductor wire/cable.
- xix. Light Point 1.5 sq.mm
- xx. Ceiling /Exhaust Fan Point 1.5 sq.mm
- xxi. Call Bell Point 1.5 sq.mm
- xxii. 6A Power Plug Point 2.5 sq.mm
- xxiii. Switch board Circuit Wiring 2.5 sq.mm
- xxiv. 16A Power Plug Point 4.0 sq.mm
- xxv. Industrial Socket with 20A DP MCB – 4.0 Sq.mm
- xxvi. 32A Industrial Socket Outlet – 6.0 Sqmm
 - a. AC Industrial Socket with 25 A DP MCB- 6.0 Sqmm
 - b. Sub Main wiring from Floor Panel to DB: -
 - a. For DB Incomer Size 25 A DP – 2RX4 Sq.mm + 1R X4 Sq.mm
 - b. For DB Incomer Size 32/40A DP – 2RX6 Sq.mm + 1RX6 Sq.mm

- c. For DB Incomer Size 63A DP – 2RX10 Sqmm + 1R X10 Sq.mm
- d. For DB Incomer Size 40A 4Pole–4RX10Sqmm+2RX6 Sq.mm
- e. For DB Incomer Size 63A 4Pole–4RX16 Sqmm+2RX6 Sq.mm
- xxvii. Execution of the work shall be carried out after obtaining necessary approval of the layout for internal electrification of all buildings, common areas and staircases. Common areas and staircase lighting shall be in group control system.
- xxviii. Modular type switches, sockets, and stepped type electronic fan regulator (two module), bellpush button along with matching mounting boxes of same make shall be used. Modular type switches & sockets shall have anti- microbial, anti-fungal & anti-bacterial properties.
- xxix. Modular Type dual USB (Universal Serial Bus) and ‘C’ Type Charger Socket outlets, 2000 mA, shall be provided in each Switch Board to be provided for Residential Buildings & Guest House Building.
- xxx. Colour coding of the conduits, switches, sockets shall be provided as per NBC 2016.
- xxxi. TV outlet point wiring shall be terminated in suitably sized **G.I. boxes**, along with **splitter boxes** installed at every floor. All splitter boxes shall be properly interconnected using conduits to form a complete and organized TV distribution system, subject to **prior approval of the Engineer-in-Charge**
- xxxii. **LED-type lighting fixtures** with **inbuilt harmonic suppression mechanisms** shall be provided for all internal and external lighting installations. All LED light fixtures shall have a **Total Harmonic Distortion (THD) of less than or equal to 10%**. The system efficacy minimum 110 lm/w at temp 0 to +45 °C and humidity 10% to 90% and CRI>80.
- xxxiii. **Exhaust fans or forced ventilation fans** of suitable size and capacity shall be provided in Residential building/service buildings such as **electrical substations, STP, fire-fighting pump rooms**, etc., in accordance with the provisions of the **National Building Code (NBC) 2016**.
- xxxiv. Suitable size & capacity (Minimum BEE 3 star) Exhaust Fans/Wall Fans etc shall be provided as directed by Engineer-In-Charge. All Ceiling fans shall be with BEE5- Star rating with Brushless DC (BLDC) motors conforming to the technical specifications.
- xxxv. Separate shafts shall be provided for laying of pipes/cable trays for Electrical, ELV, Mechanical and Fire Services.

- xxxvi. Laying of DWC HDPE/Hume pipes for road crossing or in pucca portion & CC path etc. for electric / Telephone/ LAN / Street lighting cables complete with adequate number of cable chambers shall be provided by the agency.
- xxxvii. After completing the work, necessary test results as envisaged in latest CPWD General Specifications Part-I (Internal) & Indian Electricity Rules, shall be recorded and submitted. The results shall be within the permissible limits.
- xxxviii. Aviation Lights (LED Type) shall be provided on buildings and structures as per prevalent norms & IS- Codes.
- xxxix. SS Raceways (SS 304 or SS 316) with all accessories shall be provided in Data Server room, ELV Rooms of various buildings as per requirements.
- xl. Suitable illumination with LED light fixture shall be provided on terraces of all, Residential buildings, guest house building and community centre building and all Services Buildings.
- xli. For areas having grid type false ceiling, recessed type LED Light Fixtures of size 600 mm X 600 mm shall be provided as per directions of Engineer-In-Charge unless otherwise stated. For areas having false ceiling with Gypsum board, LED Down Lighter type light fixtures of suitable capacity shall be provided as per directions of Engineer-In-Charge unless otherwise stated. Surface mounted LED Light Fixtures shall be provided in the areas without False Ceiling.
- xlii. E-Vehicle charging facilities to be provided for at least 20% of the total 4-wheeler parking space along with adequate signage as per GRIHA 3-star rating/Govt norms.
- xliii. All exterior lighting (including facade lighting should be controlled by photosensor or astronomical Time Switches.
- xliv. Automatic lighting shut-off (daylight and occupancy sensors) as per ECBC / GRIHA 3-star rating.

5.19 INTERNAL LIGHTING DESIGN & LIGHTING FIXTURE

For internal illumination, LED lighting fixtures with inbuilt harmonic suppression systems shall be provided to ensure Total Harmonic Distortion (THD) is less than 10% across all areas and buildings. The illumination levels shall conform to the latest applicable standards of the IS Code,

NBC 2016, and ECBC (latest amendments). All LED indoor lighting fixtures should be meet the following parameters:

TABLE 6: LIGHTING FIXTURES PARAMETERS

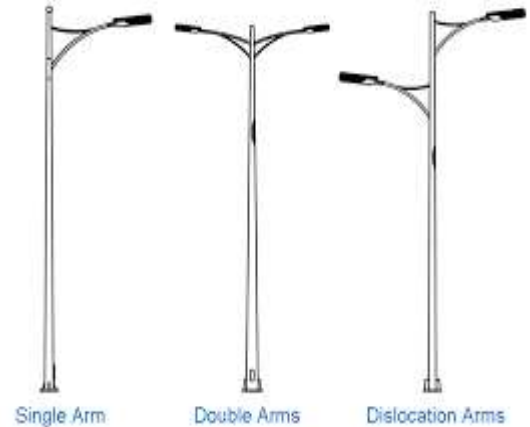
Sr.No.	Description	Parameters
1	CRI	> 80
2	THD	$\leq 10\%$
3	System efficacy	Minimum 110 lm/W at temp 0 to +45°C and humidity 10% to 90%
4	System life time (Burning hour with driver)	50000 hours @L70 as per LM 70 and testing as per LM 80.
5	Driver voltage	220-240V
6	Power factor	> 0.9
7	Efficiency	$\geq 85\%$
8	Surge protection	(L-N) 1KV, (E-L/N) 2KV
9	Colour Temperature	As per site requirement

Lighting Power Density (LPD) shall be achieved for various buildings/ areas as per lighting simulation requirements as per GRIHA & ECBC norms. Lighting to be done with the help of Dialux software & referring lux level to be as per guidelines. The EPC Contractor must obtain approval of the lighting level calculations from the Engineer-in-Charge before finalizing the lighting layout.

In terrace floor of all buildings provide bulkhead lights & some of them in emergency DB also. Lights fed from emergency DB to be shown with a different colour. Common areas and stair case 100% lighting shall be connecting though emergency DB.

5.20 EXTERNAL/ STREET LIGHTING

Road and Outdoor external lighting shall be provided conforming to IS 1944 (Part 1 & 2) 1970, CPWD specifications and National Building Code (NBC-2016) with up-to-date amendments. External street lighting shall be provided for all external areas, including surroundings Buildings/block, parking areas, landscaping and horticultural areas, roads, walkways, pathways, and entrance gates as per the approved layout drawings by Engineer-in-charge. High-efficiency LED lighting fixtures will be installed for external road



lighting, compound/landscape lighting, and solar street lighting. The lighting control (on/off operation) for external lighting will be automatically controlled with a digital timer control switch through outdoor-type feeder panels. External street lighting fixtures, including street light poles (9 metre. height) Octagonal GI poles, post- top lantern poles (4 metre height), bollards (1 metre height), and "All-in- One" integrated solar lighting poles (6 m height or as per road specifications), will cover the entire BQ Housing complex area being developed as per GRIHA 3-star rating requirements. Road & compound lighting, landscape lighting, and facade lighting will be designed in compliance with NBC and ECBC codes. Road and compound lighting will be provided with outdoor-type street light fittings. The degree of protection for outdoor LED light fixtures shall be minimum IP66. The EPC contractor shall provide, supply, fix, and install LED integrated solar street lights of the "All-In-One" hybrid type with an inbuilt charge controller, health monitoring system, and overcharging protection. These will be mounted on 6-meter street poles (as per road requirements). All LED indoor lighting fixtures should be meet the following parameters:

TABLE 7: LED LIGHTING FIXTURES PARAMETERS

Sr.No.	Description	Parameters
1	CRI	> 80
2	THD	≤ 10%
3	System efficacy	Minimum 110 lm/W at temp 0 to +45°C and humidity 10% to 90%

4	System life time (Burning hour with driver)	50000 hours @L70 as per LM 70 and testing as per LM 80.
5	Driver voltage	220-240V
6	Power factor	> 0.9
7	Efficiency	≥ 85%
8	Surge protection	(L-N) 1KV, (E-L/N) 2KV
9	Colour Temperature	As per site requirement
10.	Degree of protection of the fittings	Minimum IP 66/IP67

All-in-one Hybrid solar street light shall be meet the following parameters:

- a. 95Wp High Efficiency Monocrystalline Silicon Panel.
- b. 70W±10% - System Wattage, System lumens-11900lm±10%.
- c. CCT – 5700K ±500K, CRI >80.
- d. 36Ah LiFePO4 Battery.
- e. High Efficiency Battery with higher life of >2000 cycles.
- f. Full Power up to 8Hrs. (At fully charged battery).
- g. Saving mode up to 24Hrs. (At fully charged battery).
- h. Automatic Dusk to Dawn feature.
- i. Integral PIR sensor for motion detection.
- j. MPPT charge controller with 90% to 96% efficiency.
- k. System voltage: 12.8V
- l. Output Power: 70W±10%.
- m. Output Current at Full Power ~ 5.95A.
- n. No load loss: <10mA.
- o. IK08 housing (without motion sensor).
- p. Component level IP65 product.
- q. Aluminium Housing.

r. All in one hybrid Solar Street lights to be mounted on GI pole height 9/6 meter with all accessories as required.

Facade lighting shall be provided to adequately illuminate all Main Entry/Exit gates, guest house & community centre building of the BQ housing complex.

Lux sensor switches should be provided in common illuminations points such as street light, park light, stilt floor, stair cases etc. Further, programmable timer switch may be provided in the park and similar areas where light should auto turn off after late evening.

All Street Light Poles shall be made of Galvanized Iron (GI) Octagonal tubes. Poles will be suitable for single / double side arms or as required. Single arm street light Poles shall be provided for all single carriageway roads. Likewise, Double arm street light Poles shall be provided for all double carriageway roads or as per requirement at site. Street light Poles shall have a service window at the bottom comprising connector terminal & MCB. Poles can be mounted on civil foundation with GI Anchor bolts of suitable size & quantity. Spacing of the street light poles will be designed to achieve illumination Lux levels. Spacing of poles and illumination Lux level should be as per latest CPWD Specifications, NBC 2016, and ECBC code 2017 as amended up to date.

Outdoor-type feeder panels (IP66/IP67) with **digital timer control** will be provided for the power distribution of various circuits, including street lighting poles, Bollard, Post top lights and facade lights, etc. An **astronomical timer** will be included in the **feeder pillar**, along with a **photocell sensor**, to account for seasonal variations and automatically switch the lights on and off. This system will enable **energy savings** while ensuring **minimum security lighting** is maintained during the night.

5.21 MASTER ANTENNA TELEVISION SYSTEM (MATV)

The system shall include supply, installation, testing & commissioning of MATV System (Wiring and conduiting) in residential building & guest house building as directed by Engineer-In-Charge. A central Antenna (or satellite dish) receives the signal, and then a network of cables and amplifiers it to each TV point of Residential buildings & guest house buildings.

The scope shall be limited to supply and installation of MATV conduiting & wiring with TV Outlet sockets as per CPWD specifications and approved GFC Drawings.

The Master antenna /Dish shall be provided individually & separately for all residential building blocks. It shall be provided at the top most convenient point in respective building with a suitable room/space on the top most floor or terrace for housing the amplifier unit. From the amplifier rooms, conduits should be laid in recess/surface or in the shaft to facilitate drawing co-axial cable to individual rooms/areas. Suitable Tap Off boxes shall be provided in every room. The specifications of the same are listed below:

- a) **Conduiting-** Conduiting for MATV system shall be carried out in MS / Heavy Duty PVC Conduit.
- b) **Outlets -** All MATV outlets shall be provided with modular range of cover plate, box, and coaxial outlet. Cover plate shall match in shape & finish with other light and power accessories.
- c) **Junction Box:** Suitable size of GI box shall be provided for termination of conduit for MATV system.
- d) **MATV Line Amplifiers:** The MATV Amplifier shall be housed in a high frequency resistant Aluminium housing at the highest level (terrace level). The MATV Amplifier shall have an in-built variable equalizer and Alternator for site signal condition adjustments.
- e) **Coaxial Cables:** The coaxial cable shall be of wideband type.
- f) **Tap Off / Splitter:** These shall be of ultra-wide bandwidth and of hybrid type. These shall have a flat frequency response over the entire operating range. These shall have aluminium cast housing for high frequency radiation resistance. The Tap off shall be in one way, two way and four-way configurations.

5.22 UPS (UNINTERRUPTIBLE POWER SUPPLY)

60 KVA UPS with 30 min. battery backup suitable for 3-Phase, 50 Hz power supply shall be provided to cater the following:

- Server/ switching
- CCTV
- Voice, Networking, Internet.
- Fire Alarm System, Public Address and Voice Announcement system (PAVA).
- Any other necessary requirements if required at site.

The UPS shall have electrical and mechanical characteristics and requirements for a continuous duty, highly reliable, Microprocessor Based true Online Double conversion UPS with latest PWM IGBT technology shall be provided. The UPS System shall have minimum efficiency as per

ECBC.i.e. 91.9% at 100% load. UPS system shall have separate Power distribution system (comprising of distribution boards, incoming isolators etc.) shall be provided. UPS room and control room of suitable size with suitable air conditioning shall be provided by the EPC contractor. UPS Battery room shall be suitable ventilated.

The system shall include the interconnection of UPS Input/output power supply Panels & UPS units, UPS & Batteries Heavy-duty VRLA type Maintenance Free batteries with highest efficacy for latest sophisticated through flexible copper cables of suitable size.

5.23 SOLAR PHOTOVOLTAIC POWER GENERATION SYSTEM

The EPC Contractor shall carry out supply, installation, testing & commissioning of Direct Online Grid connected Solar Photo Voltaic Power system 375 kWp as per GRIHA/ ECBC/NBC 2016 norms. The generated power will be directly connected to the Power grid/Distribution Panel/rising mains of respective Building/ Block for load sharing during day time. The average area requirement per kWp will be 10 sq.mtr on roof top. The Solar Photo Voltaic Power System shall be provided at terrace residential building blocks, guest house and community centre building to obtain minimum GRIHA 3-star rating certification for the project. Solar cables shall be used for interconnection of equipment like Solar PV Panels and Inverter etc. Solar cables shall be suitable for outdoor installations. Given Solar quantity is minimum, EPC contractor recalculates solar requirement and approved by Engineer-in-charge as per GRIHA- 3-star rating.

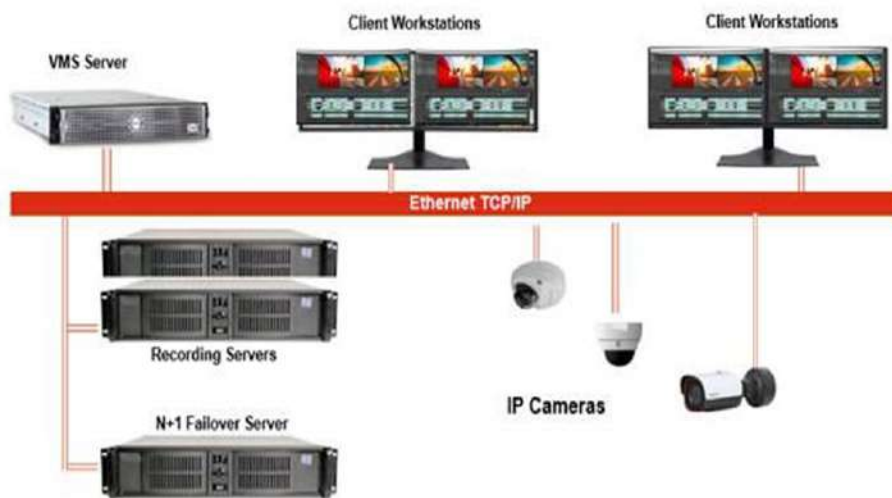
A Grid Tied Solar Rooftop Photo Voltaic (SPV) system consists of SPV array, Module Mounting Structure, Power Conditioning Unit (PCUs) consisting of Maximum Power Point Tracker (MPPT), Inverter, and Controls & Protections, interconnect cables, Junction boxes, AC & DC Distribution boxes and switches etc. should conform to the BIS or IEC or international specifications.

Grid tied SPV system is without battery and should be designed with necessary features to supplement the grid power during day time. The EPC Contractor shall use only the PV modules that are empanelled to the OEM empanelment. The total Solar PV array capacity shall be assembled with minimum 545 Wp (with minimum of 24V) Mono Crystalline MNRE approved solar modules with 144 cells with minimum 20% Module Efficiency. The modules should be tested and certified as per relevant IEC standard as per MNRE guidelines and offered module shall have a power output warranty of 90% of the rated power for 10 years.

5.24 CLOSED CIRCUIT TELEVISION SURVEILLANCE SYSTEM (CCTV)

The IP based CCTV and Surveillance System shall control and monitor in Fire control room as per NBC 2016/ relevant IS codes/Local by laws. All the buildings shall have IP-Based Dome Cameras, Bullet Cameras, PTZ Cameras for surveillance. The types of cameras & locations of various Cameras indicated here in are minimum to be provided. EPC Contractor shall ensure complete CCTV camera coverage of the intended internal & external surveillance areas. However, if required and found necessary during detailed designing, the type/rating of cameras will be upgraded.

FIGURE 13: DIAGRAM GENERAL CCTV SYSTEM



Following type of cameras shall be used in the CCTV network:

- i) Bullet/Dome camera: - These cameras shall be with minimum 5 MP Resolution with IR suitable for Day and night operations. All the cameras shall be operative on automotive manual & scheduled mode. All the cameras shall be operative on automotive manual & scheduled mode. RJ- 45 Cable connectors will be used for Network/POE connectivity with UL, FCC, EN, RoHS & BIS certifications. camera Should be supportable Quad Stream. Generally, the Dome Cameras and Bullet Cameras shall be installed at all Entry & Exit Points, main corridor, Lift lobbies & common areas of various buildings of BQ Housing complex.

- ii) PTZ outdoor cameras: - The true IP day and night viewing PTZ cameras proposed are 30 x optical zoom, P-iris with minimum 5 MP resolution & Auto focus lens with focal length 4.3 mm to 141.9mm or better. The PTZ camera shall automatically track the object to use its Pan/Tilt/Zoom feature and record 20 frames/ sec. The cameras shall be UL Listed. RJ- 45 Cable connectors will be used for Network/POE connectivity with UL, FCC, EN, RoHS & BIS certifications.
- iii) The cameras should be on minimum configuration of 5MP and above and the recorded images across all the cameras shall be stored locally for at least 30 days to facilitate investigations of a reported case. The Camera should have ONVIF Profile S&G Compliance.
- iv) All the outdoor proposed cameras shall be in IP-66 Housing with IP 66 Junction box.
- v) Industrial Grade Switches should be used for CCTV network at the edge layer.
- vi) Network connectivity for cameras having distance more than 75 meters, shall be on optical Fiber cable. Suitable provision shall be provided to connect with LAN infrastructure.
- vii) The video recording shall be non-embedded based recording server with video management software. The NVR solution shall be a centralized system deployed in the Server room. The viewing of the cameras shall be managed through IP network and can be made accessible at any location.
- viii) All the outdoor cameras shall be in IP-66 Housing with Junction box, media converter etc. is proposed in weather proof housing. The existing LAN network switches would be used for CCTV connectivity and will be connected to central CCTV server/ control room & All cameras shall have PoE connectivity ports.
- ix) Centralized Viewing at the command centre: The live camera feed should be visible in the command centre on the screens for all the cameras deployed across the housing.

- x) LED Display Monitor: Adequate number of Full HD LED monitors with high resolution screen size of 55" (diagonal) are to be provided for viewing all live/recorded camera proceedings. The resolutions of 1920(H)x1080(V) pixels with comfortable viewing angle shall be provided for all monitors suitable for operation on 220-240V AC, 50 Hz power. The display LED monitors shall be equipped with all standard features such as Video-in, Audio-in, HDMI/DVI, RJ 45 Port etc.
- xi) Individual/ dedicated LED Display Monitors shall be provided for each building/block and external areas of the campus as per requirement.
- xii) CCTV Control Room (with display monitor screens & video management server) shall be in the Security Room/ Control Room at entry gate of the BQ housing complex. The wiring inside the building shall be with CAT 6a cable in conduit and for Outdoor connectivity armoured Fiber Optic Cable shall be used. Optical Fiber cables shall be laid underground in HDPE/DWC pipes with suitable manholes for easy pulling and proper maintenance. PTZ cameras will be placed on roof top and on Street light Poles with necessary mounting arrangements in external areas. The video management server (VMS) shall have minimum 30 days storing capacity.
- xiii) Suitable hardware/ software shall be provided for streaming live CCTV signal feeds of all indoor and outdoor cameras installed in the campus to remote locations outside the campus through Internet connectivity.
- xiv) EPC contractor shall prepare floor wise layouts of IP CCTV system mentioning location of different types of cameras, location of network rack including details of POE switches, Backbone connectivity from each floor to main rack, Details and locations of LED display, NVR & Server, Main rack, All interconnecting cable details etc.

5.25 IP BASED PUBLIC ADDRESS SYSTEM (PA SYSTEM):

IP Based Public Address System shall be provided in residential building/blocks, guest house and community centre building as per NBC 2016 norms. Speakers in the Ceiling/Wall shall be provided in corridors, lift lobbies, waiting area and reception counters etc. and other common areas as per NBC 2016/relevant IS codes. The general diagrams of PA system placed as under:

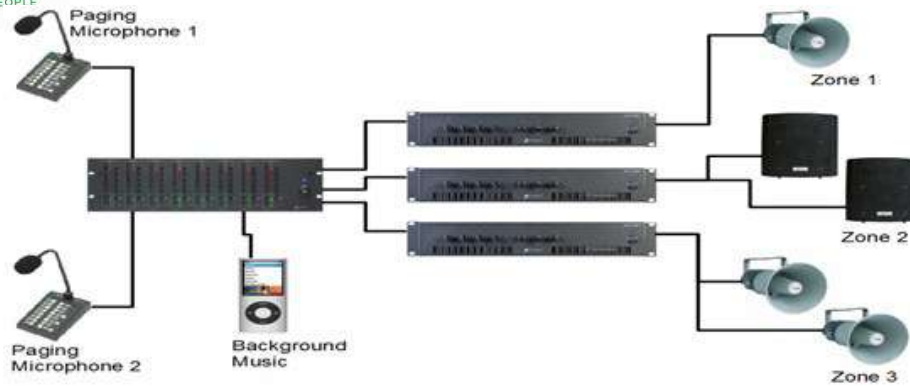


FIGURE 14: IP BASED PA SYSTEM

- a) Box type speaker shall be provided in the entrance lobby.
- b) Horn type speaker shall be provided in the basement (as per requirement).
- c) Recessed speakers in the false ceiling areas shall be provided.
- d) Proper zoning is to be done considering the user requirement, waiting areas & floors etc.
- e) IP Based Touchscreen Paging console for PA System shall be in the Fire control room with pre amplifiers, amplifiers, Pen-drive, FM Player & gooseneck microphone.
- f) PA System shall have TCP/IP based De-Centralized/distributed controller architecture with no single point of failure for entire system. All IP based Controllers shall work independently of each other.
- g) PA System shall have the facility to make announcements on all floors of a residential, guest house and community centre buildings simultaneously or selectively on individual floors and floor zones of the building.
- h) Wiring shall be done with twin twisted tinned copper, FRLS speaker cable in conduit/ Fire survival cable as per CPWD specification/ Fire norms.
- i) The PA System shall be integrated with Fire Alarm Panel and with the BMS also.
- j) All items of fire alarm system and PA system shall be from one OEM only for better system integration.

5.26 IP BASED EPABX TELEPHONY SYSTEM

- k) IP core Server based Voice communication system in (1+1) redundant mode shall be provided for guest house and community centre building. IP phones & access to call outside the housing complex through analogue trunks or PRI trunks via local service provider (e.g. MTNL/ BSNL etc.). The system shall deliver functionality of both internal communication

(intercom) and external communication.

- l) The core IP telephony server shall be deployed in the Server room and the PRI lines shall be terminated in the separate Rack in the data centre.
- m) RJ-45 Telephone socket outlets with suitable IP Phones instruments shall be provided in guest house & community centre building at convenient locations, as per the requirement.
- n) Telephone & data points shall be mainly provided in Guest house & Community centre only.
- o) CAT-6A wiring shall be provided in recessed/surface conduits/ raceways from each telephone point up to the Rack panel (Patch Panel & Switch) at each floor or nearest Rack Panel of buildings. The Maximum length of the CAT-6A cable shall be 90 M.
- p) The scope of work shall include planning, designing, supply, installation, testing and commissioning of complete of IP based voice communication system with required number of PRI Trunks lines (30 channels) with clip facility for the specified number of IP users license for lifetime.
- q) The system should be expandable to be able to cater the complete need of BQ housing complex and other buildings in future, by simply taking more IP users licenses.
- r) EPABX Room Provisioning: An air-conditioned room with adequate space must be provisioned to accommodate the EPABX system.
- s) Telephone Cabling and Wiring:
 - a. EPABX-side telephone cable MDF, along with all necessary protection (IPM Module), and field-side MDF are to be installed within the EPABX room.
 - b. Telephone cabling should be laid from the EPABX room to the Guest House and Community areas, in accordance with standard specifications.
 - c. Communication Tower will not be required if EPABX control room is placed within 100m of cabling distance. Accordingly, possibility will be explored to place the EPABX control room inside Guest House, and antenna be mounted at the terrace of Tower-D.
 - d. All the phones system should be ANALOGUE type (not IP Phone).

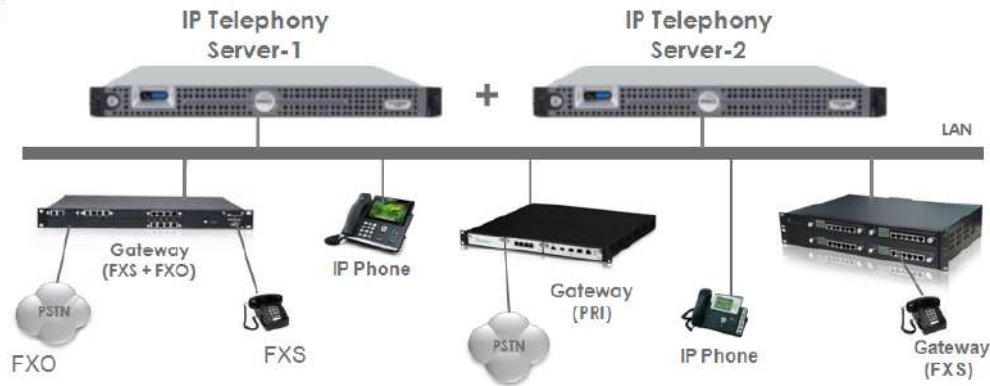


FIGURE 15: IP BASED EPABX SYSTEM



5.27 FIRE DETECTION AND ALARM SYSTEM

The EPC contractor shall be provided Addressable Intelligent fire detection and Alarm system of latest technology with Fire alarm panels, multi-Sensor detectors, smoke detectors, heat detectors, beam detectors, response indicators, manual call point and hooters, light strobe etc. It shall meet the requirement of NBC 2016/NFPA/ CPWD Specifications/ State By laws. License/Approval of Local Fire Authorities shall be provided for housing complex, guest house and community center. There shall be the proper Zoning of the housing considering the critical areas in the housing Block or in the other buildings.

There shall be independent fire panel for housing, and all fire control panels shall be interconnected with each other. Fire Detectors and devices shall be connected to Fire Control Panel of the building through necessary cables in DWC HDPE pipes. The monitoring of whole housing shall be in the Main Fire Control Room of housing Building. For Central Monitoring of all the Fire Panels, necessary devices like PC with monitor, Printer, modules & Software etc. of latest technology with minimum 1 TB hard disk shall be provided in the Control room. Fire Alarm control Panel shall not

have more than 100 devices and 100 detectors in one loop. The details of the system proposed shall be as follows:

- Addressable intelligent dual type Fire Detectors is suggested.
- Detectors should be with inbuilt short circuit isolator & automatic addressing. Detectors should be installed as per coverage defined in NBC-2016 and NFPA 72.
- Each detector & devices shall have integrated short circuit isolator.
- Suitable numbers of input/ output (C/M) relay modules are suggested for connecting other equipment like Electrical Panels, lifts, firefighting system, AHUs etc.
- Spacing between two detectors shall be as per relevant code.
- Addressable intelligent Fire Detectors shall be provided. (UL/ EN & FM approved only)
- Suitable addressable loop powered sounders/ hooters for 100db sound level are suggested.
- Addressable manual call boxes shall be provided near all exits, stair cases, lift lobbies etc. as per relevant Norms.
- The Response Indicators shall be used in the waiting areas, Corridors, Common Area, conference Hall or in large room etc.
- Microprocessor IP based fire alarm control panel (fully redundant) for number of required loops with 24 hrs. Battery backup with LCD display, printer etc. shall be in the fire control room.
- Fire Alarm Panels shall be integrated with PA system and compatible with BMS also.
- Suitable sounders /hooters with strobe for minimum 90 dB sound level.
- Addressable manual call boxes shall be provided near all exits, stair cases, lift lobbies etc. as per relevant norms.
- Two Way communication talk back system to be provided at each floor near MCP with necessary accessories in all the buildings (Residential, community centre & Guest House).
- Cabling shall be with Armoured Fire Survival copper cable in case of false ceiling. Armoured Fire survival cable shall be laid with clamps. Unarmoured fire survival copper cables shall be laid in concealed MS Conduits (as per fire norms/ CPWD specifications) of suitable size, wherever there is no false ceiling.
- Digital Voice Command Centre (DVCC) with Two Way Communication Fire Fighters Telephone Jack & Handset with necessary accessories are to be provided in all the buildings as required. DVCC shall be an integral part of fire alarm system and shall work even in case of Fire Alarm system failure.

5.28 BOOM BARRIERS

The EPC Contractor shall provide RFID based **Boom Barriers** at all **entry and exit gates** of the BQ Housing Complex. The boom barriers shall be **electro-mechanical type**, complete with all necessary cabling and accessories, with a boom length of up to 6 meters to meet site requirements and as per the directions of the **Engineer-In- Charge**. **Push button-type control** for 'open' and 'close' operations of the boom barrier shall be provided through high-torque motors, operated using a **230V AC supply**. In the event of a power failure, a provision shall be made to raise or lower the boom barrier manually. The boom length shall be designed to meet the width of the road, and the opening time shall vary from **3 seconds to 6 seconds**. The control unit shall be **IP-65 rated**, ensuring protection against rough weather conditions. General Boom Barrier Schematic as shown as below:

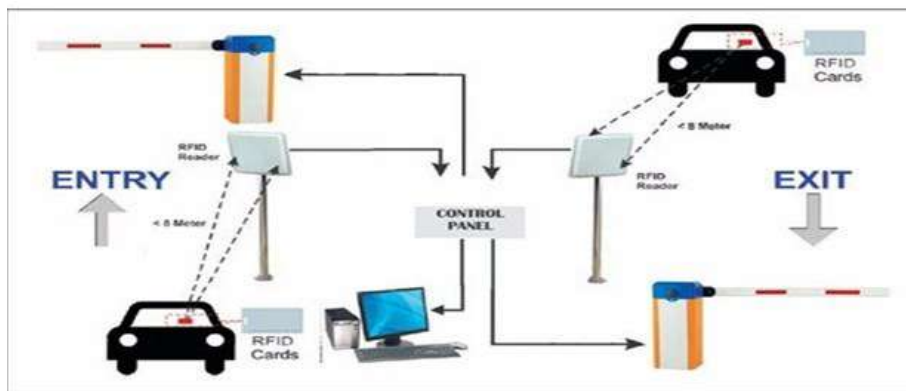


FIGURE 16: BOOM BARRIER SCHEMATIC

5.29 ESTIMATED ELECTRICAL LOAD SHEET:

TABLE 8: ELECTRICAL LOAD SHEET

Proposed BQ Housing Complex, Duliajan, Assam.						
Sr.No.	Description	Unit	Basis of load calculations (Watt/Sqm, Watt/no.)	Area in Sq.m / Qty in Nos.	Load (kW)	Remarks
1	BQ Housing (G+6) Buildings including stilt	Watt	25	26954.96	673.87	Based on CPWD Guidelines for Substation & Power Distribution System of Buildings 2019.
2	Provision for split AC (1.5 ton) in BQ Housing Building - 02 units per dwelling	watt	1310	384.00	503.04	Tentative
3	Provision for split AC (1 ton) in BQ Housing Building - 01 units per dwelling	watt	1100	192.00	211.20	Based on CPWD Guidelines for Substation & Power Distribution System of Buildings 2019.
4	Mumty of quarters	Watt	15	400	6.00	
5	Community Hall	Watt	25	1035.00	25.88	
6	ESS, Pump room, Guard room, STP room	Watt	15	636.00	9.54	
7	Community Centre HVAC System	Watt	40	1035.00	41.40	
8	Add 20% Extra	Watt			8.28	

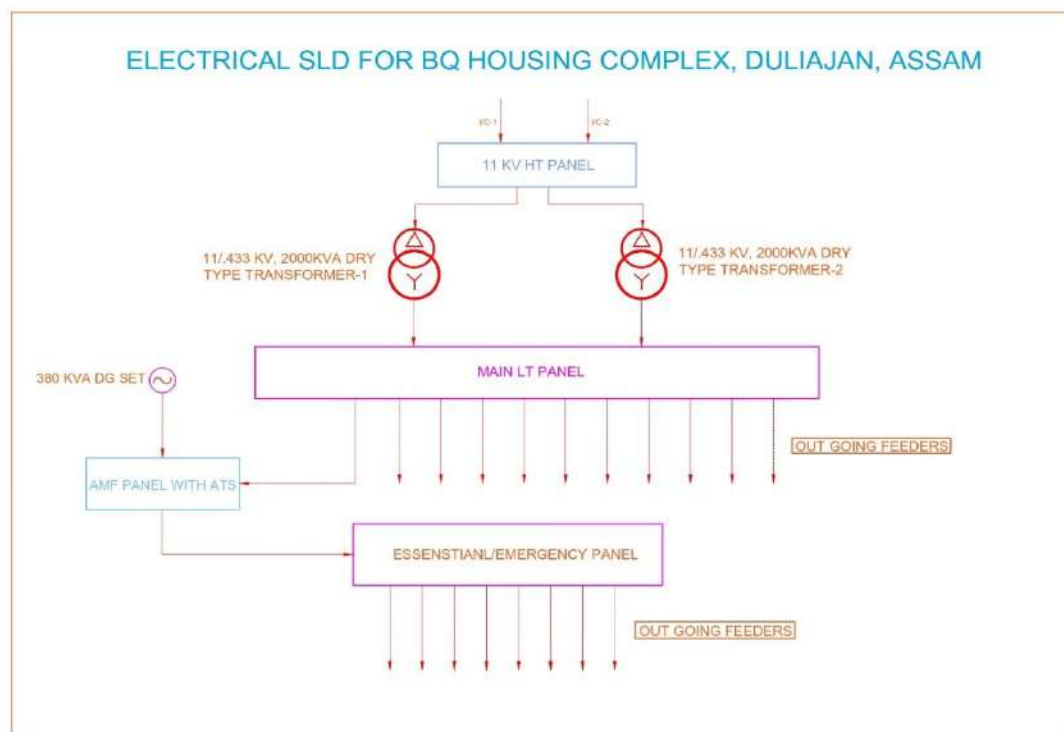
9	Guest House (G+3) including Mumty	Watt	25	1990.00	49.75	
10	MRL Lift, 15-person 1020 Kg capacity, Speed 1.5m/sec, Travel Height - G+6 (2 Nos. in each tower)	KW	15	16.00	240.00	Tentative
11	MRL Lift, 15-person 1020 Kg capacity, Speed 1.5m/sec, Travel Height - G+3 (1 No. for guest house)	KW	15	1.00	15.00	Tentative
12	Street Lighting /Approach Road Lighting	Watt	70	65	4.55	Tentative
13	STP	L/S			25.00	Tentative
14	Fire & Water Pumps	L/S			25.50	Tentative
15	Misc. Loading	L/S			5.00	Tentative
16	For electrical vehicles charging	KW	7.4	12	88.80	For each block total 1 nos of charging point considered. 1 Nos. 7.4KW.
17	For electrical vehicles charging	KW	22	4	88.00	
18	Split AC, 1.5TR, Inverter type Required for Guest House	Watt	1310	14.00	18.34	Total 38 Nos., 24 Nos of 2 Ton AC & 14 Nos. of 1.5 Ton

19	Split AC, 2.0 TR, inverter type Required for Guest House	Watt	1830	27.00	49.41	AC Required for Guest House.
	Total (kW)				2088.56	Total Connected Load
	Total Demand Load (kW)				1566.42	Demand load after considering diversity factor 0.75
	Taking 5% for future load (kW)				1644.74	
	Total Required kW (Say)				1644.74	
a)	Selection of Distribution transformer for the above load					
	Total Required kVA @0.98 Power factor				1678.31	
	Total Required (kVA @85% transformer loading)				1974.00	
	Proposed Transformer rating in kVA				2000.00	1. Transformer 2000 KVA Dry type-2 sets (1W +1S).
b)	DG Rating in KVA				380.00	DG considered only for emergency load as under: 1. Lifts load: 204KW 2. Domestic water pumps & STP: 50 KW 3. Street Lighting: 5 KW 4. Common Area Lighting of Housing, community center & guest house: - 15 KW Total: - 274 KW, Total: 322KVA at .85

			PFDG set required @ 85% loading= $322/0.85=378$ KVA Considered 380 KVA DG set for Essential load
c)	UPS Rating In KVA	60.00	Considered 60 KVA as per M/s OIL
d)	Solar Panel in KWp	375.00	Tentative

Note: The above load calculations are minimum load requirement & based on Plinth area. EPC contractor recalculates their own load as per site condition and ECBC requirements.

5.30 ELECTRICAL SLD



Note-
The

above single line diagram is for reference only, EPC contractor redesigns it as per site condition and ECBC requirements.

FIGURE 17: ELECTRICAL SLD

LIST OF APPLICABLE INDIAN STANDARDS FOR ELECTRIFICATION WORK WITH UPTO DATE AMENDMENTS.

TABLE 9: LIST OF APPLICABLE IS CODES

Sr. No.	STANDARDS	TITLE
1.	IS: 732 – 1989	Code of practice for electrical wiring Installations.
2.	IS: 8061 – 1976	Code of practice for design, installation, and Maintenance of service lines up to and including 650V
3.	IS: 8884 – 1978	Code of practice for installation of electric bells and call system.
4.	IS: 5578 – 1985	Guide for marking of insulated conductor.

5.	IS: 11353 – 1985	Guide for uniform system of marking and identification of conductors and apparatus terminals.
6.	IS: 5728 – 1970	Guide for short-circuit calculations.
7.	IS: 3646 (Part – 1) 1992	Code of practice for interior illumination: Principles for good lighting and aspects of design.
8.	IS: 3646(Part – 2) 1992	Code of practice for interior illumination: Schedule of illumination and glare index.
9.	IS: 1646 – 1997	Code of practice for Fire Safety of Building (General): Electrical Installation.
10.	IS: 10118(Part – 1) – 1982	Code of practice for selection, installation and maintenance of switchgear and control gear: General.
11.	IS: 10118(Part – 2) – 1982	Code of practice for selection, installation and Maintenance of switchgear and control gear.
12.	IS: 10118(Part – 3) – 1982	Code of practice for selection, installation and maintenance of switchgear and control gear: Installation.
13.	IS: 10118(Part – 4) – 1982	Code of practice for selection, installation and maintenance of switchgear and control gear: Maintenance
14.	IS: 3043 – 2018	Code of practice for earthing.
15.	IS: 5216(Part – 1) – 1982	Guide for safety procedures and practices in electrical work: General.
16.	IS: 6875(Part – 1) – 1973	Control switches (switching devices for control and auxiliary circuits including contractor relays) for voltages up to and including 1000 V AC and 1200 DC: General requirements and tests.
17.	IS: 4064(Part – 1) – 1978	Airbreak switches, Airbreak dis-connectors, air - Break switch dis-connectors and fuse-combination units for voltages not exceeding 1000 V AC or 1200 DC: General requirements.

18.	IS: 8828 – 1978	Miniature Airbreak circuit breakers for voltages not exceeding 1000 volt.
19.	IS: 13032 – 1991	Miniature circuit breaker boards for voltages up to and including 1000 volts AC.
20.	IS: 12640 – 1988	Residual current operated circuit breakers.
21.	IS: 2959 – 1985	Contactors for voltages not exceeding 1000V AC or 1200 V DC.
22.	IS: 8623(Part – 1) – 1977	Factory built assemblies of switchgear and control gear for voltages up to and including 1000 V AC and 1200 V DC: General requirements.
23.	IS: 8623(Part – 2) – 1980	Factory assemblies of switchgear and control gear for voltages up to and including 1000 V AC and 1200 V DC: Requirements for bus bar trunking system (bus ways).
24.	IS: 694 – 1990	PVC Insulated cables for working voltages Up to and including 1100 V.
25.	IS: 1554(Part – 1) – 1988	PVC insulated (heavy duty) electric cables for working voltages up to and including 1100 V.
26.	IS: 3961(Part – 5) – 1968	Recommended current ratings for cables: PVC insulated light duty cables.
27.	IS: 9537(Part – 1) – 1980	Conduits for electrical installations: General requirements.
28.	IS: 9537(Part – 2) – 1981	Conduits for electrical installations Rigid steel Conduits.
29.	IS: 3480 – 1966	Flexible steel conduits for electrical wiring.
30.	IS: 2667 – 1988	Fittings for rigid steel conduits for electrical wiring.
31.	IS: 3837 – 1976	Accessories for rigid steel conduits for electrical wiring.

32.	IS: 5133(Part – 1) – 1969	Boxes for enclosure of electrical accessories Steel and cast-iron boxes.
33.	IS: 371 – 1979	Ceiling roses.
34.	IS: 3854 – 1988	Switches for domestic and similar purposes.
35.	IS: 4615 – 1968	Switch socket outlets (non-interlocking type).
36.	IS: 4160 – 1967	Interlocking switch socket outlet.
37.	IS: 1293 – 1988	Plugs and socket outlets of rated voltage Up to and including 250 volts and rated current up to and including 16 amperes.
38.	GRIHA Manual	ECBC Guidelines and BEE guidelines.
39.	IEC 61439 Part – 1	Main & Sub Distribution Boards (Electrical Panels)
40.	IS: 2713 (Part – I to III) – 1980	Specification for Tubular, Street Poles for Overhead Power Lines
41.	IS: 15652 – 2006	Specification for Insulating Mats for Electrical Purposes
42.	IS: 9537 (Part – I to III) – 1983	Specification for Conduits for Electrical Installation Part-I: General Requirement.
43.	IS: 13947 (Part – 2)	Specification for Low Voltage Switchgear and control Gear: Circuit Breakers.
44.	IS: 12640 – 1988	Residual Current operated Circuit Breakers Specification
45.	IS: 7098 (Part – I to II)	LT PVC XLPE Power Cables with Copper / Aluminium Conductor (650/1100volt)
46.	IS: 2309 – 1989 & IS: 62305	Code of practice for the protection and allied Structures against lightning.
47.	IS:11171 -1985	Specification for Dry Type Transformer
48.	IS:7752(Part-1)-1975	Guide for improvement of power factor in consumer installation: Low and medium supply voltages.

49.	IS:10118(Part-1)-1982	Code of practice for selection, installation and maintenance of switchgear and control gear: General.
50.	IS:10118(Part-2)-1982	Code of practice for selection, installation and Maintenance of switchgear and control gear.
51.	IS:10118(Part-3)-1982	Code of practice for selection, installation and maintenance of switchgear and control gear: Installation.
52.	IS:10118(Part-4)-1982	Code of practice for selection, installation and maintenance of switchgear and control gear: Maintenance.
53.	IS:10028 (Part-I)- 1985	Code of practice for selection, installation, and maintenance of Transformers: Part-I (Selection)
54.	IS:10028 (Part-II)- 1981	Code of practice for selection, installation, and maintenance of Transformers: Part-II (Selection)
55.	IS:5728 - 1970	Guide for short-circuit calculations.
56.	IS:3043 – 2018	Code of practice for earthing.
57.	IS:5216(Part-1)-1982	Guide for safety procedures and practices in electrical work: General.
58.	IS:4237 – 1983	General requirements for switchgear and control gear for voltages not exceeding 1000 V AC or 1200 V DC.
59.	IS:6875(Part-1)-1973	Control switches (switching devices for control and auxiliary circuits including contractor relays) for voltages up to and including 1000 V AC and 1200 DC : General requirements and tests.
60.	IS:4064(Part-1)-1978	Airbreak switches, Airbreak dis-connectors, air- Break switch dis-connectors and fuse-combination units for voltages not exceeding 1000 V AC or 1200 DC: General requirements.
61.	GRIHA Manual	ECBC Guidelines and BEE guidelines.

62.	IS:10987: 1992	Code of Practice for Design, Fabrication, Testing & Installation of Underground / Above Ground Horizontal cylinder steel Storage tank for Petroleum Products.
63.	IS: 15652 -2006	Specification for Insulating Mats for Electrical Purposes
64.	IS: 13947 (Part- 2)	Specification for Low Voltage Switchgear and control Gear: Circuit Breakers.
65.	IS: 7098 (Part-I to II)	LT PVC XLPE Power Cables with Copper / Aluminum Conductor (650/1100volt)

5.31 LIFTS

The EPC Contractor shall carry out design, supply, installation, testing & Commissioning of Lift Works including lift licence of all lifts. Passenger cum Bed lifts/stretchers lifts and Passenger Lifts shall be provided for Residential housing blocks and Guest house building in the complex. The installation shall be carried out as per statutory rules & regulations stipulated by CPWD/NBC-2016 and IS Codes that govern the installations of the lift. The voltage and frequency of the supply shall subject to variation permissible under Indian Electricity Act and Rules.

TABLE 10: LIFT DETAILS

Passenger Lifts/Bed lifts (Machine room Less type)			
Sr.No.	Name of Building	Details of Passenger Lifts	Quantity
1.	Residential Building-G+6 (08 Block)	15 passengers lift, Speed - 1.5m/sec., Stop - 7 stops	08 Nos.
2.	Residential Building-G+6 (08 Block)	15 passengers/Stretchers lift, Speed -1.5m/sec., Stop - 7 stops	08 Nos.
3.	Guest House Building-G+3	15 passenger/stretchers Lift, Speed- 1 m/sec., Stop- 4 stops	01 Nos.

The work shall be executed as per CPWD General Specifications for Electrical Works (Part III Lifts & Escalators -2003), NBC-2016, ECBC-2017 (Latest amendments), as per relevant IS as per directions of Engineer-in-charge.

IP-based CCTV camera (1 No.) shall be provided in all Lift Cars of all buildings for surveillance/monitoring of lift control panel. The CCTV cameras, installed in the lifts, shall be connected/ integrated with the IP based CCTV system with suitable cabling, hardware items etc. as required. Suitable warning mentioning “You are under CCTV surveillance” shall be posted inside the lift car. These cameras may be located at ceiling level and at any location which cannot be easily accessed, noticed, or tampered with. The cable used for connecting in car camera to monitoring system, shall be of trailing grade. Ordinary co-axial or twisted pair wires shall not be tied along with trailing cables. Specialized trailing cables shall be used in lift applications. Wireless CCTV cameras complete with transmitter/ receiver & all accessories as required, may also be provided alternatively.

- Lift Well, Car Size, Lift Pit Depth, Overhead, and Clear Entrance Width & Height dimensions shall conform to NBC 2016 / IS Code as amended up to date. All lifts shall be Machine room less type.
- Anti-skid granite or Stainless Steel 304 chequered plate flooring of suitable thickness shall be provided in all the lifts.
- Car enclosure & doors shall be made of SS 304 sheet of required thickness.
- All lift shall have necessary provisions & door opening as required for physically challenged person.
- Power supply to each elevator shall be connected to emergency source.
- Lift Car operating Panel shall be equipped with Braille buttons. Automatic rescue device and emergency lighting shall be provided in each elevator supported by independent rechargeable batteries.
- All Lifts shall have provision of integrated with BMS System for normal operation status & operation in case of fire. Lift shall be compatible with Modbus /BACnet communication protocol or as per OEM requirement.

5.32 SAFETY REQUIREMENTS

This section covers the requirements of items to be provided in the Electrical rooms for compliance with statutory regulations, safety, and operational needs.

Safety provisions shall be generally in conformity with appendices (A) and (C) of CPWD General Specifications of Electrical Works (Part I-Internal), 2013. In particular, following items shall be provided:

(a) Insulation Mats

Insulation mats conforming to IS 15652: 2006 shall be provided all around the HT/LT panels in all electrical rooms located in Substation.

(b) First Aid Charts and First Aid Box

Charts (one in English, one in Hindi, one in regional language), displaying methods of giving artificial respiration to a recipient of electrical shock shall be prominently provided at appropriate place. Standard first aid boxes containing materials as prescribed by St. John Ambulance brigade or Indian Red Cross should be provided in sub-station & ground floor of each building at a clearly visible and easily accessible location for all occupants.

(c) Danger Plate

Danger Plates shall be provided on HV and MV equipments. MV danger notice plate shall be 200 mm x 150 mm made of mild steel at least 2 mm thick vitreous enameled white on both sides and with the descriptions in signal red colour on front side as required. Notice plates of other suitable materials such as stainless steel, brass or such other permanent nature material shall also be accepted with the description engraved in signal red colour.

(d) Fire Extinguishers

Portable CO₂ conforming to IS 2878: 1976/ chemical conforming to IS 2171: 1976 extinguishers, ABC (Cartridge type) shall be installed in the sub-station at suitable places. Other extinguishers recommended for electric fires may also be used.

(e) Fire Buckets

Fire buckets conforming to IS 2546: 1974 shall be installed with the suitable stand for storage of water and sand.

(f) Tool Box

A Standard tool box containing necessary tools required for operation and maintenance shall be provided in the sub-station.

(g) Caution Board

Necessary number of caution boards such as “Man on Line” ‘Don’t Switch on’ etc. shall be available in the sub-station.

(h) Key Board

A keyboard of required size shall be provided at a proper place containing castle keys, and all other keys of sub-station and allied area.

(i) Evacuation Plan

Floor Evacuation plans to be provided for each floor of Residential towers, community centre & guest house.

5.33 ELECTRIC VEHICLE CHARGING STATION:

Internal EI of PRIVATE CAR PARKING, EV Charging Points (EV Charger – 16 Nos):

- EV Fast charging point 4 Nos, 22 KW each.
- Normal charger points 12 Nos, 7.4 KW each.
- Minimum requirements of charges point as specified in the guidelines dated 14.12.2018 of Ministry of power are placed as below:
- E-Vehicle parking and charging facilities to be provided as per norms of the total 4-wheeler parking space along with adequate signage.

10.4.2 All other buildings (including Group Housing)

Any PCS installed at Public/Private areas or building premises of any category that caters to commercial mode of charging of EVs shall be deemed as a Public Charging Station and shall have to install the minimum requirements of chargers as specified in the Guidelines dated 14.12.2018 of Ministry of Power (refer Annexure IV for MoP Guidelines. However, in order to provide sufficient charging points for the EV share in all vehicles (refer clause 3 of the *Explanatory Note- Annexure III*), ratio of types of chargers is recommended in the table below -

Table 2: Charging Infrastructure requirements for PCS (commercial use)

Building Type	Any building type			
Ownership of Station	Service provider			
Connection and Metering	Commercial Metering and Payment			
Types of Charger	as per min. requirements specified in MoP Guidelines (refer Annexure IV)			
Additional chargers	PCS service providers shall install additional number of kiosk/chargers beyond the minimum specified requirements to meet the ratio of charging points as prescribed below (by the type of vehicles).			
Norms of Provisions for charging points	4Ws 1 SC - each 3 EVs 1 FC - each 10EVs	3Ws 1 SC - each 2 EVs	2Ws 1 SC - each 2 EVs	PV (Buses) 1 FC - each 10 EVs

Note:

- Charging bays shall be planned currently at 20% capacity of all vehicles including 2Ws and PVs(cars)

DESIGN BASIS REPORT FOR HVAC SYSTEM

6 DESIGN BASIS REPORT FOR HVAC SYSTEM

6.1 INTRODUCTION

The project aims to construct the BQ Housing Complex for OIL India Limited in Duliajan, Assam, featuring a comprehensive development that includes a seven-story (G+6) residential building, a four-story (G+3) guest house, and a two-story (G+1) community center, all integrated within the complex to cater to the housing and recreational needs of the residents.

6.2 SCOPE OF WORK

This report is intended to cover engineering and technical details relating to the following heads in Mechanical services:

HVAC System

- Mechanical Ventilation
- Pressurization (Lift Well)
- Comfort air conditioning system.
- Emergency Ventilation
- HVAC system design for **Mechanical Ventilation, Lift Well Pressurization, Comfort Air Conditioning, and Emergency Ventilation** in a commercial building, using specified technologies: **VRV** and **Split AC** for cooling, **propeller and inline fans** for ventilation, **axial flow fans** for lift well pressurization, and **axial fans** for emergency ventilation at **12 ACPH**.
- **Mechanical Ventilation:** Ensures indoor air quality per ASHRAE 62.1, using propeller fans and inline fans. Ducts are galvanized steel.
- **Lift Well Pressurization:** Prevents smoke ingress during fires (NFPA 92), maintaining 50 Pa pressure using axial flow fans. Fire-rated ducts, pressure relief dampers, and fire alarm-linked controls ensure safety, with backup power.
- **Comfort Air Conditioning:** Maintains 22-26°C and 40-60% RH (ASHRAE 55) using VRV systems for multi-zone cooling and Split AC for Typical Floor.
- **Emergency Ventilation:** Removes smoke at 12 ACPH (NFPA 92) using axial fans for fresh/exhaust air. Fire-rated ducts, smoke dampers, and fire alarm activation ensure safety, with makeup air and backup power.

- This design ensures occupant comfort, safety, and energy efficiency, tailored to the building's needs.

6.3 HVAC SYSTEM

6.3.1 OBJECTIVE

This concept design report aims to briefly touch upon the Design Aspects of HVAC works for the project. The object of air conditioning design is to create safe and comfort environment in respect to temperature, humidity, dust, fresh air, and odour etc. By keeping the all the design condition and heat load calculations,

6.3.2 SCOPE OF WORK

The Heating, Ventilation and Air-Conditioning (HVAC) system shall comprise of following:

- a. Variable Refrigerant Volume.
- b. Ceiling Suspended Duct-able Units (DUs) for open office and cabins.
- c. Cassette Units for Lab
- d. Motor control centers.
- e. Refrigerant Piping and Condensate Drain Piping inclusive of all fittings, pipe, channel, braided wire supports, vibration isolators, strainers etc.
- f. Sheet metal ducts, external insulation, acoustic lining, canvas connections, silencers, volume control dampers, smoke dampers and braided wire supports as required.
- g. Supply and return air registers, grills and diffusers.
- h. Insulation of pipes with protective coating.
- i. Vibration isolators
- j. Central Controller for VRF with provision of BMS integration.
- k. Fire sealant around ducts / pipes and cables when passing through wall / floor and other openings.
- l. Wiring and earthling from MCC panels to various refrigeration, air conditioning and mechanical ventilation equipment, control wiring and interlocking.

- m. Cutting holes, chases and the like through all types of non-structural walls, and finishing for all services crossings, including sealing, frame work, fire proofing, providing sleeves, cover plates, making good structure and finishes to an approved standard.
- n. Balancing, testing and commissioning of the entire HVAC and mechanical ventilation installation.
- o. Test reports, list of recommended spares, as-installed drawings, operation and maintenance manual for the entire HVAC installation.
- p. Complete co-operation with independent commissioning agent and corrective actions on snag list points prepared by such independent commissioning agent.
- q. Multiple trainings of technical staff.

6.3.3 TERMINOLOGY

6.3.3.1 Air Conditioning

The process of treating air so as to control simultaneously its temperature, humidity, purity, distribution and air movement and pressure to meet the requirements of the conditioned space.

6.3.3.2 Coil By Pass

The coil by pass factor is the percentage of air passing over the coil that is not affected by the coil.

6.3.3.3 SHGC

The SHGC is the fraction of incident solar radiation admitted through a window both directly and absorbed and subsequently released. SHGC is expressed as a number between 0 to 1. The lower a window solar heat coefficient the less solar heat is transmitted.

6.3.3.4 Relative Humidity

Ratio of the actual water vapour in the air as compared to the maximum amount of water that may be contained at its dry bulb temperature. When the air is saturated, dry bulb, wet bulb and dew point temperatures are all equal.

6.3.3.5 Dry-Bulb Temperature

The temperature of air as registered by an ordinary thermometer.

6.3.3.6 Wet-Bulb Temperature

The temperature registered by a thermometer whose bulb is covered by a wetted wick and exposed to a current of rapidly moving air.

6.3.3.7 Dew Point Temperature

The temperature at which condensation of moisture begins when the air is cooled at same pressure.

6.3.3.8 Humidity

It is the amount of water vapor present in a certain volume of air.

6.3.3.9 Enthalpy

A thermal property indicating the quantity of heat in the air above an arbitrary datum in kilojoules per kg of dry air (or in Btu per pound of dry air).

6.3.3.10 Psychrometry

Psychrometry is the science involving thermodynamic properties of moist air and the effect of atmospheric moisture on materials and human comfort. It also includes methods of controlling thermal properties of moist air.

6.3.3.11 Psychrometric Chart

A Psychrometric chart graphically represents the thermodynamic properties of moist air. If two properties are known, all the other properties can be determined with the help of psychrometric chart.

6.3.3.12 Positive Ventilation

The supply of outside air by means of a mechanical device, such as a fan.

6.3.3.13 Atmospheric Pressure

The pressure of air exerted on the surface of earth by the atmospheric column is called atmospheric pressure. At sea level, the atmospheric or barometric pressure is 760mm column of mercury (29.92 in Hg/ 406.8-inch water column/ 101.325 Kpa).

Generally atmospheric pressure is used as a datum for indicating the system pressures in air-conditioning and accordingly, pressures are mentioned above the atmospheric pressure or below the atmospheric pressure considering the atmospheric pressure to be zero. A 'U' tube manometer will indicate zero pressure when atmospheric pressure is measured.

6.3.3.14 Indoor Air Quality (IAQ)

Indoor air quality refers to the nature of conditioned air that circulates throughout the space/ area where one works or lives, i.e. the air we breathe when we are indoors. IAQ refers not only to comfort which is affected by temperature, humidity and odours but also to harmful biological contaminants and chemicals present in the conditioned space.

Bad Indoor Air Quality can be a serious health hazard. Carbon dioxide (CO₂) has been recognized by ASHRAE as the surrogate ventilation index or the only measurable variable for the indoor air contaminants.

6.3.3.15 Thermal Transmittance

Thermal transmission through unit area of the given building unit divided by the temperature difference between the air or some other fluid on either side of the building unit in 'steady state' conditions.

6.3.3.16 Thermal Energy Storage

Storage of 'Cold Energy' sensible, latent or combination for use in central system for air-conditioning or refrigeration is called thermal energy storage. It uses a primary source of refrigeration for cooling and storing 'Cold Energy' for reuse at peak demand or for backup as planned.

6.3.3.17 Shade Factor

The ratio of instantaneous heat gain through the shading device to that through a plain glass sheet of 3mm thickness.

6.3.3.18 Sensible Heat Factor (SHF)

Sensible heat factor is the ratio of sensible heat to total heat, where total heat is the sum of sensible and latent heat.

6.3.3.19 Supply Air

The air that has been passed through the conditioning apparatus and taken through the duct system and distributed in the conditioned space is termed as supply air.

6.3.3.20 Return Air

The air that is collected from the conditioned space and returned to the conditioning equipment is termed as return air.

6.3.3.21 Re-Circulated Air

The return air that has been passed through the conditioning apparatus before being re-supplied to the space is called re-circulated air.

6.3.3.22 Duct System

A continuous passageway for the transmission of air which in addition to the ducts, may include duct fittings, dampers, plenums and grilles & diffusers.

6.3.3.23 Plenum

An air compartment or chamber to which one or more ducts are connected and which forms part of a distribution system.

6.3.3.24 Supply and Return Air Grilles & Diffusers

Grilles and diffusers are the devices fixed in the air-conditioned space for distribution of conditioned supply air and return of air collected from the conditioned space for re-circulation.

6.3.3.25 Fire Damper

A closure which consists of a normally held open damper installed in an air distribution system or in a wall or floor assembly and designed to close automatically in the event of a fire in order to maintain the integrity of the fire separation.

6.3.3.26 Smoke Damper

A smoke damper is similar to fire damper. However, it closes automatically on sensing presence of smoke in air distribution system or in conditioned space.

6.3.3.27 Fire Separation Wall

The wall provides complete separation of one building from another or part of a building from another part of the same building to prevent any communication of fire of any access or heat transmission to wall itself which may cause or assist in the combustion of materials of the side opposite to that portion which may be on fire.

6.3.3.28 Refrigerant

The fluid used for heat transfer in a refrigerating system, which absorbs heat at a low temperature and low pressure of the fluid and rejects heat at a higher temperature and higher pressure of the fluid, usually involving changes of state of the fluid.

6.3.3.29 Global Warming Potential (GWP)

Global Warming can make our planet and its climate less hospitable and more hostile to human life. It is, therefore, necessary to reduce emission of greenhouse gases such as Co₂,

Sox, NOx and refrigerants. The potential of a refrigerant to contribute to Global Warming is called its GWP. Long atmospheric life time of refrigerants results in Global Warming unless the emissions are controlled.

6.3.4 REFERENCE STANDARDS

- National Building Code of India -2016
- ECBC -2019
- Relevant Bureau of Indian Standards (BIS) Codes
- ASHRAE (American Society of Heating Refrigerating & Air Conditioning Engineers) Handbooks
 - Application -2019
 - Refrigeration -2018
 - Fundamentals -2017
 - Systems & Equipment -2016
- ISHRAE (Indian Society of Heating Refrigerating & Air Conditioning Engineers) HVAC Handbook-2017
- Duct construction standards as per relevant BIS codes & SMACNA standards.
- Air filters as per ASHRAE Standard 52.1 -1992
- Indoor Air Quality as per ASHRAE Standard 62.1 - 2019
- ASHRAE Standard 90.1-2007
- Motors, cabling, wiring and accessories as per BIS codes / IE Rules / IS codes
- Recommended practices of Equipment suppliers

6.3.5 ABBREVIATIONS

DBT	Dry Bulb Temperature
WBT	Wet Bulb Temperature
BF	By Pass Factor
DP	Dew Point
SHF	Sensible Heat Factor
RH	Relative Humidity
T	Temperature
NBC	National Building Code
SHGC	Solar Heat Gain Coefficient
U Value	Over All Heat Transfer Coefficient

CFM	Cubic Feet Per Minute
DX	Direct Expansion
F	Degree Fahrenheit

6.3.6 BASIS OF DESIGN

6.3.6.1 OUTDOOR DESIGN CONDITION

Site Location	Duliajan, Assam.
Geographic location	Latitude 17°27' N & Longitude 78°28' E
Altitude	545 M above mean sea level.

Outside Design Condition

Summer

Dry Bulb Temperature	:	102.9 Deg. F
Wet Bulb Temperature	:	71.9 Deg. F
Relative Humidity	:	21.7%

Monsoon

Dry Bulb Temperature	:	92.9 Deg. F
Wet Bulb Temperature	:	78.1 Deg. F
Relative Humidity	:	52.2%

Winter

Dry Bulb Temperature	:	56.80Deg. F
Mean coincident Wet Bulb Temperature	:	52.8Deg. F
Relative Humidity	:	77.3%

6.3.6.2 INDOOR DESIGN CONDITIONS

Following indoor design conditions for split/centrally air conditioned (VRF/VRV) space shall be adopted.

TABLE 11: INDOOR DESIGN CONDITIONS FOR HVAC

SPACE	INDOOR CONDITIONS		NOISE CRITERIA	FRESH AIR
	Temp. (°F)	Relative Humidity	NC	AS PER ASHRAE 62.1-2019
DORMITORY	73.4±2°F	55%	40	5 CFM / person + 0.06 CFM / Sq. Ft.
DINNING ROOM	73.4±2°F	55%	40	7.5 CFM / person + 0.18 CFM / Sq. Ft.
ROOM-1	73.4±2°F	55%	40	5 CFM / person + 0.06 CFM / Sq. Ft.
MULTIPURPOSE RM	73.4±2°F	55%	40	7.5 CFM / person + 0.18 CFM / Sq. Ft.
KIDS SPORTS ROOM	73.4±2°F	55%	40	0.3 CFM / Sq. Ft.
INDOOR SPORTS ROOM	73.4±2°F	55%	40	0.3 CFM / Sq. Ft.
INDOOR GYM ROOM	73.4±2°F	55%	40	0.3 CFM / Sq. Ft.
MEETING HALL	73.4±2°F	55%	40	5 CFM / person + 0.06 CFM / Sq. Ft.
COMMON HALL	73.4±2°F	55%	40	5 CFM / person + 0.06 CFM / Sq. Ft.
BEDROOM	73.4±2°F	55%	40	5 CFM / person + 0.06 CFM / Sq. Ft.
LIVING RM	73.4±2°F	55%	40	5 CFM / person + 0.06 CFM / Sq. Ft.

Study Room	73.4±2°F	55%	40	5 CFM / person + 0.06 CFM / Sq. Ft.
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6.3.6.3 MECHANICAL VENTILATION

TABLE 12: MECHANICAL VENTILATION

Toilet (Private)	10 ACPH
Toilet (Public)	10 ACPH
Kitchen	35 ACPH
Store	6 ACPH
Record room	6 ACPH
Emergency Smoke Ventilation	12 ACPH

6.3.6.4 (Assembly U value)

TABLE 13: ASSEMBLY U VALUE

CLIMATE ZONE	Description	Unit		Remarks
		Maximum U-factor (BTU/Hr. Sq. Ft. Deg. F)	Maximum SHGC	
Composite	Façade	1.13	0.56	Single glass
	Wall Exposed to Sun	0.35	-	230 mm thick
	Wall Partition	0.40	-	230 mm thick
	Floor Partition	0.40	-	150 mm thick
	Roof Exposed to Sun	0.12	-	Insulated Roof
	Roof Shaded	0.07	-	Insulated Roof

Note: The above values are to be fine-tuned with energy simulation outputs and based on the actual selection of Glass by Architect or Client at the detailed design stage.

6.3.6.5 Occupant Density/Lighting Load/Equipment Load:

TABLE 14: OCCUPANT DENSITY/LIGHTING LOA/EQUIPMENT LOAD

SPACE	Occupancy Sq. Ft/Person	Equipment Load W/Sq. Ft	Lighting Load W/Sq. Ft
DORMITORY	63	1.0	1.2
DINNING ROOM	23	1.0	1.2
ROOM-1	132	1.0	1.2
MULTIPURPOSE RM	31	1.0	1.2
KIDS SPORTS ROOM	32	1.0	1.4
INDOOR SPORTS ROOM	44	1.0	1.3
INDOOR GYM ROOM	34	1.5	1.4
MEETING HALL	15	1.5	1.4
COMMON HALL	33	1.0	1.5
BEDROOM	70	1.5	1.1
LIVING RM	64	1.0	1.1
STUDY RM	36	1.0	1.1

6.3.7 AIR CONDITIONING LOAD AND TYPE OF SYSTEM

HEAT LOAD SUMMARY OF COMMUNITY CENTRE

TABLE 15: HEAT LOAD SUMMARY OF COMMUNITY CENTRE

HEAT LOAD SUMMARY OF COMMUNITY CENTRE																							
S. No.	Description	Area (Sqft)	Height(Ft)	Occ. (Sqft/Per)	Equip. Load (W/Sqft)	Light Load (W/Sqft)	No. of Persons	Fresh Air (CFM)	Coolin g Load (TR)		Heatin g Load (KW)	Deh. CFM (Summer)	IDU Selection						ODU Selection				
									Summer	Monsoon			CFM	TR	Type	Motor (KW)	Nos	Total (KW)	Cap. HP	Nos.	TOTAL HP	INPUT POWER KW	TOTAL (KW)
											Winter												
A	GROUND FLOOR																						
1	COMMON HALL	3368	11.2	37	1.0	1.5	90	652	13.83	11.21	13.50	6098	812	1.6	4-WAY CASSETTE	0.2	10	2	26	1	26	19.5	20
2	TFA	0	11.2	0	0.0	0	0	652	1.98	3.46	-6.00	101	635	4	TFA UNIT	0.4	1	0.40					
3	SOCIETY OFFICE	180	11.2	45	1.0	1.1	4	31	1.24	0.97	-1.20	544	794	2.00	SPLIT HIWALL UNIT	2.60	1	2.6					
4	ATM	114	11.2	57	1500.0	1	2	17	2.38	2.11	0.50	1154	TENANT SCOPE										
	Total	3662					96	1352	19.4	17.8	6.8	7896					12	5	26	1	26	19.5	20
B	FIRST FLOOR																						
1	KIDS SPORTS ROOM	568	11.6	33	1.0	1.4	17	170	4.27	3.93	1.80	1486	865	2.60	4-WAY CASSETTE	0.20	2	0.4	28	1	28	21	21
2	INDOOR SPORTS ROOM	833	11.6	49	1.0	1.3	17	250	5.50	5.05	0.30	1991	1183	3.20	4-WAY CASSETTE	0.30	2	0.6					
3	INDOOR GYM ROOM	908	11.6	36	1.5	1.4	25	272	7.51	7.01	2.80	2774	830	2.00	4-WAY CASSETTE	0.20	4	0.8					
4	MEETING HALL	326	11.6	18	1.5	1.4	18	110	3.01	2.73	0.80	1194	812	1.60	4-WAY CASSETTE	0.20	2	0.4					
	Total	2635					77	802	20.3	18.7	5.7	7445					10	2		1	28		21
	G.Total	6297					173	2154	40	36	13	15341					22	7		2	54		41

HEAT LOAD SUMMARY OF GUEST HOUSE BUILDING

TABLE 16: HEAT LOAD SUMMARY OF GUEST HOUSE BLOCK

HEAT LOAD SUMMARY-GUEST HOUSE BLOCK																		
S. No.	Description	Area (Sqft)	Height(Ft)	Occ. (Sqft/Per)	Equip. Load (W/Sqft)	Light Load (W/Sqft)	No. of Persons	Fresh Air (CFM)	Cooling Load (TR)		Heating Load (KW)	Deh. CFM (Summer)	HVAC Unit Selection					
									Summer	Monsoon			Winter	CFM	TR	Type	Motor (KW)	Nos
A	GROUND FLOOR																	
1	RECEPTION AREA	339	10.3	56	1.0	0.5	6	50	2.77	2	-1.80	1266	540	1.50	SPLIT HIWALL	2.0	2	3.9
2	DORMITORY-1	440	10.3	63	1.0	1.2	7	61	2.53	2	-0.90	1121	540	1.50	SPLIT HIWALL	2.0	2	3.9
3	DORMITORY-2	440	10.3	63	1.0	1.2	7	61	2.48	1.9	-1.00	1095	540	1.50	SPLIT HIWALL	2.0	2	3.9
4	DINNING ROOM	902	10.3	23	1.0	1.2	40	462	7.34	7	-1.80	2677	540	1.50	SPLIT HIWALL	2.0	4	7.8
	Total	2121					60	635	15.1	12.8	-5.5	6159					10	19.5
B	FIRST FLOOR																	
1	DORMITORY-3	440	10.3	63	1.0	1.2	7	61	2.53	2	-0.90	1121	540	1.50	SPLIT HIWALL	2.0	2	3.9
2	DORMITORY-4	440	10.3	63	1.0	1.2	7	61	2.48	1.9	-1.00	1095	540	1.50	SPLIT HIWALL	2.0	2	3.9
3	ROOM-1	264	10.3	132	1.0	1.2	2	26	1.59	1.1	-1.20	746	794	2.00	SPLIT HIWALL	2.6	1	2.6
4	ROOM-2	264	10.3	132	1.0	1.2	2	26	1.49	1.1	-0.90	695	794	2.00	SPLIT HIWALL	2.6	1	2.6
5	ROOM-3	209	10.3	104	1.0	1.2	2	23	1.46	1	-1.50	682	794	2.00	SPLIT HIWALL	2.6	1	2.6
6	MULTIPURPOSE RM	629	10.3	31	1.0	1.2	20	263	4.52	4.2	-1.30	1720	540	1.50	SPLIT HIWALL	2.0	4	7.8
7	STAFF ROOM	207	10.3	103	1.0	1.2	2	22	1.45	1	-1.40	681	794	2.00	SPLIT HIWALL	2.6	1	2.6
	Total	2452					42	483	15.5	12.2	-8.2	6741					12	26.0
C	SECOND FLOOR																	

1	ROOM-4	264	10.3	132	1.0	1.2	2	26	1.69	1.2	-1.70	795	794	2.00	SPLIT HIWALL	2.6	1	2.6
2	ROOM-5	264	10.3	132	1.0	1.2	2	26	1.59	1.1	-1.20	746	794	2.00	SPLIT HIWALL	2.6	1	2.6
3	ROOM-6	264	10.3	132	1.0	1.2	2	26	1.59	1.1	-1.20	746	794	2.00	SPLIT HIWALL	2.6	1	2.6
4	ROOM-7	264	10.3	132	1.0	1.2	2	26	1.49	1.1	-0.90	695	794	2.00	SPLIT HIWALL	2.6	1	2.6
5	ROOM-8	209	10.3	104	1.0	1.2	2	23	1.46	1	-1.50	682	794	2.00	SPLIT HIWALL	2.6	1	2.6
6	ROOM-9	264	10.3	132	1.0	1.2	2	26	1.70	1.2	-1.70	802	794	2.00	SPLIT HIWALL	2.6	1	2.6
7	ROOM-10	264	10.3	132	1.0	1.2	2	26	1.56	1.1	-1.00	732	794	2.00	SPLIT HIWALL	2.6	1	2.6
8	ROOM-11	264	10.3	132	1.0	1.2	2	26	1.56	1.1	-1.00	732	794	2.00	SPLIT HIWALL	2.6	1	2.6
9	ROOM-12	264	10.3	132	1.0	1.2	2	26	1.56	1.1	-1.00	732	794	2.00	SPLIT HIWALL	2.6	1	2.6
10	ROOM-13	207	10.3	103	1.0	1.2	2	22	1.45	1	-1.40	681	794	2.00	SPLIT HIWALL	2.6	1	2.6
	Total	2524					20	251	15.7	11.1	-12.6	7343					10	26.0
D	THIRD FLOOR																	
1	ROOM-14	264	10.3	132	1.0	1.2	2	26	1.69	1.2	-1.70	795	794	2.00	SPLIT HIWALL	2.6	1	2.6
2	ROOM-15	264	10.3	132	1.0	1.2	2	26	1.59	1.1	-1.20	746	794	2.00	SPLIT HIWALL	2.6	1	2.6
3	ROOM-16	264	10.3	132	1.0	1.2	2	26	1.59	1.1	-1.20	746	794	2.00	SPLIT HIWALL	2.6	1	2.6
4	ROOM-17	264	10.3	132	1.0	1.2	2	26	1.49	1.1	-0.90	695	794	2.00	SPLIT HIWALL	2.6	1	2.6
5	ROOM-18	209	10.3	104	1.0	1.2	2	23	1.46	1	-1.50	682	794	2.00	SPLIT HIWALL	2.6	1	2.6
6	ROOM-19	264	10.3	132	1.0	1.2	2	26	1.70	1.2	-1.70	802	794	2.00	SPLIT HIWALL	2.6	1	2.6
7	ROOM-20	264	10.3	132	1.0	1.2	2	26	1.56	1.1	-1.00	732	794	2.00	SPLIT HIWALL	2.6	1	2.6
8	ROOM-21	264	10.3	132	1.0	1.2	2	26	1.56	1.1	-1.00	732	794	2.00	SPLIT HIWALL	2.6	1	2.6
9	ROOM-22	264	10.3	132	1.0	1.2	2	26	1.56	1.1	-1.00	732	794	2.00	SPLIT HIWALL	2.6	1	2.6
10	ROOM-23	207	10.3	103	1.0	1.2	2	22	1.45	1	-1.40	681	794	2.00	SPLIT HIWALL	2.6	1	2.6
	Total	2524					20	251	15.7	11.1	-12.6	7343					10	26.0
	Grand Total	9622					142	1621	61.9	47.3	-38.9	27586					42	97.5

6.3.8 SYSTEM PROPOSED

6.3.8.1 COMPARISION OF DIFFERENT AC SYSTEM AVAILABLE

TABLE 17: COMPARISON OF DIFFERENT AC SYSTEMS

S No.	System Characteristic	Variable Speed/Flow/Frequency/ Capacity Unitary Systems (Inverter Split ACs/Package ACs)	Variable Speed Multi or Modular Systems (VRF)	Water Cooled DX Units	Central Systems
i)	Temperature	Minimal uniform and effective control possible	Reasonably uniform and effective control Possible	Reasonably uniform and effective control possible	Uniform and effective control possible
ii)	Relative humidity	Effective control not possible	Effective control not possible	Effective control not possible	Effective control possible
iii)	Space pressure	Effective control not possible	Effective control not possible	Effective control not possible	Effective control possible
iv)	Capacity requirements	Capacity to suit zone peak, no diversity	Capacity to suit zone peak, limited diversity can be considered	Capacity to suit zone peak, no diversity	Allows the design engineer to consider HVAC load diversity factors, accordingly reduce installed equipment capacity
v)	Redundancy	Does not have the benefit of back-up or standby equipment	It has the benefit of partial back-up or standby equipment	Does not have the benefit of back-up or standby equipment	Back-up or standby equipment can easily be accommodated

vi)	Facility management	Allows minimal provision to maximize performance using good facility management techniques in operation	Limited possibilities to maximize performance using good facility management techniques in operation	Limited possibilities to maximize performance using good facility management techniques in operation	Allows to maximize performance using good facility management techniques in operation
vii)	Spatial requirements	No plant/equipment room required. Compromises building elevation	No plant/equipment room required. Outside units can be located on roof or on adjacent ground. Very small shaft desirable for refrigerant piping	No plant/equipment room required. Cooling Towers can be located on roof or on adjacent ground. Shaft desirable for water piping	Equipment rooms/spaces and accessible shaft required for chilled water piping
viii)	Electric supply	Distributed electric supply required	Zone-wise distributed electric supply	Distributed electric supply required.	Minimal distribution cost by centralized supply near the substation
Ix	Initial cost	Minimum initial cost	Moderate initial cost but marginally higher than variable speed unitary product	Higher initial cost.	Even with HVAC diversity, a central system may not be less costly than decentralized HVAC systems
X	Operating cost	Strategic scheduling of	Strategic scheduling of equipment can	Strategic scheduling of	More energy-efficient primary

		multiple pieces of equipment can save reasonable operating cost, but higher peak energy requirement	save operating cost better than unitary product, but higher peak energy requirement	equipment can save operating cost better than unitary product, but higher peak energy requirement	equipment and multiple pieces of HVAC equipment allow staging of operation to match building loads while maximizing operational efficiency.
xi)	Maintenance cost	Maintenance cost marginally higher than fixed speed	Comparatively less maintenance cost	Maintenance cost marginally higher VRF system.	Comparatively higher since centralized equipment room requires operator with no access to occupant workspace, but with fewer pieces of HVAC equipment to service

xii)	Reliability	Reliable equipment but low service life	Reliable equipment but moderate service life	Reliable equipment but moderate service life	Reliable equipment with much longer service life
xiii)	Flexibility	Has to be placed at fixed locations	Can be placed at distributed locations	Can be placed at distributed locations	Flexibility available in terms of alternative locations
xiv)	Level of control	Limited control level available	Moderate control level available	Moderate control level available	Close control level available

From above table, it is concluded that Variable Refrigerant Volume is more reliable for the proposed project as the same is cost effective, long life and least operating cost.

6.3.8.2 Variable Refrigerant Volume (VRV) System

Variable Refrigerant Volume (VRV) System shall be designed to provide year-round thermal environmental control for Open Office, Office Cabins, Conference Room, Passage, and Lab etc. as VRV having following advantages.

In a VRV system, multiple indoor fan coil units may be connected to one outdoor unit. The outdoor unit has one or more compressors that are inverter driven, so their speed can be varied by changing the frequency of the power supply to the compressor. As the compressor speed changes, so does the amount of refrigerant deliver by the compressor.

VRV systems are available either as heat pump systems or as heat recovery systems for those applications where simultaneous heating and cooling is required in addition to providing superior comfort.

VRV systems offer design flexibility, energy savings, and cost-effective installation.

In a VRV system, since the speed of the compressor can be varied, the compressor does not cycle on and off, but operates continuously for longer periods. The required refrigerant flow is supplied to the indoor fan coil and once the set point is reached, the refrigerant flow is adjusted to maintain the room temperature smoothly without fluctuation. In addition to having distinct set points, the indoor unit fan speeds and louver positions can be changed to provide additional comfort in the space. The extended maximum piping length gives more flexibility when designing the system. Layout changes can be made easily because the capacity of the indoor units can be up to 130% that of the outdoor units. New compressor technology eliminates the need for piping calculations, which shortens the time needed for design. Outdoor units can be placed on the roof where they have no effect on the design of the building interior.

6.3.8.3 ELECTRICAL POWER REQUIREMENT (AC SYSTEM)

TABLE 18: ELECTRICAL POWER REQUIREMENT (AC SYSTEM)

S. No	Description	Electrical Power Requirement (KW) (Approx.)
1	GUEST HOUSE BLOCK	96.2
2	TYPICAL FLOOR EACH FLOOR	36.4
	TYPICAL FLOOR EACH TOWER	218.4
3	COMMUNITY CENTRE	49
	TOTAL	400

6.3.9 VENTILATION

6.3.9.1 Toilets and Pantry

Toilets and Pantries are mechanically ventilated. Propeller/ inline fans shall be provided for the ventilation. Fresh air shall be drawn from adjoining AC areas through undercut which is provided in the toilet/pantry doors. The system shall consist of fans, electrical panel, power cabling, control wiring and earthing. Ventilation fans shall be provided for exhausting equivalent amount of air.

6.3.9.2 Emergency Smoke Ventilation

As defined in the National Building Code of India, mechanical ventilation for emergency smoke ventilation & 12 ACPH in case of fire or distress call. The system shall comprise of ducted ventilation system:

Exhaust air shall be expelled with the exhaust air fan. The exhaust fan shall be axial flow fans, axial flow fan shall run at higher speed

6.3.9.3 Lift Well Pressurization

It is proposed to install one numbers of tube axial flow fans of required capacity, one number on the terrace to pressurize lift well. In the lift well air shall be directly fed through axial flow fan connected to a small piece of duct.

All pressurization fans shall run in case of fire only. Panel of pressurization fans shall be integrated with the fire alarm panel.

6.3.9.4 Lift Lobby Pressurization

Lift lobbies are not provided at each floor; therefore, hence pressurization system is not applicable.

6.3.9.5 Staircase Pressurization

All the staircases shall have natural connection with ambient through open able window / architecture louvers.

6.3.10 DESIGN PARAMETERS FOR MAJOR EQUIPMENTS

Duct-able Units and Its Components		
Maximum velocity across pre-filters	:	150 M / Min.
Maximum face velocity across cooling coils	:	150 M / Min.
Maximum fan outlet velocity for fans above 300 mm dia.	:	450 M / Min.
Maximum fan speed for fans up-to 300 mm dia.	:	1450 RPM
Maximum fan speed for fans above 300 mm dia.	:	1000 RPM
Maximum motor speed	:	1450 RPM
Centrifugal Fans		
Maximum fan outlet velocity for fans up-to 450 mm dia.	:	450 M / Min.
Maximum fan outlet velocity for fans above 450 mm dia.	:	500 M / Min.

Duct Design		
Maximum flow velocity	:	365 M / Min.

Maximum Friction	:	1 cm WG/100 M run or (0.1 inch per 100 Ft)
Maximum Velocity at supply air outlet	:	150 M / Min.

VRV / VRF Outdoor Unit

The performance rating of the VRV / VRF Outdoor Unit shall be based on the following design parameters:

COP	:	More than or equal to 4.00
Refrigerant	:	R-410 / R-407C
Air Suction Temperature to condenser	:	45.0 ° C

6.3.11 BRIEF OUTLINE SPECIFICATIONS FOR MAJOR EQUIPMENT

6.3.11.1 GI Propeller Fan

GI propeller fan shall have blades made of pressed steel of aero-foil design, motor and blades assembly mounted in a cast iron / sheet steel frame with steel brackets, rubber mounts between the mounting frames and brackets etc.

6.3.11.2 Box Type Inline Fan

Inline fans shall have housing of hot rolled GSS sheet metal construction, fan wheel forward curved type with statically and dynamically balanced, direct driven single / three phases motor, acoustically lined from inside etc.

6.3.11.3 Air Distribution System

All ducts shall be factory fabricated from galvanized steel sheets of various thicknesses ranging from 26 gauges to 18 gauges according to duct sizes and in accordance with SMACNA standards. Galvanized steel sheets shall be produced by hot dip process. Grilles / diffusers shall be powder coated extruded aluminium construction and shall be provided as per the requirement of interior design.

6.3.11.4 VRV / VRF Outdoor Unit

VRV / VRF Outdoor Unit specified in this section are of Air-Cooled Type, inverter scroll multiple compressors (minimum one inverter scroll compressor in one module) on 415 V motor. Refrigerant shall be R-410a / R-407C. The unit shall consist of compressors, refrigerant charged, motor, oil separator, condenser, factory mounted microprocessor-based panel,

interconnecting refrigerant piping provision, electronic expansion valve, controls and accessories to make it compact & efficient unit.

6.3.11.5 VRV / VRF Indoor Units

Indoor unit shall be single skin construction draw-through type comprising of filter section, coil section and fan section, drain pan with uniform slope from all sides leading to drain pan ensuring no stagnation of condensate water.

4-way flow VRV/VRF Cassette Type Indoor ceiling mounted unit equipped with synthetic washable media pre-filter, fan section with low noise fan/dynamically balanced blower, multispeed motor, coil section with DX Copper coil, electronic expansion valve, outer cabinet, drain pump, grill, necessary supports, vibration isolation, cord less remote control etc., suitable for operation on single phase 230 V $\pm$ 10%, 50Hz AC supply, complete, as required. The unit shall have automatic force shut down provision in case of fire on receiving signal from BMS System. The cooling capacity of indoor unit will be at air inlet conditions of 27 Degree C DB and 19 Degree C WB temperature.

VRF/VRV ceiling mounted duct-able type Indoor unit equipped with washable synthetic media pre-filter, fan section with low noise fan/dynamically balanced blower, multispeed motor, coil section with DX copper coil, electronic expansion valve, corded remote control, outer cabinet, vibration isolators, drain pan, other necessary supports etc., suitable for operation on single phase AC supply 230 V $\pm$ 10%, 50 Hz complete as required. The unit shall have automatic force shut down provision in case of fire on receiving signal from BMS System. The cooling capacity of indoor unit will be at air inlet conditions of 27 Degree C DB and 19 Degree C WB temperature.

6.3.11.6 VRV / VRF Refrigerant Piping

Refrigerant piping shall be constructed from soft seamless up to 19.1mm and hard drawn copper refrigerant pipes for above 19.1mm with copper fittings and silver-soldered joints. The thickness of copper piping shall not be less than 20 gauges for pipes up to 19.1mm and 18 gauges for bigger sizes.

6.3.11.7 Condensate drain piping

Drain Piping shall be galvanized steel class 'B' (medium class) confirming to relevant BIS codes. All jointing in the condensate drain pipe system shall be by screwed joints or by screwed flanges.

6.3.11.8 Insulation of AC Ducts

Insulation material for ducts, refrigerant & condensate pipes shall be closed cell XLPE with aluminium facing. Refrigerant pipes with thermal insulation shall be provided with protective coating against mechanical damages.

6.3.12 VENTILATION CALCULATIONS WITH EQUIPMENT SELECTION

a) TYPICAL FLOOR

TOILETS VENTILATION CALCULATION																
Sr. No.	Description	AREA (Sq. ft)	Height (ft)	Ventilation Rate (ACPH)	Air quantity (CFM)	Supply Fan CFM	Type of Fan	Qty.	DIA (MM)	Motor(kw)	Exhaust Fan CFM	Type of Fan	Qty.	DIA (MM)	Motor (kw)	Total Electrical load (kw)
A	TYPICAL FLOOR															
1	LD-C. TOILET	34.4	10	10	57.4	THROUGH UNDERCUT					100	PF	1	150	0.1	0.1
2	LD-A. TOILET	41.1	10	10	68.5	THROUGH UNDERCUT					100	PF	1	150	0.1	0.1
1	LT-C. TOILET	34.4	10	10	57.4	THROUGH UNDERCUT					100	PF	1	150	0.1	0.1
2	LT-A. TOILET	41.1	10	10	68.5	THROUGH UNDERCUT					100	PF	1	150	0.1	0.1
1	RT-C. TOILET	34.4	10	10	57.4	THROUGH UNDERCUT					100	PF	1	150	0.1	0.1
2	RT-A. TOILET	41.1	10	10	68.5	THROUGH UNDERCUT					100	PF	1	150	0.1	0.1
1	RD-C. TOILET	34.4	10	10	57.4	THROUGH UNDERCUT					100	PF	1	150	0.1	0.1
2	RD-A. TOILET	41.1	10	10	68.5	THROUGH UNDERCUT					100	PF	1	150	0.1	0.1
TOTAL		302.1											8.0			0.8
NOTE:																
1	LD=LEFT DOWN															
2	LT=LEFT TOP															
3	RD=RIGHT DOWN															
4	RT=RIGHT TOP															

A) COMMUNITY CENTRE

COMMUNITY CENTRE VENTILATION CALCULATION
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Sr. No.	Description	AREA (Sq. ft)	Height (ft)	Ventilation Rate (ACPH)	Air quantity (CFM)	Supply Fan CFM	Type of Fan	Qty.	Fan Static (MMWG)	Motor(kw)	Exhaust Fan CFM	Type of Fan	Qty.	Fan Static (MMWG)	Motor (kw)	Total Electrical load (kw)
A	GROUND FLOOR															
1	M+F TOILET	376.6	10	10	627.7	THROUGH UNDERCUT					650	IF	1	15	0.2	0.2
	TOTAL	376.6											1.0			0.2
B	FIRST FLOOR															
1	M+F TOILET	376.6	10	10	627.7	THROUGH UNDERCUT					650	IF	1	15	0.2	0.2
2	RECORD ROOM	376.6	11.15	6	420.0	THROUGH UNDERCUT					450	IF	1	10	0.2	0.2
	TOTAL	753.2											2.0			0.4
	G.TOTAL	1129.8											3.0			0.6
	NOTE:															
	IF=INLINE FAN															

b) GUEST HOUSE

GUEST HOUSE VENTILATION CALCULATION																		
Sr. No.	Description	AREA (Sq. ft)	Height (ft)	Ventilation Rate (ACPH)	Air quantity (CFM)	Supply Fan CFM	Type of Fan	Qty.	DIA	Fan Static (MMWG)	Motor(kw)	Exhaust Fan CFM	Type of Fan	Qty.	Fan Static (MMWG)	DIA	Motor (kw)	Total Electrical load (kw)
A	GROUND FLOOR																	
1	DORMITORY-1 TOILET	136.2	10	10	227.0	THROUGH UNDERCUT						225	IF	1	10	-	0.15	0.15

THE INFRASTRUCTURE PEOPLE

2	DORMITORY-2 TOILET	136.2	10	10	227.0	THROUGH UNDERCUT						225	IF	1	10	-	0.15	0.15
3	KITCHEN	495.0	10.3	35	2983.1	3000	AW	1	-	50	1.5	3300	AS	1	60	-	3.75	3.75
4	STORE	124.8	10.3	6	129.0	THROUGH UNDERCUT						400	IF	1	15	-	0.2	0.2
5	STORE/UTILITY	250.2	10.3	6	258.5	THROUGH UNDERCUT												
	TOTAL	1142.4						1.0						4.0				4.3
B	FIRST FLOOR																	
1	DORMITORY-1 TOILET	136.2	10	10	227.0	THROUGH UNDERCUT						225	IF	1	10	-	0.15	0.15
2	DORMITORY-2 TOILET	136.2	10	10	227.0	THROUGH UNDERCUT						225	IF	1	10	-	0.15	0.15
3	ROOM-1 TOILET	42.0	10	10	69.9	THROUGH UNDERCUT						100	PF	1	-	150	0.1	0.1
4	ROOM-2 TOILET	42.0	10	10	69.9	THROUGH UNDERCUT						100	PF	1	-	150	0.1	0.1
5	ROOM-3 TOILET	40.4	10	10	67.3	THROUGH UNDERCUT						100	PF	1	-	150	0.1	0.1
6	STAFF ROOM TOILET	39.8	10	10	66.4	THROUGH UNDERCUT						100	PF	1	-	150	0.1	0.1
	TOTAL	436.5						0.0						6.0				0.7
C	SECOND FLOOR																	
1	ROOM-4 TOILET	42.0	10	10	69.9	THROUGH UNDERCUT						100	PF	1	-	150	0.1	0.1
2	ROOM-5 TOILET	42.0	10	10	69.9	THROUGH UNDERCUT						100	PF	1	-	150	0.1	0.1
3	ROOM-6 TOILET	42.0	10	10	69.9	THROUGH UNDERCUT						100	PF	1	-	150	0.1	0.1
4	ROOM-7 TOILET	42.0	10	10	69.9	THROUGH UNDERCUT						100	PF	1	-	150	0.1	0.1
5	ROOM-8 TOILET	40.4	10	10	67.3	THROUGH UNDERCUT						100	PF	1	-	150	0.1	0.1
6	ROOM-9 TOILET	42.0	10	10	69.9	THROUGH UNDERCUT						100	PF	1	-	150	0.1	0.1
7	ROOM-10 TOILET	42.0	10	10	69.9	THROUGH UNDERCUT						100	PF	1	-	150	0.1	0.1
8	ROOM-11 TOILET	42.0	10	10	69.9	THROUGH UNDERCUT						100	PF	1	-	150	0.1	0.1
9	ROOM-12 TOILET	40.4	10	10	67.3	THROUGH UNDERCUT						100	PF	1	-	150	0.1	0.1
10	ROOM-12 TOILET	39.8	10	10	66.4	THROUGH UNDERCUT						100	PF	1	-	150	0.1	0.1
	TOTAL	414.3						0.0						10.0				1.0

NOTE:
IF=INLINE FAN
PF=PROPELLER FAN
AW=AIR WASHER
AS=AIR SCRUBBER

6.3.13 EMERGENCY SMOKE VENTILATION CALCULATIONS

SMOKE VENTILATION CALCULATIONS COMMUNITY CENTRE																				
Sr. No.	Description	Area (Sqft)	Height (ft)	Ventilation Rate (ACPH)	Air quantity (CFM)	Supply Air Fan							Exhaust Air Fan							Total Electrical load (KW)
						Supply Fan CFM	Type of Fan	Fan Quantity (Nos)	Fan Static (MMWG)	Motor (RPM)	Fan Dia. (MM)	Motor KW	Exhaust Fan CFM	Type of Fan	Fan Quantity (Nos)	Fan Static (MMWG)	Motor (RPM)	Fan Dia. (MM)	Motor KW	
	EMERGENCY CASE																			
A	FIRST FLOOR																			
1	COMMON HALL	3131	11.2	12	6984	3500	AF	2	25	1450	560	1.1	3500	AF	2	25	1450	560	1.1	4.4
	TOTAL	3131						2							2					4.4
	AF = Axial Fan																			

6.3.14 LIFT WELL PRESSURIZATION TYPICAL TOWER

TABLE 19: LIFT WELL PRESSURIZATION

<u>Lift Well Pressurization (Ground to 6th floor)</u>		
Total Floor	7 Nos.	
Door Size	2.1	1.0
No of Landing	: 7 Nos. (GF-6th Floor)	
No of doors	7	
Leakage through the door, Q	Q= 0.839 *√50*Leakage Area	
Leakage Area	0.042 Sqmt	
Q (Through one door)= 0.839 *√50 * leakage area	0.246 Cum/s	
Q1 Through 6 Nos. Doors	1.477 Cum/s	
Air flow through the one door open @ velocity 0.75 mtrs per second		
Q2	1.575 Cum/s	
Total Air quantity = Q1 + Q2	3.052 Cum/s	
TOTAL CFM	6462 CFM	
Add 15 % factor of safety	7431 CFM	
1 Nos. 7500 CFM EACH @ 40 mmwg SP, 4.0 KW Motor, Dia=710, Tube axial fan for pressurization		
Total numbers of fans-2 Nos. (2 Nos. of Lifts)		

6.3.15 EXCLUSIONS FROM HVAC SCOPE OF WORK

- Civil work for foundations, trenches, sumps, drain channels and gratings (Civil)
- Thermal Insulation for all roof terraces above air-conditioned areas exposed to sun - equivalent to 50 mm thick extruded polystyrene (Civil).
- False ceiling and boxing (Interiors)
- Supply and fixing of wooden frames for grilles / diffusers (Interiors).
- Power supply and earthing to all HVAC system panels (Included in Electrical).
- Power socket within 2 meters of each split A/C unit, exhaust fan and each fan coil unit (Included in Electrical).
- Drainage from AHU rooms, AWU rooms and Duct-able Units (Included in Plumbing).

DESIGN BASIS REPORT FOR PLUMBING & FIREFIGHTING SYSTEMS

7 PLUMBING & FIREFIGHTING SYSTEMS

7.1 PREAMBLE

7.1.1 SCOPE OF WORK

The objective of this report is to give an overview of services designed for the Workmen Housing Complex (proposed project) located at Duliajan, Assam. The designs of Engineering Services undertaken includes following:

- Plumbing System.
- Fire Fighting System.

7.1.2 PROJECT OBJECTIVES

The project objective includes eco-friendly energy efficient design, location of services installation to ensure easy maintenance, state of the art technology, fast track installation and compliance to all statutory regulations.

7.1.3 BROAD CONCEPT OF SERVICES:

The Services Systems for the project have been conceptualized based on experience and acceptable international design standards. Effort shall be made to conceal all services and still provide access to these for accommodating changes in requirement in future. Conservation of energy, optimization of resources, eco-friendliness and State of the art technology shall be the key factors in the design concept to ensure least downtime and reduce maintenance hassles. Every effort shall be made to design, layout and install equipment in locations, which will tend to encourage routine preventive maintenance by providing easy access for operation personnel. Manual isolation will be provided to enable servicing, expansion or renovation of any part of the system without interrupting the services in adjacent areas.

7.1.4 ENERGY CONSERVATION FEATURES:

- Low water consuming sanitary fixtures.
- Water Transfer system.
- Pumps & equipment selected based on “best” energy efficiency point and guidelines issued by IGBC, India.
- Sensor operated flushing system for urinals in Public Toilets.
- Flow restrictors.

7.1.5 DESIGN PHILOSOPHY AND DESIGN CONSIDERATION

The design philosophy of the Plumbing/ Sanitary and Fire Fighting is to use the latest state of the Art Technology for the design of system that will work reliably and efficiently to meet the code requirements and Client's operational requirements in close co-ordination with the Client/Architect, Structure Consultants and other related Consultants.

The Services shall be designed having provisions for proper access for maintenance of system equipment and to ensure sustainable, reliable and efficient operation.

TABLE 20: DESIGN CONSIDERATIONS FOR FIRE FIGHTING & PLUMBING SYSTEM

Design Considerations	What it Means
Reliability	To allow for sufficient equipment and service redundancies to meet the operational need of the operator.
Maintainability	Provision for shut down or repairs of sub-systems with minimum disruption to the operation of the building, and allow for ease of access for maintenance, repair and replacement.
Flexibility and adaptability	To meet the changing need in the sub-division of commercial spaces and ever-evolving information technologies for comfort and secured living.
Expandability	To meet future needs with designated equipment having spare capacities.
Energy Efficiency	Efficient equipment and design features for optimum efficiency during normal and off-peak operation.
Cost Effectiveness	Optimizing the performance of the equipment and system.
Modular System Design	Enhances the build ability and allows flexibility in maintenance and replacement. Minimizes the stocking of parts.
State-of-the –Art Technology and Building Intelligence	Fully integrated central control and monitoring of the building sub system with maintenance management and facilities management.

7.1.6 SPACE REQUIREMENT

The space /Area requirement for plant rooms & other equipment including shafts for the various PHE and Firefighting services shall be worked out as per the intended requirements of tender document for successful commissioning of facility in co-ordination with the Architects.

7.1.7 INTEGRATED DESIGN APPROACH WITH CONSULTANT TEAM

Discussions and meeting shall be held to consider the most economic, buildable and maintainable design of the building services.

7.1.8 HIGH-END DEVELOPMENT

The planning and provisions are designed to meet the requirement of a High-end Residential development.

7.2 PLUMBING SYSTEM

7.2.1 REFERENCE STANDARDS

- National Building Code of India Part IX 2016
- American Society of Plumbing Engineers(ASPE)
- Design Data Book Volume I to IV
- Institute of Plumbing Engineers UK Design Data Book – 2002
- International Plumbing Code 2022 Edition
- Uniform Plumbing Code of India 2022 Edition
- Energy Conservation Building Code 2017 (Revised 2008)
- Green Building IGBC Green Homes Ver. 3.0 (Residential)

7.2.2 APPROACH TO PLANNING

- 1 Requirement of adequate and equal pressure of cold water in toilets, floor pantries, kitchen and other designated areas. Cold-water makeup supply to water bodies shall also be ensured.
- 2 The water storage tank capacity should be sufficient to guarantee a one-day supply of water. To ensure a continuous water supply, a storage tank with a capacity of one day (1.0 day) is recommended, along with an overhead tank (OHT) having a capacity of half a day.
- 3 Recycling of treated wastewater (from sewage treatment plant) for flushing & for horticulture water use.
- 4 Implementation of requirements of MoEF (Ministry of Environment and forest) relating to rain Water harvesting, water conservation, use etc.

- 5 Levels of roads / pavements and other services in the area.
- 6 Drainage and water supply provision for Landscape layout.
- 7 Water conservation using low flow fixtures.

7.2.3 SYSTEM REQUIREMENTS

- 1 It is proposed to supply treated water for Domestic usage to entire project. Water treatment plant is located at site as per convenient location. Water treatment plant (WTP) to ensure that the chemical and bacteriological parameters of water supply in the Housing Complex are in accordance with World Health Organization (WHO) standards. Since, quality of water from bore well is Non-potable, WTP shall be very basic, comprising of filters and hypo dosing units. However, WTP system design shall be verified for suitability prior to installation in accordance to latest water analysis report.
- 2 Domestic and flushing water supply through gravity feed from overhead tanks for making water available at all toilets, kitchen, wash areas, wash basins etc.
- 3 Sewage and sullage collection & conveyance system based on ASPE (American Society of Plumbing Engineers) standard and applicable guidelines by NBC.
- 4 Storm / rain water drainage system from the roof terrace and various levels of the building, including balcony drains, planter drains, fountain drains, saucer drains along road side/ by means of draining and surface run-off water to rain water recharge pits for ground water recharging.
- 5 Provision for individual Geysers will be planned in each toilet for meeting hot water demand of the building.
- 6 Sewage treatment plant for treatment of sewage & sullage waste. The plant shall comprise of preliminary, secondary (chemical & biological) and tertiary treatment units. The treated effluent shall be recycled and reused for flushing and for landscape requirement of the Housing Complex. As per guidelines by MoEF, it will be ensured to provide high flow UV System in tertiary treatment plant of STP.

7.2.4 DESIGN CALCULATIONS

Water requirement for the project is designed according to NBC considering water demand as 135 liters per capita per day accordingly OH & U.G tanks are designed.

The scheme has been designed considering 4 persons per 1 bedroom dwelling unit, 5 persons per 2 bed room dwelling unit and 6 people for 3-bedroom dwelling unit and 7 people for 4 or more bedrooms dwelling units.

TABLE:- POPULATION ESTIMATION OF RESIDENTIAL BUILDING

S. No.	Description	Occupancy / Population as per NBC-2016 VOLUME-2, PART-9 Section-I Clause 4.1
1.	4 BHK Apartment / Pent house	7 Person / Dwelling unit
2.	3 BHK Apartment	6 Person / Dwelling unit
3.	2 BHK Apartment	5 Person / Dwelling unit

Note:

Table: - WATER REQUIREMENT / CONSUMPTION AS PER NBC-2016 VOLUME-2,

S. No.	Description	Water Requirement as per NBC-2016, Volume-2, Part-9 2016 VOLUME-2, PART-9 Section-I			Remarks
		Domestic	Flushing	Total	
1.	Residential Apartment	90 LPCD	45 LPCD	135 LPCD	Clause 4.1.2
2.	Guest House	135 LPCD	45 LPCD	180 LPCD	Clause 4.1.2
3.	Community Centre	25 LPCD	20 LPCD	45 LPCD	Clause 4.1.2
4.	Floating Population/Visitors	5 LPCD	10 LPCD	15 LPCD	Clause 4.1.2
5.	Irrigation Water Requirement	8.0 liter/Sqm/Day (NBC 2016 Part-9 Section-I)			Clause 4.1.5.2

- Water requirement is given in Annexures attached.
- In case of drip irrigation, the irrigation water requirement will reduce. Landscape consultant/EPC Contractor shall provide actual water requirement at later stage.

DETAILED WATER CALCULATIONS ARE GIVEN BELOW IN TABLE-1

TABLE 21: WATER DEMAND CALCULATIONS

S.NO.	Description	Population	LPCD Standard	Water Demand (Domestic)	Water Demand (Flushing)
1	Housing	960 (192*5)	135(90 Domestic+45 Flushing)	86400	43200
2	Guest House				
3	Guest Room	23 Rooms (2 Pax/room)	180/pax(135 Domestic + 45 Flushing)	6210	2070
3.1	Dormitory	4 Dormitory (6 Beds in Each)	135/pax(90 Domestic + 45 Flushing)	2160	1080
3.2	Dining	80 Pax	35 for kitchen (25 domestic+10 flushing)	1200	480
3.3	Housekeeping + Staff	10 Pax	45/Pax (25 Domestic + 20 Flushing)	275	220
	Guest Room			9845	3850
4	Community Center	100	45(25Domestic+20 Flushing)	2500	2000
5	Guard Room	10	45(25Domestic+20 Flushing)	250	200
6	Irrigation/Green Area	6000	8L/sqm		48000
Net Water Requirement				98,995	97,250
Total Water Demand				1,96,245	

TABLE- (WASTE WATER CALCULATIONS)

Total Water Requirement	196 KLD
Domestic Water Demand	99 KLD
Flushing Water Demand	97 KLD
Waste Water Generated (90% of Total Water Requirement)	$196 \text{ KLD} - 48 \text{ KLD} = 148 \text{ KLD} \times 0.9 = 133 \text{ KLD}$
Say	135 KLD

TABLE - (WATER BALANCE CALCULATIONS)

S.NO.	DESCRIPTION	ROUND-OFF
A.	DOMESTIC WATER	
1.	Domestic Water Requirement	Say 99 KL/DAY
B.	RECYCLED WATER	
1.	Flushing Water Requirement	Say 49 KL/DAY
2.	Gardening and Landscape W.R @8litres/sqm./day as per NBC i.e. $6000 \times 8 = 48000 \text{ Litres}$	Say 48 KL/DAY
	TOTAL	Say 97 KL/DAY

CONCLUSION

CONCLUSION		REF. No.	Remarks
Water Balance			
Domestic Water Requirement	99 KL	A	
Flushing Water Requirement	49 KL	B	
Total Waste Water Generated (90% of A + 90% of B)	135 KL	C	Clause 3.5 of CPHEEO Manual on Sewerage Page 3-3
Treated Water Requirement from STP for Flushing (B)	49 KL	D	
Total Water requirement for Irrigation Purpose	48 KL	E	
Recycled Water Generated from STP @90% of C	121 KL	F	
Total Recycled Water Requirement D+E	97 KL	G	
Surplus Water Generated from STP During Non-Rainy Days F-G	24 KL	H	
Surplus Water Generated from STP During Rainy Days F-D	72 KL	I	

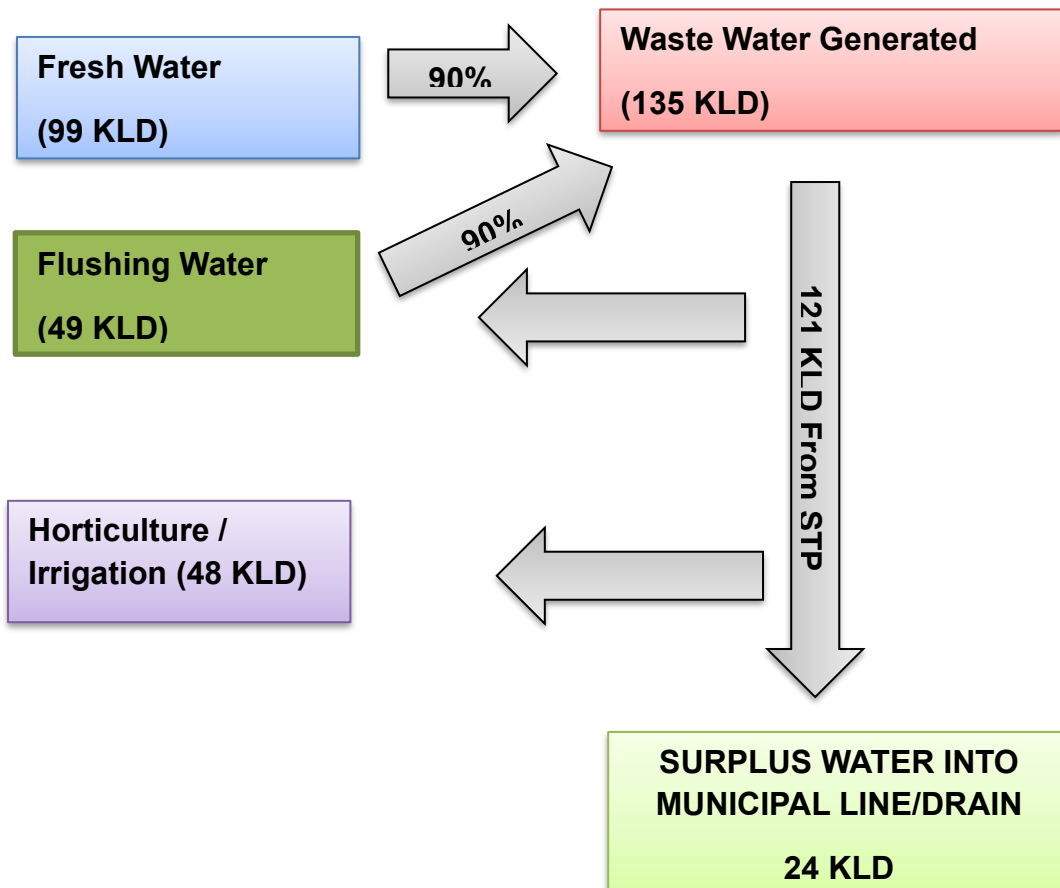


FIGURE 18: WATER BALANCE DIAGRAM IN NON-RAINY DAYS

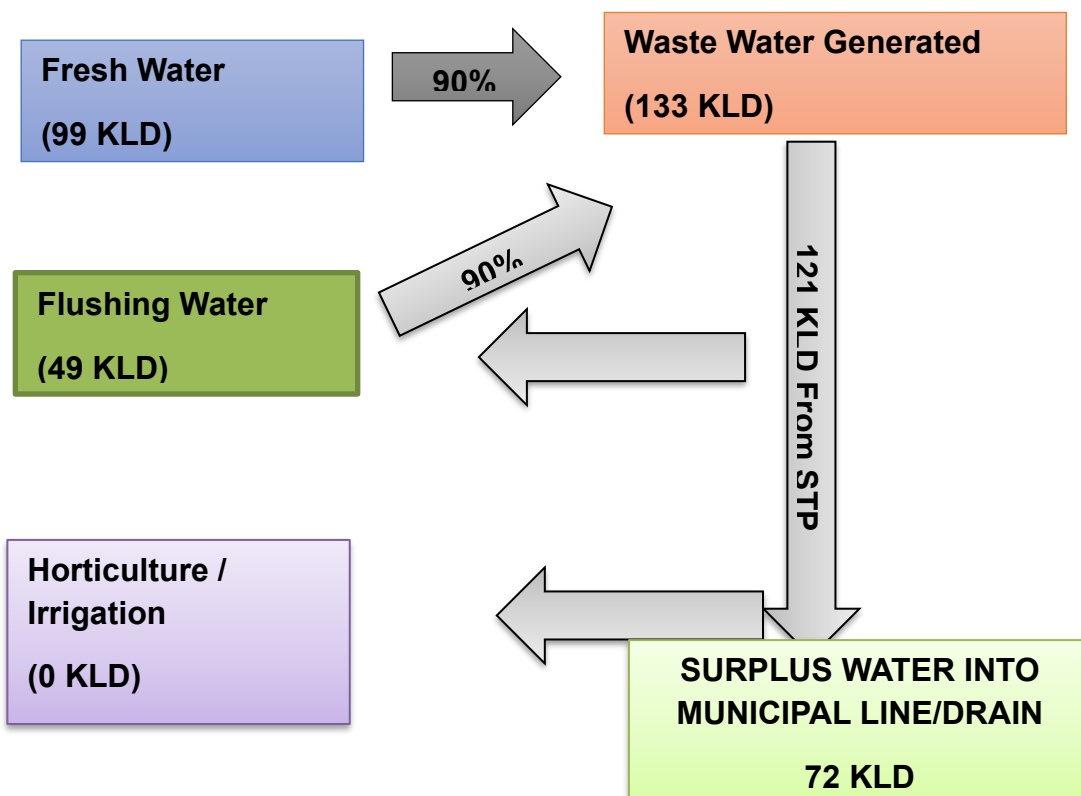


FIGURE 19: WATER BALANCING IN RAINY DAYS

7.2.5 WATER STORAGE

It is proposed to provide UGT& OHT (Table Below) water storage capacity based on ultimate requirement water calculations. The incoming main from Tanker / bore well water supply line shall be led into raw water reserve tanks; thereafter the water shall be stored in domestic water storage tanks. The water storage capacities of the tanks are as follows in Table based on the tower wise calculations as per the details given below:

TABLE 22: WATER STORAGE CALCULATIONS

S.No.	Item Description	Daily Requirement	Factor of storage
1	Fresh Water Requirement	99,000	1.5 (Underground); 0.5 (Overhead) Clause 4.4.10 of NBC 2016 Part 9 Section 1
2	Flushing Water	49,000	1.5 (Underground); 0.5 (Overhead)
3.	Fire Protection – Ground level	NA (HOUSING) 50,000 (GUEST HOUSE)	1
4.	Fire Protection – Roof Level	25000/Block (HOUSING) + 10,000 GUEST HOUSE+20000 (community)	1
5	Treated water (For irrigation)	48,000	1 (for Underground)

7.2.6 CAPACITY OF UNDERGROUND & OVERHEAD WATER STORAGE TANK

TABLE 23: CAPACITY OF UNDERGROUND & OVERHEAD WATER STORAGE TANK

S. No.	Tank type	Under Ground (L)	Overhead Tank (L)
1	Fire Water (HOUSING BLOCKS)	-	25000 on Each Housing Block, 8 nos. Blocks Total 2,00,000 Ltrs
2	Fire Water (GUEST HOUSE)	50,000	10,000
3	Community Fire Water	-	20,000

S. No.	Tank type	Under Ground (L)	Overhead Tank (L)
4	Community Domestic Water & Flushing		1,250 Domestic Water & 1,000 Flushing Water
5	Raw Water/Domestic Water	1,48,500	-
6	Flushing Water	73,500	-
7	Treated water Storage (For irrigation)	48,000	-
8	Domestic Water (HOUSING BLOCKS)	-	5,400 Ltrs Per Block Total 43200 for 8 Blocks
9	Flushing Water (HOUSING BLOCKS)	-	2,700 Ltrs Per Block Total 21600 for 8 Blocks
10	Domestic Water (Guest House)		5,000
11	Flushing Water (Guest House)		2,000
12.	Domestic Tank (Guard Room)		250
13.	Flushing Tank (Guard Room)		200
	Total	3,20,000	3,04,500

7.2.7 SOURCE OF WATER

The daily potable domestic water requirement for the Complex shall be supplied by OIL at site.

7.2.8 WATER TYPE & TREATMENT

Water treatment plant shall be provided in accordance with the various requirements of use. The various type / quality of water and their water treatment plant are as follows:

Domestic Water: Filtered Water supplied by the OIL shall be used. Water shall be used for domestic usage in toilets (wash basin), Kitchen & Pantry (for dishwashing etc.), water body, filter back wash and other areas where direct human contact / use is envisaged. Water softener is proposed to be installed based on water sample report to supply soft water to sanitary faucets.

Drinking Water: These shall be arranged and provided by individual resident consisting of RO water unit.

Recycled Water from STP: Water recovered from tertiary sewage treatment plant shall be used as make-up water for Flushing and for external irrigation.

Flushing Water: Flushing water for the housing Complex shall be sourced from Centralized STP of the complex. The treated effluent for flushing shall be obtained after tertiary treatment at STP. The treated effluent shall be odourless and shall be in acceptable chemical / bacteriological parameters as per local pollution control board laws. Introduce blue color in this water to avoid human contact.

Irrigation Water: Irrigation water for landscape use within the complex shall be of domestic quality and shall be sourced from the in-house STP and shall have acceptable parameters as per governing requirement.

7.2.9 WATER DISTRIBUTION

Water distribution for domestic water supply for the Complex shall be designed on principle of zoning to ensure availability of minimal residual pressure at all user outlets. The water distribution system for domestic water supply and flushing shall be through gravity. Smart Water Meters as per bore size to be installed for individual Towers, other Buildings at terrace level (for domestic/fire-fighting & flushing tanks) prior to inlet & also at Underground Water Tank (inlet & outlet) & STP (Outlet-Irrigation & flushing), RWH (if required) etc. for all distribution lines.

Domestic and flushing water tanks at the terrace levels shall receive water through Hydro-pneumatic system installed at plumbing plant room and STP respectively to transfer the water. However, flushing water tank shall have the provision to receive water from adjacent domestic water tank in case of shortfall of STP treated water.

Underground water storage tank and pump room housing pumps, water treatment plant, transfer pumps shall be as proposed in the tender drawings.

All inlets, outlets, washouts, vents, ball cocks, overflows control valves and all such other piping connections including level indicator shall be provided for underground water storage tanks.

Full way gate valves of approved make shall be provided as close to the underground tank as practicable on every outlet pipe from the storage tank, except the overflow pipe. Overflow and vent pipes shall terminate with mosquito proof grating.

The overflow pipe shall be so placed as to allow the discharge of water being readily seen. The overflow pipe shall be of size as indicated. A stop valve shall also be provided in the inlet water connection to the tank. The outlet pipes shall be fixed approximately 75mm above the bottom of the tank towards which the floor of the tank is sloping to enable the tank to be nearly emptied for cleaning. The pipe sizing shall be based on fixture unit calculation as per IS standard. The pipe size of riser shall be restricted nearest and not more than 3" size to optimize on capital cost, and for ease of installation and maintenance. However, the maximum velocity in the water supply piping shall not exceed 2.4 m/second. Pipe lines used for cold domestic water and flushing water shall be of CPVC Pipes. All hot water lines shall be carried out in CPVC Pipes. Colour coding for flushing, domestic, irrigation and hot water supply piping shall be ensured for clear identification of the piping. The landscape water supply system comprising of pumping, piping, water supply nozzles and control equipment shall be designed and built by the appointed irrigation water supply vendor. Water for irrigation of lawns shall be sourced from recycled treated water from STP. Colour coding for flushing and domestic water supply piping shall be ensured for clear identification of the piping.

PUMP ROOM LOCATION: - Pump room location is designed according to the requirements of water supply of all towers, buildings/infrastructure as required.

Pump room is located at site for entire complex as shown in tender drawing so that water is supplied any different entire complex throughout water supply pipe originated from pump room.

7.2.10 APPURTENANT

Following appurtenant shall be included in the design of water supply system for efficient functioning:

7.2.10.1 Domestic Air Vents:

Automatic air vent shall be provided on cold water risers to eliminate possibility of air locking and to ensure efficient water flow / pressure availability at the user outlets.

7.2.10.2 Flow Restrictors:

Appropriate flow restrictions shall be provided for economizing on water consumption. The flow resistors shall be typically sized for following flow / discharge.

TABLE 24: FLOW RESISTORS DETAILS

S.No.	Type of Faucet	Normal Flow Rates	Low Flow Rates
1	Wash Basin	6.3 LPM	4 LPM
2	Shower	10 LPM	8 LPM
3	Kitchen Sink	6.3 LPM	4 LPM
4	WC Flushing	3/6 LPF	2/4 LPF
5	Urinal Flushing	3.0 LPF	1.5 LPF

7.2.10.3 Ball Valve:

Full bore gun-metal ball valve shall be provided for isolation of cold-water supply for the designated area. Further chrome-plated angle valve shall be provided for cold water isolation to wash basin faucets and for WC cistern water supply.

7.2.10.4 Pressure Gauge:

Pressure Gauge shall be provided on cold water supply piping for efficient balancing and monitoring of the system.

7.2.10.5 Pressure Reduce Valves:

Pressure reducing valves are provided at the inlet of all toilets/Kitchen alternatively in shaft also whichever is economical where pressure is exceeding 3 Kg/sqcm.

7.2.10.6 Hot water system

Provision for electric hot water geysers shall be made in the kitchen and all toilets of each apartment to meet future hot water requirements. Plumbing for hot water shall also be done in all the toilets of the residential complex.

7.2.10.7 WATER METERING:

Smart water meter shall be provided for following services/Area: -

- Municipal water supply / Connection.
- Borewell
- Recycled / Treated water for Flushing & Irrigation water supply.
- Captured Rainwater for reuse in the project.
- Buildings for bulk metering.

However, final metering strategy shall be provided/identified by the Green Building Consultant appointed by EPC Contractor.

7.2.11 SEWAGE AND STORM WATER DRAINAGE

The following parameters/ site conditions shall be kept in mind when designing the sewage and storm water drainage system:

- Natural slope of the area;
- Layout of different facility in the complex;
- Sub-soil water table;
- Soil condition;
- Provision of Sewage lifting station;
- Provision of venting arrangement for manholes;
- Construction of manholes & laying of pipes considering ground condition;
- Termination of vent cowl at terrace level;
- Provision of adequate slope for horizontal header in the under slung pipes especially for public toilets.

7.2.11.1 Sewage & waste drainage

The soil and waste shall be carried down in two pipes vented pipes. Two pipes drainage systems shall be adopted as per IS standard. The sanitary, waste & vent system shall be water tight and gas tight designed to prevent escape of foul gas and odour from various fixtures.

Vent system shall be designed to facilitate escape of gases and odour from all parts of sanitary and waste system to the atmosphere at a point above the building and to allow admittance of air to all part of the system, so that siphonage, aspiration or back pressure conditions do not cause loss of seal at traps.

It is proposed to use uPVC pipes for soil and waste drainage within toilets, and DWC pipes for external soil and waste drainage. The soil and waste piping within toilets shall be routed above the false ceiling, and horizontal branches shall be connected to vertical stacks located within associated pipe shafts. Proper coordination with other services shall be ensured in consultation with the Architect. Care shall be taken to avoid pipe runs in electrical switch rooms, PABX rooms and other critical areas.

Provision for cleaning and rodding eyes shall be made at strategic locations to allow the system maintenance. Provision of drip tray is also proposed for critical areas to avoid dripping of leaking pipes.

7.2.12 DESIGN CRITERIA

The system shall be designed as per following design criteria stipulated in the “Manual for Sewerage & Treatment” published by the Central Public Health and Environment Engineering Organization, Ministry of Urban Development, Govt. of India, IS-SP/35(S&T)-1987 and National and International practices on the subject.

Flow of sewage	:	90% of water supply (peak flow)
Peak flow	:	3 times average flow
Min. diameter of pipe	:	1500 mm dia.
Min. velocity in pipe	:	0.75 mps Clause 4.5.3.5.2.2 of NBC 2016 Part 9 Section 2 Page 37
Max. Velocity in pipe	:	2.4 m / second Clause 4.5.3.5.2.4 of NBC 2016 Part 9 Section 2
Flow conditions in pipes :		
Pipes.	:	0.50 full running Clause 4.5.3.5.2.2 of NBC 2016 Part 9 Section 2
Min. depth of pipe below ground level:		
For branches	:	0.6 m
For other:		1.0M

Formula for calculation and design of sewer lines

Manning's Formula $Q = A * 1.486 * R^{(2/3)} * S^{(.5)/n}$

$$V = R^{(2/3)} * S^{(.5)/n}$$

n=Roughness coefficient

R=hydraulic radius S=slope

Q= discharge

7.2.13 SANITARY FIXTURES & FITTINGS

- The selection of sanitary fixtures and fittings shall be carried out in consultation with Interior Designer/Architect. However, based on experience, the following selection is proposed.
- All sanitary wares shall be white vitreous china. Water closets (European pattern) shall be wall hung/pedestal type as required. Water closet shall be provided with dual flush type cistern.
- All wash basins throughout the complex shall be of white vitreous china under counter, oval wash basin / flat back wash basins with pedestal, and hot and cold-water single lever mixer fitting, waste coupling with CP bottle trap fixed to the outlet. Inlet connections shall be connected by CP angle stop cocks below the counter.
- Shower in the toilets shall have a cold & hot water single lever mixer arrangement and CP spout shall be provided. The shower arm and rose shall be of throw away type and the spray from the rose shall be adjustable.
- All sinks in kitchens shall be of stainless steel with single / double drain board and cold / hot water mixer fittings. CP waste coupling and bottle trap shall be provided in the pantries.
- Suitable accessories such as toilet paper holder, towel rails, soap dish shall be provided at suitable locations.

7.2.14 SEWAGE TREATMENT PLANT

Sewage generated from the project shall be treated in a centralized Sewage Treatment Plant (STP) of 135 KLD capacity, packaged type constructed using FRP (Fibre Reinforced Plastic) structure and based on MBBR (Moving Bed Biofilm Reactor) technology, having tertiary treatment facility integrated with an ultrafiltration unit. The treatment process shall ensure that the quality of the treated effluent meets the prescribed standards for reuse in various non-potable applications such as flushing, gardening etc. Treated water tank should have sensor to monitor the treatment performance within the desired parameters, pump room, plant room. The STP to be installed above ground.

Design Parameters

Following parameters is required for treated water reuse

TABLE 25:DESIGN PARAMETERS FOR STP

Parameters	Permissible limits (As Per NGT Effluent Discharge Standards for STP Application Number 1069/2018)
Ph	6.5 – 8.5
BOD	Less than 5 Mg/L
Suspended Solids	Less than 10 Mg/L
COD	Less than 50 Mg/L
Oil & Grease	Less than 10 Mg/L
Detergents	Less than 10 Mg/L

Sewage network shall collect the sewage from all units, and flow by gravity to the proposed sewage treatment plant. Following are the benefits of providing the Sewage Treatment Plant in the present circumstances: Reduced net daily water requirements, source for Horticultural purposes by utilization of the treated waste water. Reduced dependence on the public utilities for water supply and sewerage systems. Sludge generated from the Sewage Treatment Plant shall be rich in organic content and an excellent fertilizer for horticultural purposes.

A. Waste water Details (As Per CPHEEO Manual on Sewerage 2013 Chapter 5 Clause 5.1.4.1 Table 5.4)

(a) Daily load	:	135 KLD
(b) Duration of flow to STP	:	24 hours
(c) Temperature	:	Maximum 32oC
(d) pH	:	2.2
(e) Colour	:	Mild
(f) T.S.S. (mg/l)	:	300-350 mg/l
(g) BOD (mg/l)	:	250-300 mg/l
(h) COD (mg/l)	:	450-500 mg/l

B. Final discharge characteristics As Per NGT Effluent Discharge Standards for STP Application Number 1069/2018)

(a) pH	:	6.5 to 8.5
(b) Oil & Grease	:	<10 mg/l
(c) B.O.D.	:	<5 mg/l
(d) C.O.D.	:	<50 mg/l
(e) Total Suspended Solids	:	<10 mg/l

C. Treatment Technology

The technology is based on attached growth aerobic treatment followed by clarification by a tube settler. Lime will be dosed in for suppression of foaming tendencies. The clarified water will be filtered in a pressure sand filter after dosing of coagulant (alum) for removal of unsettled suspended impurities. This water will be passed through an activated carbon filter for removal of organics. The filtered water from ACF is then chlorinated & stored in the flushing tank.

The attached growth moving bed bio reactor (MBBR) process combines the biological processes of attached & suspended growth.

The waste water after screening is collected in an equalization tank. The equalization tank is required for preventing surges in flow & facilitating equalization of characteristics over the entire quantity of effluent in a given time. A provision for pre-aeration is made in the equalization tank in order to ensure mixing & to prevent the sewage from going septic.

The equalized sewage is then pumped into the MBBR reactor for biological processing. The water enters the bottom of the reactor & flows up through the media which grossly enhances the hydraulic retention time & provides a large surface area for growth of biological micro – organisms. The MBBR reactor is aerated by fine pore sub – surface diffusers which provide the oxygen for organic removal. The synthetic media floats on the water & the air agitation ensures good water to micro-organism contact.

The MBBR treatment is an attached growth type biological treatment process where in, the majority of biological activity takes place on the surface of the PVC media.

Continuous aeration ensures aerobic activity on the surface of the media. Micro – organisms attach themselves on the media & grow into dense films of a viscous jelly like nature. Waste water passes over this film with dissolved organics passing into the bio-film due to concentration gradients within the film. Suspended particles & colloid may get retained on this sticky surface where they are decomposed into soluble products. Oxygen from the aeration process in the waste water provides oxygen for the aerobic reactions at the bio-film surface. Waste products from the metabolic processes diffuse outward & get carried away by the waste water or air currents through the voids of the media.

The aerated effluent passes into a tube deck settler for clarification. The theory of gravity tube settler system is that the carrier fluid maintains laminar flow in the settling media at specified maximum viscosity. These two parameters of a carrier fluid, flowing through a hydraulic configuration, will determine the velocity gradients of the flow, the height of boundary layer at the inclined surface and the residence time within the media.

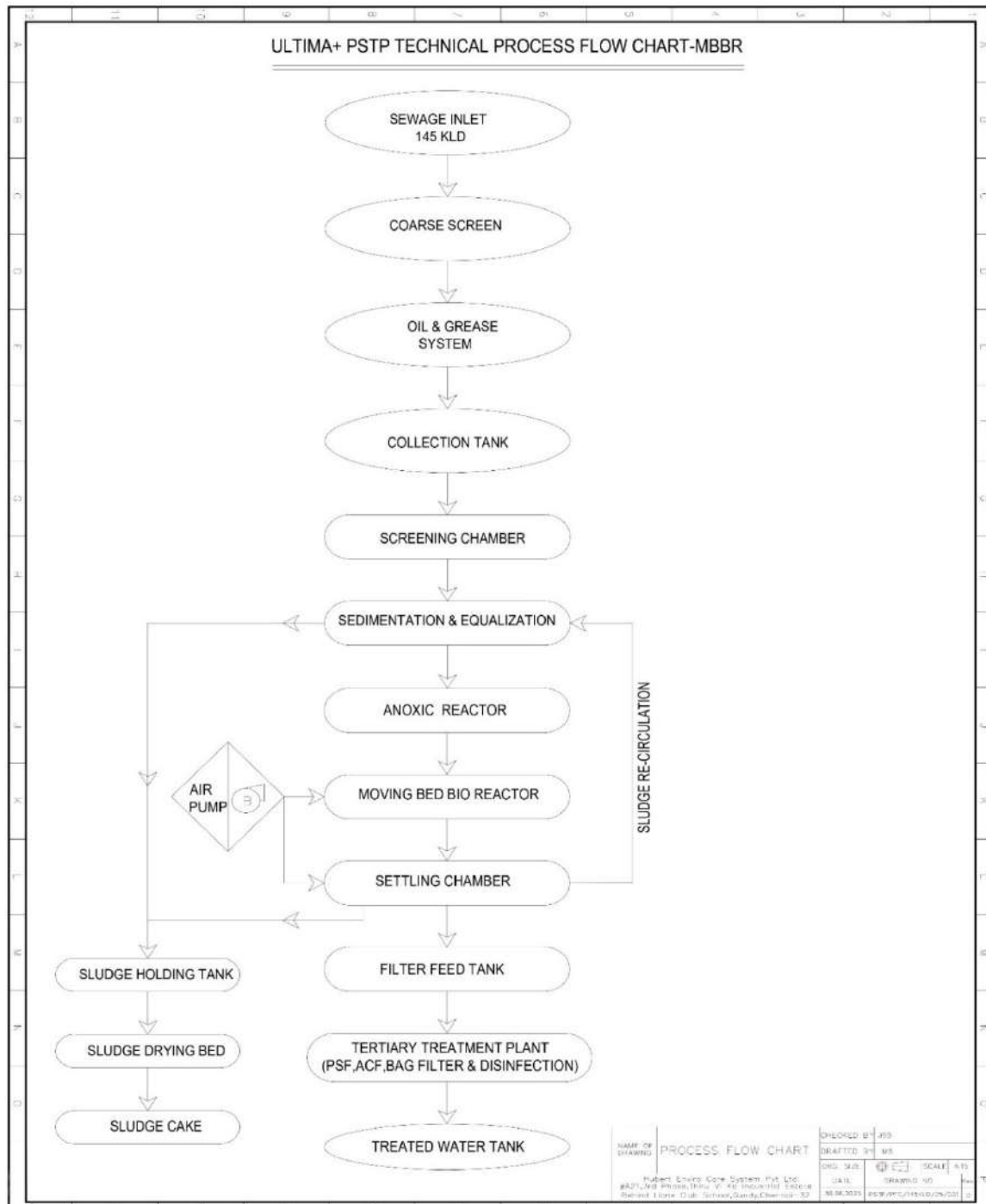
The carrier fluid must be viscous Newtonian, exhibiting a Reynolds number of less than 1000 and preferably, a number under 400. The laminar flow, through the inclined tubes, will produce velocity gradients sufficiently large to form an adequate boundary layer, where the velocity of fluid approaches zero. Boundary layers are necessary in functioning tube settlers, to allow suspended solids to separate from the viscous carrier fluid. Under gravitational forces, they will settle to the hydraulic surface of the tube and subsequently from the clarifier media.

Since the tubes are inclined at 60 degrees, solids settled on the tubes are continually discharged down. This downward rolling action increases particle contact and hence further agglomeration, which increases the sludge settle ability. Studies show that these agglomerated sludge particles can have a settling rate in excess of ten times the settling rate of the individual floc particles in the influent. These heavy agglomerated masses quickly slide down the 60-degree inclined tube and settle at the bottom of the tank.

At the bottom of the Tube deck, where the sludge leaves the Tube surface, the larger agglomerated captures smaller particles in the upcoming stream. This solid contact phenomenon greatly enhances the capture efficiency.

Stages of Treatment: The treatment process consists of the following stages:

- Equalization
- Bio- Degradation
- Clarification & Settling
- Filtration



Typical process flow diagram of Sewage Treatment Plant

7.2.15 STORM WATER DRAINAGE

Separate and independent rain water drainage system shall be provided for collecting rain water from terrace, paved area, lawns and roads. Independent rain water down takes of appropriate size and number shall be provided in close coordination with Architect. Surface/Sub Surface drainage system shall be provided for open-to-sky courtyard/lawn.

The final disposal shall be in rain water harvesting pits to recharge the ground water with overflow connection to external municipal storm drain.

The storm water disposal system for the premises shall be self-sufficient to avoid any collection/stagnation and flooding of water. The amount of storm water run-off depends upon many factors such as intensity and duration of precipitation, characteristics of the tributary area and the time required for such flow to reach the drains. The covered drains shall be located near the carriage way along either side of the roads. Taking the advantage of road camber, the rainfall run off from roads shall flow towards the drains. Saucer drains and gully traps with gratings planned for areas and connected to covered drain channel. Storm water from various blocks/buildings shall be connected to adjacent drain channel by a pipe through catch basins. Therefore, it has been rainwater harvesting pits proposed at selected locations, which will catch the maximum run-off from the area.

- 1) Since the existing topography is congenial to surface disposal, a network of storm water drain is planned adjacent to roads. All building roof water will be brought down through rain water pipes.
- 2) Proposed storm water system consists of pipe drain, catch basins, saucer drains, covered drain channel and seepage pits at regular intervals for rain water harvesting and ground water recharging.

Inside the recharge pit, a recharge bore is constructed having 200 mm diameter and 30 m depth. The bottom of the recharge structure will be kept 2 m above the water level. At the bottom of the recharge well, a filter media is provided to avoid choking of the recharge bore. Design specifications of the rain water harvesting plan are as follows:

- Catchments/roofs would be accessible for regular cleaning.

- The roof will have smooth, hard and dense surface which is less likely to be damaged allowing release of material into the water. Roof painting has been avoided since most paints contain toxic substances and may peel off.
- All gutter ends will be fitted with a wire mesh screen and a first flush device would be installed. Most of the debris carried by the water from the rooftop like leaves, plastic bags and paper pieces will get arrested by the mesh at the terrace outlet and to prevent contamination by ensuring that the runoff from the first 10-20 minutes of rainfall is flushed off.
- No sewage or wastewater would be admitted into the system.
- No wastewater from areas likely to have oil, grease, or other pollutants has been connected to the system.

TABLE 26: CALCULATIONS FOR STORM WATER LOAD

Rainwater Harvesting Calculation			
ROOF TOP AREA		5400	sqm
GREEN AREA		6000	sqm
PAVED AREA		17600	sqm
Last 5 Years Highest Rainfall. (See Image in Next page: Source IMD)		As/ IGBC	
Month	Year	Rainfal l	
July	2018	431.80	mm
July	2019	555.30	mm

June	2020	563.60	mm
June	2021	371.40	mm
June	2022	515.00	mm
Average Highest Month Rainfall		487.42	mm
Criteria for One-day Rainfall for 351-500mm for 4 Points (IGBC Green Homes Rating System Version 3)		= 10%	
One Day Rainfall @10% (IGBC Green Homes Rating System Version 3)		48.742	
Runoff Coefficient of Roof Top (NBC 2016 Plumbing Services Section 2 Clause 4.5.11.2)		0.90	
Runoff Coefficient of Green Area (NBC 2016 Plumbing Services Section 2 Clause 4.5.11.2)		0.15	
Runoff Coefficient of Paved Area (NBC 2016 Plumbing Services Section 2 Clause 4.5.11.2)		0.80	
Runoff Load (NBC 2016 Plumbing Services Section 2 Clause 4.5.11.2)	Area(sq.m)*rainfall intensity (cum/hr)*runoff coefficient		
Runoff Load Roof Top		236.89	cum
Runoff Load Green Area		43.87	cum
Runoff Load Paved Area		686.29	cum
Total Runoff Load		967.05	cum

Considering 30 Mins of Retention Time the Total Load to be Recharged (As/ SEAC)		483.53	cum
Selection of Recharge Well			
Rectangular			
Length		7.00	m
Width		4.50	m
Depth		2.50	m
Volume		78.75	cum
Desilting Chamber			
Length		1.50	m
Width		1.00	m
Hight		1.00	m
Volume		1.5	cum
Total Volume of a single Recharge pit & Desilting Chamber		80.25	cum
Nos. of Recharge Pit		6.03	Nos.
Say		6.00	Nos.

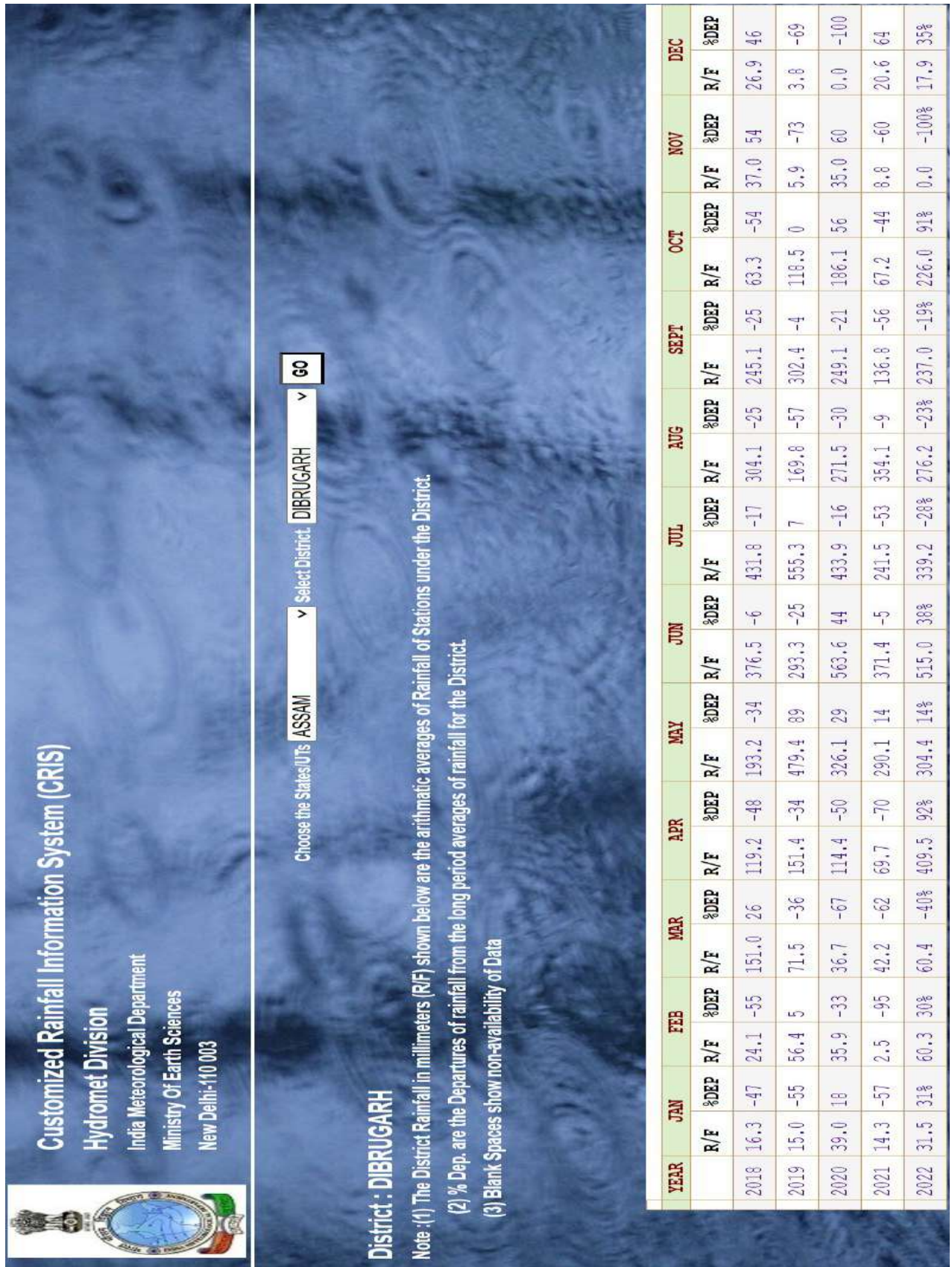


FIGURE 20: RAINFALL DATA, IMD

Sl.No.	Average Peak Month Rainfall (in mm)	One-day Rainfall (% of Average Peak Month Rainfall)			
		1 Points	2 Points	3 Points	4 Points
1.	Upto 250	6%	9%	12%	15%
2.	251 – 350	5%	7.5%	10%	12.5%
3.	351 – 500	4%	6%	8%	10%
4.	501 – 700	3%	4.5%	6%	7.5%
5.	701 & above	2%	3%	4%	5%

FIGURE 21: IGBC CRITERIA FOR 4 POINTS

7.2.15.1 FRP COMPOSITE VESSEL FILTER/ MS

The filter shall be constructed of FRP material with inner shell of integrated polyethylene, polypropylene and other material as per manufacturer's standard or MS.

The inter distribution system and the underbed draw off system shall be of Hub & Lateral type of UPVC / MS material. The filter shall be provided with manhole cover, hand hole, flanged outlet for piping / valve connection and adequate tripod with skid self-supporting structure for making the installation complete. The filter shall also be provided with vacuums breaking connection / accessories to avoid any collapse of internal lining. All filters shall be provided with manhole for media loading and lifting lugs.

The filter shall be suitable for 1.5 times the working pressure of pump. Each filter shall be provided with following accessories: -

- Inlet and outlet pressure gauge
- Sampling cock / valve on raw water inlet and filter water outlet
- Individual drain connection with gunmetal full way valve for each filter.
- Air release valve with connecting pipe

7.2.15.2 PIPES, HANGERS, SUPPORTS & CLAMPS

All vertical pipes shall be fixed by galvanized clamps and galvanized angle brackets truly vertical. Branch pipes shall be connected to the stack at the same angle as that of the fittings. No collars shall be used on vertical stacks. Each stack shall be terminated at top with a Maxi vent.

Horizontal pipes running along ceiling shall be fixed on galvanized structural adjustable clamps of special design as shown on the drawings. Horizontal pipes shall be laid to uniform slope and the clamps adjusted to the proper levels so that the pipes fully reset on them.

Contractor shall provide all sleeves, openings, hangers, inserts shall be provided during the construction by making such provisions in the structure as necessary.

All pipes clamps, supports and hangers shall be galvanized. Factory made prefabricated clamps shall be preferred. Contractor may fabricate the clamps of special nature and galvanize them after fabrication but before installation. All nuts, bolts, washers and other fasteners shall be factory galvanized.

7.2.15.3 VALVES

Isolation Valve

All valves (gate, globe, check, safety) shall be of gun metal suitable for the particular service as specified. All valves shall be of the particular duty and design as specified. Valves shall either be of screwed type or flanged type, as specified, with suitable flanges and non-corrosive bolts and gaskets. Tail pieces as required shall be supplied along with valves. Gate, globe and check valves shall conform to ASTM specifications.

Sluice valves, where specified shall be flanged sluice valves of cast iron body. The spindle, valve seat and wedge nuts shall be gunmetal. They shall generally have non-rising spindle and shall be of the particular duty and design as specified. The valves shall be supplied with suitable flanges, non-corrosive bolts and gaskets.

Ball valves with floats to be fixed in storage tanks shall consist of cast brass lever arm having copper balls (26 SWG) screwed to the arm integrally. The copper ball shall have bronze welded seams. The closing/ opening mechanism incorporating the piston and

cylinder shall be non-corrosive metal and include washers. The size and construction of ball valves and float shall be suitable for desired working pressure operating the supply system. Where called for brass valves shall be supplied with brass hexagonal back nuts to secure them to the tanks and a socket to connect to supply pipe.

TABLE 27: VALVES DETAILS

S. No.	Type of Valve	Size	Construction	Ends
a.	Isolating Valve	15 mm to 50 mm	Gun Metal	Screwed
		65 mm and above	Gun Metal	Flanged
b.	Sluice Valve & Butterfly Valve	65 mm and above	Cast Iron	Flanged
c.	G.M. non return valve	15 mm to 50 mm	Gun Metal	Screwed
		65 mm and above	Gun Metal	Flanged
d.	Flap Type – Non return valve	65 mm and above	Cast Iron	Flanged

All valves shall be suitable for the working pressure involved.

Pressure Reducing Valve Set

Each pressure reducing valve shall contain loading neoprene diaphragm and a full floating, self-aligning, ignition resistant seat and shall be of the single stage, pressure reduction type with provision for manually adjusting the delivery pressure. The valve shall be failed safe to the lowest pressure.

Valves shall be capable of operating and maintaining automatically the respective delivery pressure and flow rates as indicated and shall not be liable to creep. Valves shall also be capable of maintaining the pre-set downstream pressure under static condition.

7.2.15.4 Pressure Gauge

The pressure gauge shall be constructed of die cast aluminium and stove enamelled. It shall be weather proof with an IP 55 enclosure. It shall be stainless steel Burden tube type pressure gauge with a scale range from 0 to 16 Kg / cm square. Each pressure gauge shall have a siphon tube connection. The shut off arrangement shall be by Ball Valve.

7.2.15.5 Water Fittings

All Gunmetal fittings such as gate, globe, check & safety valves shall be fitted in pipe line with unions on both ends of the valve for easy replacement. The joints between fittings and pipes shall be leak- proof when tested to desired pressure rating. The defective fittings and joints shall be replaced.

TABLE 28: DRINKING WATER CHARACTERISTICS (IS: 10500)

S.NO.	PARAMETERS	ACCEPTABLE LIMITS AS PER IS 10500
1.	Turbidity (NTU)	1.0
2.	Colour (hagen)	5.0
3.	Taste & Odour	Unobjectionable
4.	pH	6.5 to 8.5
5.	Total Dissolved Solids (mg/l)	600
6.	Total Hardness as CaCo ₃	200
7.	Chlorides as chlorine (mg/l)	250
8.	Sulphate (as SO ₄) (mg/l)	200
9.	Fluorides (mg/l)	1
10.	Nitrates (mg/l)	45
11.	Calcium (as Ca.) (mg/l)	75
12.	Magnesium (mg/l)	<30
13.	Iron (mg/l)	0.3
14.	Copper (mg/l)	0.05
15.	Zinc (mg/l)	5
16.	Lead (mg/l)	0.05
17.	Alkalinity (mg/l)	200
18.	Boron (mg/l)	1
19.	Cyanide (mg/l)	0.05
20.	Chromium (mg/l)	0.05
21.	Arsenic (mg/l)	0.05
22.	Anionic Detergents as MBAS (mg/l)	0.2
23.	Phenolic compounds (as C ₆ H ₅ OH) (mg/l)	0.001

24.	Bacteriological standard (E coli) (mg/l)	Shall not be detectable
25.	Total Suspended solids (mg/l)	<30

TABLE 29: LIST OF MATERIALS

LIST OF MATERIALS		
1	System Designed	Two Pipe (Stack) System as recommended in code of Practice for soil and waste pipes above ground (IS: 5329- 1964)
2	Minimum Dia Adopted	
a)	All soil, waste and branches Pipes will be uPVC SWR pipe & fitting	150mm/100mm/75mm
b)	All Main soil & waste vertical stacks	100mm
c)	Vents Pipes if required	75mm OD
d)	Wash Basin Waste connection to floor traps	32mm OD
e)	Floor Drain	75mm OD
3	Vertical Pipe (Exposed & Approachable)	All soil, waste and vent pipes shall be running vertically in vertical shafts as per architectural drawing
4	Access door junctions for cleaning purpose	Each Connection from the fixtures.
5	Clean out Plugs	Where two or three fixtures are connected to a single horizontal pipe leading to a vertical stack

		(in toilets), provided at starting point. The cleanout plugs shall have access from the ceiling or shall be flush with the finished floor level
6	Water Seal for All traps	Minimum 50mm
7	Methodology of conveying horizontal soil & waste pipes / sinking of slab (Option – I)	At ceiling level below, by hanging the pipes with M.S structural supports/dash fasteners and proper clamping etc. with proper slope (1:60). All structural Beams at ceiling level should be inverted. App. 100mm sunken area required for connection of Wash Basin & Floor Drain etc.
8	Material	
a)	All soil, waste and vent pipes and fittings.	• UPVC PIPE
b)	Waste pipe from sinks, wash basins and urinals etc.	• CPVC SDR 11 in internal toilets and GI on terrace and shaft water supply pipes medium class conforming to IS: 1239
c)	Rain Water Pipe	• UPVC PIPE
d)	External Sewage Piping	Double Wall Corrugated HDPE pipes
e)	External Storm Piping	RCC NP2 Pipes and On road Crossing NP3 Pipe will be used
f)	Manhole Cover and Frame	FRP Cover with frame/ SFRC
	From source Connection s UGT	GI Pipe - medium class conforming to IS: 1239 and with malleable iron fittings/ PE-100 Conf. to IS: 4984

	Water Supply Pumping Main	HDPE Pipe – Class-III of PE80 grade conforming to IS: 4984 with fittings.
	Piping for bore well Connection	HDPE Pipe – Class-III of PE80 grade conforming to IS: 4984 with fittings.
	Valves	Upto 40 mm dia. Brass Ball Valve and 50mm dia. And above butterfly valve conforming to IS: 13095

TABLE 30: COMPONENTS OF STP

i.	Filtration	Pressurized through Multi grade and filters / Activated carbon filter
ii.	Softening; Water softener and brine tank (if source of water having total Hardness exceeds 200 mg/Liter As per IS 10500)	
iii.	Bacteriological treatment by Chlorination/UV treatment.	

PROCESS FLOW DIAGRAM FOR WATER TREATMENT PLANT

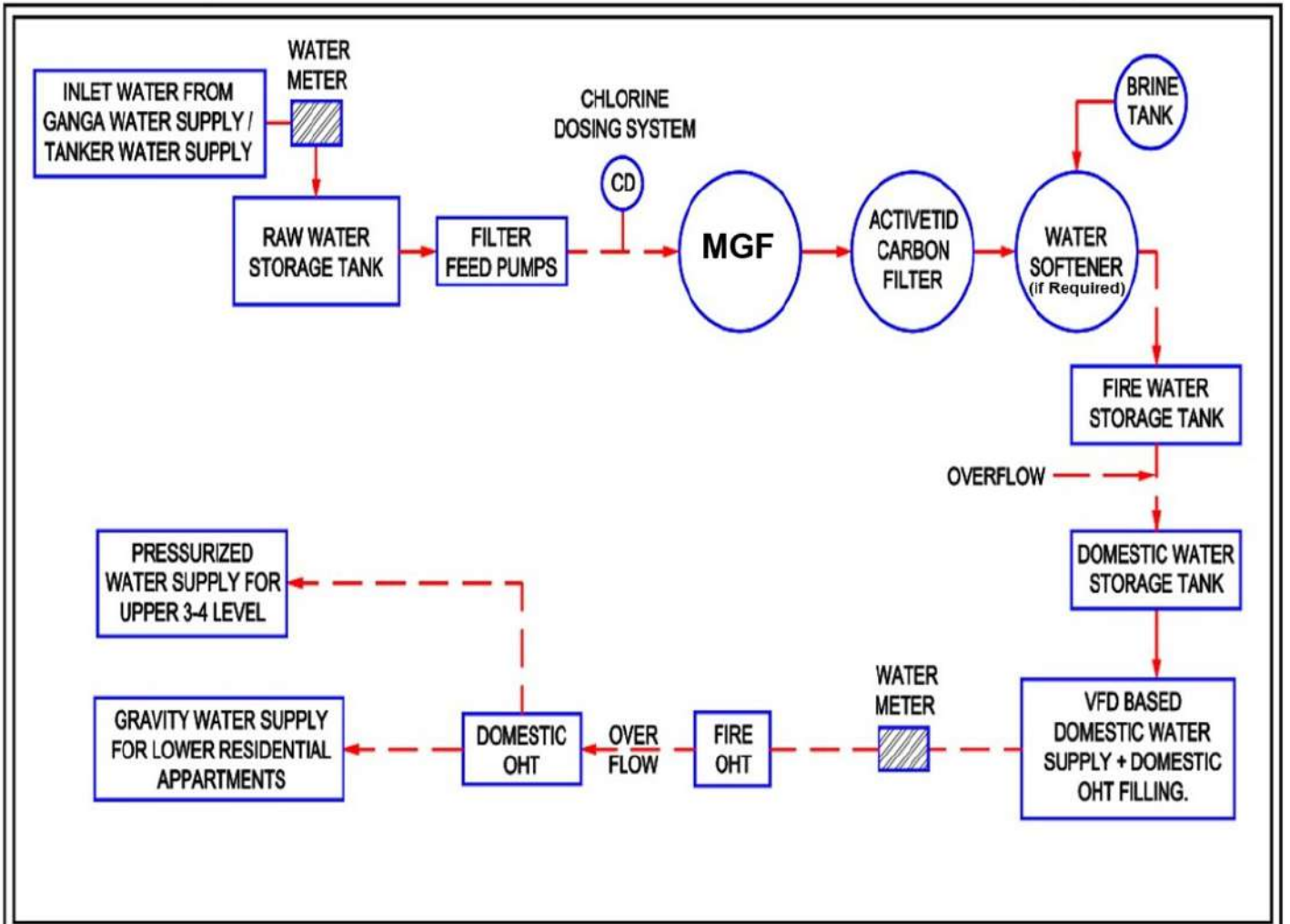


FIGURE 22: PROCESS FLOW DIAGRAM FOR WATER TREATMENT PLANT

REFERENCE FOR OCCUPANCY/ POPULATION CALCULATION

Table 3 Occupant Load (Clauses 4.3 and 4.4.2.1)		
Sl No.	Group of Occupancy	Occupant Load Factor (m ² /person) (see Note 1)
(1)	(2)	(3)
i)	Group A: Residential	12.50
ii)	Group B: Educational	4.00
iii)	Group C: Institutional (see Note 2):	
	a) Indoor patients area	15.00
	b) Outdoor patients area	10.0
iv)	Group D: Assembly:	
	a) Concentrated use without fixed seating	0.65
	b) Less concentrated use without fixed seating (see Note 3)	1.40
	c) Fixed seating	see Note 4
	d) Dining areas and restaurants with seating and table	1.80
v)	Group F: Mercantile:	
	a) Street floor and sales basement	3.00
	b) Upper sales floor	6.00
	c) Storage/warehouse, receiving and the like	20.00
vi)	Group E: Business	10.00
vii)	Group G: Industrial	10.00
viii)	Group H: Storage (see Note 5)	30.00
ix)	Group J: Hazardous	10.00

NOTES

1 Gross area shall be the floor area as defined in 2.35. All factors expressed are in gross area unless marked net.

2 Occupant load in dormitory portions of homes for the aged, orphanages, insane asylums, etc, where sleeping accommodation is provided, shall be calculated at not less than 7.5 m² gross floor area/person.

3 These shall include gymnasium, table tennis room, billiard room and other gaming rooms, library, swimming pool and like.

4 In case of assembly occupancy having fixed seats, the occupant load shall be determined by multiplying the number of seats by 1.2.

5 Car parking areas under occupancy other than storage shall also be 30 m² per person.

a) Residential buildings:

Accommodation	Population Requirements
1 bedroom dwelling unit	4
2 bedroom dwelling unit	5
3 bedroom dwelling unit	6
4 bedroom dwelling unit and above	7

NOTES

1 The above figures consider a domestic household including support personnel, wherever applicable.

2 For plotted development, the population may be arrived at after due consideration of the expected number and type of domestic household units.

3 Dwelling unit under EWS category shall have population requirement of 4 and studio apartment shall have population requirement of 2.

b) For communities with: 100 to 135 lphd
population 20 000 to 100 000
together with full flushing system

FIGURE 23: REFERENCE FOR OCCUPANCY/POPULATION CALCULATION

ANNEXURE: A REFERENCE FOR WATER REQUIREMENT & POPULATION CALCULATION

REFERENCE FOR WATER REQUIREMENT CALCULATION

Table 1 Water Requirements for Buildings Other than Residences <i>(Clause 4.1.2)</i>				
Sl No.	Type of Building	Domestic Per Day litre	Flushing Per Day litre	Total Consumption Per Day litre
(1)	(2)	(3)	(4)	(5)
i)	Factories including canteen where bath rooms are required to be provided	30 per head	15 per head	45 per head
ii)	Factories including canteen where no bath rooms are required to be provided	20 per head	10 per head	30 per head
iii)	Hospital (excluding laundry and kitchen) (see Note 2):			
	a) Number of beds not exceeding 100	230 per head	110 per head	340 per head
	b) Number of beds exceeding 100	300 per head	150 per head	450 per head
	c) Out patient department (OPD)	10 per head	5 per head	15 per head
iv)	Nurses' homes and medical quarters	90 per head	45 per head	135 per head
v)	Hostels	90 per head	45 per head	135 per head
vi)	Hotel (up to 3 star) excluding laundry, kitchen, staff and water bodies	120 per head	60 per head	180 per head
vii)	Hotel (4 star and above) excluding laundry, kitchen, staff and water bodies	260 per head	60 per head	320 per head
viii)	Offices (including canteen)	25 per head	20 per head	45 per head
ix)	Restaurants and food court including water requirement for kitchen:			
	a) Restaurants	55 per seat	15 per seat	70 per seat
	b) Food court	25 per seat	10 per seat	35 per seat
x)	Clubhouse	25 per head	20 per head	45 per head
xi)	Stadiums	4 per head	6 per head	10 per head
xii)	Cinemas, concert halls and theatres and multiplex	5 per seat	10 per seat	15 per seat
xiii)	Schools/Educational institutions:			
	a) Without boarding facilities	25 per head	20 per head	45 per head
	b) With boarding facilities	90 per head	45 per head	135 per head
xiv)	Shopping and retail (mall)			
	a) Staff	25 per head	20 per head	45 per head
	b) Visitors	5 per head	10 per head	15 per head
xv)	Traffic terminal stations (see Notes 3 and 4)			
	a) Airports	40 per head	30 per head	70 per head
	b) Railway stations (Junctions) with bathing facility	40 per head	30 per head	70 per head
	c) Railway stations (Junctions) without bathing facility	30 per head	15 per head	45 per head
	d) Railway Stations (Intermediate) with bathing facility	25 per head	20 per head	45 per head
	e) Railway Stations (Intermediate) without bathing facility	15 per head	10 per head	25 per head
	f) Interstate bus terminals	25 per head	20 per head	45 per head
	g) Intrastate Bus Terminals/Metro Stations	10 per head	5 per head	15 per head
NOTES 1 For calculating water demand for visitors, consumption of 15 litre per head per day may be taken. 2 The water demand includes requirement of patients, attendants, visitors and staff. Additional water demand for kitchen, laundry and clinical water shall be computed as per actual requirements. 3 The number of persons shall be determined by average number of passengers handled by stations, with due considerations given to the staff and vendors who are using these facilities. 4 Consideration should be given for seasonal average peak requirements. 5 The hospitals may be categorized as Category A (25 to 50 beds), Category B (51 to 100 beds), Category C (101 to 300 beds), Category D (301 to 500) and Category E (501 to 750 beds).				

FIGURE 24: REFERENCE FOR WATER REQUIREMENT CALCULATION

ANNEXURE: B

REFERENCE FOR UG & OHT STORAGE TANK CAPACITY

In case of intermittent water supply, the following guidelines should be adopted for calculation of capacity:

- 1) When the supply is from main OHT, the capacity of individual OHT may be taken as half a day demand.
- 2) In case of all the other buildings (apartments, hotels, hospitals, and commercial), the capacity of UGT and OHT may be taken as one and a half days and half a day demand. Wherever raw water and treated water are stored in separate UGTs, the combined storage capacity shall be of one and a half days demand.
- 3) In case of sewage treatment plant, for treated water storage in respect of flushing, one day flushing demand shall be stored in UGT or treated effluent storage tank which is part of the plant, and half a day demand shall be stored in OHT.

For additional requirement of water storage for fire fighting purposes, reference may be made to Part 4 'Fire and Life Safety' of the Code.

NOTE — Minimum requirements for calculation of capacity of these storage tanks are as follows:

- a) In case only OHT is provided, it may be taken as 33.33 to 50 percent of one day's requirement;
- b) In case only UGT is provided, it may be taken as 50 to 150 percent of one day's requirement; and
- c) In case combined storage is provided, it may be taken as 66.6 percent UGT and 33.33 percent OHT of one day's requirement.

4.4.11 Where the water supply distribution system is catering for separate potable water and flushing water supply, and where reclaimed or recycled treated water is being used within the building, it shall be considered as dual water supply system. In such cases, storages for flushing/reclaimed/recycled water shall be separated proportionately. Adequate measures and precautions shall be taken for physical segregation for sanitary purposes and to avoid cross-contamination in the distribution system and to prevent back contamination of water supply sources (see 4.5).

4.4.12 When only one communication pipe is provided for water supply to a building, it is not necessary to have separate storage for flushing and sanitary purposes for health reasons. In such cases when only one storage tank has been provided, tapping of water may be done at two different levels (the lower tapping for flushing) so that a part of the water will be exclusively available for flushing purposes.

4.5 Protection of Water Supply

4.5.1 General

The water supply system shall be designed, installed and maintained in such a manner so as to prevent

contamination from non-potable liquids, solids or gases being introduced into the potable water supply system through cross-connections or any other connection to the system.

4.5.2 Plumbing Fixtures

The water supply lines and fittings for plumbing fixtures shall be installed so as to prevent back flow and shall provide required back flow protections in accordance with 4.11.

The devices, appurtenance intended for special function such as sterilization, processing, distillation, etc, shall be provided with back flow protection devices.

The water supply for hospital fixtures shall be protected against backflow with a reduced pressure principle back flow assembly, an atmospheric or spill resistant vacuum breaker assembly, or an air gap. Vacuum breakers for bed pan washer hoses shall not be located less than 1 525 mm above floor. Vacuum breakers for hose connections in health care or laboratory areas shall not be less than 1 800 mm above floor.

4.5.3 Cross-Connection Control

Cross-connections shall be prohibited, except where approved back flow prevention assemblies/devices are installed to protect the potable water supply (see 4.11).

Potable water outlets and combination stop and waste valves shall not be installed underground or below grade. Freeze proof yard hydrants that drain the riser into the ground are considered to be stop and waste valves.

Back flow prevention can be achieved by means of providing proper air gap, reduced pressure principle back flow prevention assemblies, back flow preventer with intermediate atmospheric vent, barometric loop, pressure vacuum breaker assemblies, atmospheric type vacuum breakers, double check back flow prevention assemblies, spill resistant pressure vacuum breaker, dual check back flow preventer, etc.

4.5.4 Identification of Non-Potable Water Systems

Where non-potable water systems are installed, the piping conveying the non-potable water shall be identified either by colour marking, metal tags or tapes in accordance with the relevant standards and good engineering practices.

4.6 Materials, Fittings and Appliances

4.6.1 Standards for Materials, Fittings and Appliances
All materials, water fittings and appliances shall conform to Part 5 'Building Materials' of the Code.

4.6.2 Materials for Pipes

Pipes may be of any of the following materials:

- a) Cast iron, vertically cast or centrifugally (spun) cast;
- b) Steel (internally lined or coated with bitumen or a bituminous composition, and out-coated with cement concrete or mortar, where necessary);

FIGURE 25: REFERENCE FOR UG & OHT STORAGE TANK CAPACITY

FIRE FIGHTING & FIRE PROTECTION SYSTEM

7.3 FIRE FIGHTING

7.3.1 INTRODUCTION

HOUSING BLOCK

Type of the Building	-- Group A4 (Apartment Houses)
Max. Heights of buildings	-- Height Between 15 to 35 Meters (Categories as per NBC)
Pressure Zoning	-Towers will be designed only for one zone.

COMMUNITY

Type of the Building	-- Group D (Assembly Building)
Max. Heights of buildings	-- Less Than 10m in Height and upto 300 Person (Categories as per NBC)
Pressure Zoning	-Building will be designed only for one zone.

GUEST HOUSE

Type of the Building	-- Group A5 (Hotel Building)
Max. Heights of buildings	-- Less Than 15m in Height and Area Exceeding 300Sqm and up to 1000 Sqm (Categories as per NBC)
Pressure Zoning	-Building will be designed only for one zone.

- This report describes the main features of the Planning of Fire Fighting System at design stage.
- The entire fire safety installation shall be compliant with the NBC-2016, relevant BIS codes / standard and as per the direction of local fire authorities for the entire Complex to ensure the highest safety standard and uniformity of system. Further, before property is opened to public, the fire protection shall be fully operated and tested under simulated conditions to demonstrate compliance with the standards, codes and guidelines

7.3.2 NBC REQUIREMENT AND PROPOSALS

The building falls under Residential building under Apartment houses A-4 having height between 15 to 35 meters for apartment buildings.

As per the NBC-2016, Table 7, page 41 & 42, following are the basic minimum requirement for Fire Fighting Installation of residential towers having heights between 15 to 35 meters:

HOUSING BLOCKS.

TABLE 31: FIRE FIGHTING REQUIREMENTS FOR HOUSING BLOCK

Sl. No.	Type of installation	Requirements (as per NBC-2016)	Proposal for the Project
1.	Fire Extinguishers	Required	Provided – 1 set of 2 nos. fire extinguishers within the Fire Hydrant Cabinet of following type - 1 no. 4.5 Kg CO2 (having 90% base material) fire extinguisher. 1 no. 9 kg DCP (Having 90% Base material) fire extinguisher. In addition to above 4.5 Kg CO2 gas & 9 Kg ABC Powder type (having 90% base material) extinguishers will also be provided inside Lift machine room, parking areas, electrical panel rooms, Pump rooms, security rooms and ESS / DG areas.
2.	Hose Reel	Required	Provided – 20 mm dia and 30-meter-long hose reel drum with 5 mm shut off nozzle to be provided with each hydrant valve on each floor inside fire hydrant cabinet. The hose reel shall be connected directly to the riser.

4.	Wet Riser	Not Required	Not Provided.
5.	Down Comer	Required	Provided. 100 mm dia down comer has been proposed along with followings inside Fire Hose Cabinet (FHC) : (a) single headed 63 mm dia hydrant valve (b) 20 mm dia 30 meter long hose reel with nozzle (c) 2 nos. 63 mm dia. 15 meter long hose pipes (d) one branch pipe with nozzle (e) Fire man's axe.
6.	Yard Hydrants	Not Required	Not Provided.
7.	Automatic Sprinkler System	Not Required	Not Provided.
8.	Manual Fire Alarm System	Required.	Provided. Manual call points and electronic hooters are provided on all floors near all exit locations and other suitable locations. The public address and talk back system are also provided at suitable locations, if height of building is above 24 meters.
9.	Automatic Fire Detection & Alarm System	Not Required.	Not Provided.

			However, Smoke and heat detectors are provided in all electrical shafts, electrical panel / meter rooms, lift machine rooms, pump rooms etc. to provide early warning from a probable risk area, if building height is above 30 meters.
10.	Under Ground Water Tank	Not Required	Not Provided.
11.	Terrace Tank	25 KL	Provided – 25 KL over each block terrace
12.	Ground Level Fire Pumps	Not Required.	Not Provided.
13.	Terrace Level Fire Pump	Required – of capacity 900 lpm with minimum pressure of 3.5 kg/sq cm.	Provided – of capacity 900 lpm having pressure of 3.5 kg/sq cm

The building falls under Residential building under Hotels A-5 having height Less Than 15 meters and Floor Area Exceeding 300sqm but not more 1000sqm on any of the floor for Hotel buildings.

As per the NBC-2016, Table 7, page 41 & 42, following are the basic minimum requirement for Fire Fighting Installation of Hotel buildings height Less Than 15 meters and Floor Area Exceeding 300sqm but not more 1000sqm on any of the floor:

GUEST HOUSE

TABLE 32: FIRE FIGHTING REQUIREMENTS FOR GUEST HOUSE

Sl. No.	Type of installation	Requirements (as per NBC-2016)	Proposal for the Project

1.	Fire Extinguishers	Required	Provided – 1 set of 2 nos. fire extinguishers within the Fire Hydrant Cabinet of following type - 1 no. 4.5 Kg CO2 (having 90% base material) fire extinguisher. 1 no. 9 kg DCP (Having 90% Base material) fire extinguisher. In addition to above 4.5 Kg CO2 gas & 9 Kg ABC Powder type (having 90% base material) extinguishers will also be provided inside Lift machine room, parking areas, electrical panel rooms, Pump rooms, security rooms and ESS / DG areas.
2.	Hose Reel	Required	Provided – 20 mm dia and 30-meter-long hose reel drum with 5 mm shut off nozzle to be provided with each hydrant valve on each floor inside fire hydrant cabinet. The hose reel shall be connected directly to the riser.
3.	Wet Riser	Not Required	Provided. 100 mm dia wet riser has been proposed along with followings inside Fire Hose Cabinet (FHC): (a) single headed 63 mm dia hydrant valve (b) 20 mm dia 30-meter-long hose reel with nozzle (c) 2 nos. 63 mm dia. 15-meter-long hose pipes (d) one branch pipe with nozzle (e) Fire man's axe.

4.	Down Comer	Not Required	Not Provided.
5.	Yard Hydrants	Not Required	Not Provided.
6.	Automatic Sprinkler System	Not Required	Not Provided.
7.	Manual Fire Alarm System	Required.	Provided. Manual call points and electronic hooters are provided on all floors near all exit locations and other suitable locations. The public address and talk back system are also provided at suitable locations, if height of building is above 24 meters.
8.	Automatic Fire Detection & Alarm System	Required.	Provided Smoke and heat detectors are provided in all guest rooms, dining hall, kitchen, electrical shafts, electrical panel / meter rooms, lift machine rooms, pump rooms etc. to provide early warning from a probable risk area..
9.	Under Ground Water Tank	Required- 50KL	Provided

			50 KL in two compartments of 25KL Each Underground.
10.	Terrace Tank	10 KL	Provided – 10 KL
11.	Ground Level Fire Pumps	Required. 1620LPM Main Fire Pump Electrical, 180 LPM Jockey and 1620LPM Diesel Engine Driven Fire Pump with minimum pressure of 3.5 kg/sq. cm.	Provided. 1 No. 1620LPM Main Electrical Fire Pump. 1 No. 180 LPM Electrical Jockey Pump 1 No. 1620 LPM Diesel Engine Driven Fire Pump with minimum pressure of 3.5 kg/sq. cm.
12.	Terrace Level Fire Pump	Required – of capacity 450 lpm with minimum pressure of 3.5 kg/sq. cm.	Provided – of capacity 450 lpm having pressure of 3.5 kg/sq. cm

The building falls under Assembly Building D1-D5 having height less than 10m and up to 300 persons for assembly buildings.

As per the NBC-2016, Table 7, page 45, following are the basic minimum requirement for Fire Fighting Installation of assembly building having height less than 10m and up to 300 persons:

COMMUNITY.

TABLE 33: FIRE FIGHTING REQUIREMENT FOR COMMUNITY CENTRE

Sl. No.	Type of installation	Requirements (as per NBC-2016)	Proposal for the Project
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1.	Fire Extinguishers	Required	Provided – 1 set of 2 nos. fire extinguishers within the Fire Hydrant Cabinet of following type - 1 no. 4.5 Kg CO2 (having 90% base material) fire extinguisher. 1 no. 9 kg DCP (Having 90% Base material) fire extinguisher. In addition to above 4.5 Kg CO2 gas & 9 Kg ABC Powder type (having 90% base material) extinguishers will also be provided inside Lift machine room, parking areas, electrical panel rooms, Pump rooms, security rooms and ESS / DG areas.
2.	Hose Reel	Required	Provided – 20 mm dia and 30-meter-long hose reel drum with 5 mm shut off nozzle to be provided with each hydrant valve on each floor inside fire hydrant cabinet. The hose reel shall be connected directly to the riser.
3.	Wet Riser	Not Required	Not Provided.
4.	Down Comer	Required	Provided. 100 mm dia down comer has been proposed along with followings inside Fire Hose Cabinet (FHC) : (a) single headed 63 mm dia hydrant valve (b) 20 mm dia 30 meter long hose reel with nozzle (c) 2 nos. 63 mm dia. 15 meter long hose pipes (d) one branch pipe with nozzle (e) Fire man's axe.
5.	Yard Hydrants	Not Required	Not Provided.

6.	Automatic Sprinkler System	Not Required	Not Provided.
7.	Manual Fire Alarm System	Required.	Provided. Manual call points and electronic hooters are provided on all floors near all exit locations and other suitable locations. The public address and talk back system are also provided at suitable locations, if height of building is above 24 meters.
8.	Automatic Fire Detection & Alarm System	Not Required.	Not Provided. However, Smoke and heat detectors are provided in all electrical shafts, electrical panel / meter rooms, lift machine rooms, pump rooms etc. to provide early warning from a probable risk area, if building height is above 30 meters.
9.	Under Ground Water Tank	Not Required	Not Provided.
10.	Terrace Tank	20 KL	Provided – 20 KL over each block terrace
11.	Ground Level Fire Pumps	Not Required.	Not Provided.
12.	Terrace Level Fire Pump	Required – of capacity 450 lpm with minimum pressure of 3.5 kg/sq cm.	Provided – of capacity 450 lpm having pressure of 3.5 kg/sq cm

Note:

- a. Pressure gauge to be provided at hydraulically remotest point to ensure the required 3.5 kg/cm² pressure in the fire water line.
- b. Synthetic Fire hose type-III(63mm dia & 15m) to be provided as per IS:636.
- c. All the fire fighting equipments shall be ISI marked and certificate for the same shall be provided.
- d. A First Aid Box shall be provided on the ground floor of each building, at a clearly visible and easily accessible location for all occupants.
- e. Floor evacuation plans to be provided for each floor of Residential Housing Blocks, Community centre & Guest House.
- f. IG-541 gas-based fire suppression system and fire detection system to be installed in the substation shall be provided.

7.3.3 MANUAL FIRE EXTINGUISHER AND GAS FLOODING SYSTEM

- (i) Hand held type extinguisher shall be according IS: 15683. The location of extinguisher shall be at easily accessible place.

Class of Fire	Description	Suitable Types of Appliances
A	Fire in Ordinary combustibles (Wood , Fiber , Rubber Plastic, paper and the like)	Gas Expelled water Type
B	Fire in flammable liquids, paints, grease, solvents and the like	Mechanical Foam type and Dry Powder type
C	Electrical Fire	Carbon Dioxide and Dry Powder type

- (ii) Manual gas flooding for electrical panel will be provided as required by NBC of India and as directed by Local Fire Brigade.

National Building Code of India recommends compartmenting of parking area. Therefore, we are proposing water curtain around the periphery of every 3000sq.m. area to comply with statutory requirement.

In addition, there shall be requirement of a minimum of two compartments if the floor plate size is equal or less than 3000sq.m. area. However, such requirement of minimum two compartments shall not be required, if the floor plate is less than 750sq.m. Area.

Each compartment will be controlled by one deluge valve, one solenoid valve activated by Flow switch and a set of window nozzles connected to distribution header to create a barrier for fire and smoke. This system will be fed by a separate pump which will be activated automatically.

REFERENCE FOR FIRE FIGHTING SYSTEM

PART 4 FIRE AND LIFE SAFETY

42

Table 7 Minimum Requirements for Fire Fighting Installations
[Clauses 4.9(a), 4.9(c), 4.9(e), 5.1.1(a), 5.1.1(d), 5.1.2.1, 5.1.3(a), E-7, H-2(f) and Table 2]

Sl No.	Type of Building Occupancy	Type of Installation								Water Supply (litre)		Pump Capacity (litre/min)	
		Fire Extinguisher	First Aid Hose Reel	Wet Riser	Down Corner	Yard Hydrant	Automatic Sprinkler System	Manually Operated Electronic Fire Alarm Systems (see Note 1)	Automatic Detection and Alarm System (see Note 2)	Under-ground Static Water Storage Tank Combined Capacity for Wet Riser, Yard Hydrant and Sprinklers per Set of Pumps	Terrace Tank over Respective Tower Terrace	Pump Near Under-ground Static Water Storage Tank (Fire Pump) with Minimum Pressure of 3.5 kg/cm ² at Remote Location	At the Terrace Tank Level with Minimum Pressure of 3.5 kg/cm ²
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
c)	Dormitories (A-3) and Apartment Houses (A-4)												
1)	Less than 15 m in height	R	R	NR	NR	NR	R (see Note 4)	NR	NR	NR	5 000 (5 000) (see Note 6)	NR	450 (450) (see Note 6)

Table 7 — (Continued)

Sl No.	Type of Building Occupancy	Type of Installation								Water Supply (litre)		Pump Capacity (litre/min)	
		Fire Extinguisher	First Aid Hose Reel	Wet Riser	Down Corner	Yard Hydrant	Automatic Sprinkler System	Manually Operated Electronic Fire Alarm Systems (see Note 1)	Automatic Detection and Alarm System (see Note 2)	Under-ground Static Water Storage Tank Combined Capacity for Wet Riser, Yard Hydrant and Sprinklers per Set of Pumps	Terrace Tank over Respective Tower Terrace	Pump Near Under-ground Static Water Storage Tank (Fire Pump) with Minimum Pressure of 3.5 kg/cm ² at Remote Location	At the Terrace Tank Level with Minimum Pressure of 3.5 kg/cm ²
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
2)	15 m and above but not exceeding 35 m in height	R	R	NR	R	NR	R (see Note 4)	R (see Note 8)	NR	NR	25 000	NR	900
3)	Above 35 m in height but not exceeding 45 m in height	R	R	NR	NR	NR	R (see Note 4 and Note 9)	R	NR	75 000	25 000	(see Note 10)	NR
4)	Above 45 m in height but not exceeding 60 m in height	R	R	R	NR	R	R	R	NR	150 000	10 000	(see Note 11)	NR

FIGURE 26: REFERENCE FOR FIRE FIGHTING SYSTEM

Table 7 — (Continued)

Sl No.	Type of Building Occupancy	Type of Installation										Water Supply (litre)		Pump Capacity (litre/min)	
		Fire Extinguisher	First Aid Hose Reel	Wet Riser	Down Comer	Yard Hydrant	Automatic Sprinkler System	Manually Operated Electronic Fire Alarm Systems (see Note 1)	Automatic Detection and Alarm System (see Note 2)	Under-ground Static Water Storage Tank Combined Capacity for Wet Riser, Yard Hydrant and Sprinklers per Set of Pumps	Terrace Tank over Respective Tower Terrace	Pump Near Underground Static Water Storage Tank (Fire Pump) with Minimum Pressure of 3.5 kg/cm ² at Remote Location	At the Terrace Tank Level with Minimum Pressure of 3.5 kg/cm ²		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)		
2)	15 m and above but not exceeding 35 m in height	R	R	NR	R	NR	R (see Note 4)	R (see Note 8)	NR	NR	25 000	NR	900		
3)	Above 35 m but not exceeding 45 m in height	R	R	R	NR	NR	R (see Note 4 and Note 9)	R	NR	75 000	5 000	(see Note 10)	NR		
4)	Above 45 m in height but not exceeding 60 m in height	R	R	R	NR	R	R	R	NR	150 000	10 000	(see Note 11)	NR		
5)	Above 60 m in height	R	R	R	NR	R	R	R	R	200 000	10 000	(see Note 12 & Note 13)	NR		
d)	Hotels (A-5)														
1)	Less than 15 m in height														
i)	Floor area not exceeding 300 m ² on any of the floor	R	R	NR	NR	NR	R (see Note 4)	R	NR	NR	5 000 (5000)	NR	450 (450)		
ii)	Floor area exceeding 300 m ² but not more 1 000 m ² on any of the floor	R	R	R (see Note 7)	NR	NR	R (see Note 4)	R	R	10 000 for every 500 m ² floor area subject to minimum of 50 000 (see Note 7)	10 000 (5000) (see Note 6)	(see Note 14 in case of provision of wet riser)	450 (450) (see Note 6)		
iii)	Floor area exceeding 1 000 m ² on any of the floor	R	R	R (see Note 15)	NR	R	R	R (see Note 1)	R	(see Note 15)	10 000 (see Note 4)	(see Note 15 and Note 14)	NR		

PART 4 FIRE AND LIFE SAFETY

Table 7 — (Continued)

Sl No.	Type of Building Occupancy	Type of Installation								Water Supply (litre)		Pump Capacity (litre/min)	
		Fire Extinguisher	First Aid Hose Reel	Wet Riser	Down Comer	Yard Hydrant	Automatic Sprinkler System	Manually Operated Electronic Fire Alarm Systems (see Note 1)	Automatic Detection and Alarm System (see Note 2)	Under-ground Static Water Storage Tank Combined Capacity for Wet Riser, Yard Hydrant and Sprinklers per Set of Pumps	Terrace Tank over Respective Tower	Pump Near Underground Static Water Storage Tank (Fire Pump) with Minimum Pressure of 3.5 kg/cm ² at Remote Location	At the Terrace Tank Level with Minimum Pressure of 3.5 kg/cm ²
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
2)	10 m and above but not exceeding 15 m in height	R	R	R	NR	R	R (see Note 4)	R	R	100 000	5 000 (5 000) (see Note 6)	(see Note 10)	NR
3)	15 m and above but not exceeding 24 m in height	R	R	R	NR	R	R	R	R	150 000	10 000 (see Note 6)	(see Note 11)	NR
4)	24 m and above but not exceeding 30 m in height	R	R	R	NR	R	R	R	R	200 000	20 000	(see Note 11)	NR
ASSEMBLY BUILDINGS (D) (see Note 16)													
a)	Buildings (D-1 to D-5)												
1)	Less than 10 m in height												
i)	Up to 300 persons	R	R	NR	R	NR	R (see Note 4)	R	NR	NR	20 000 (5 000) (see Note 6)	NR	450 (450) (see Note 6)

STANDARDS

National Building Code, 2016 as per latest amendment

IS : 1641 : Fire safety of building – general principals of fire grading

IS : 1642 : Fire safety of building – details of constructions

IS : 1644 : Fire safety of building – Exit requirements

IS : 1646 : Fire safety of building – Electrical installations

IS : 2189 : Automatic fire detection and alarm system

IS : 2190 : Selection of fire extinguishers

IS : 3844 : Installation of internal fire hydrant and hose reels on premises

IS : 9668 : Provision of water supply for Fire fighting

IS : 13039 : Provision of external hydrant system

IS : 15105 : Design of automatic sprinkler system

IS : 9972 : Specifications of automatic sprinklers heads

IS : 884 : Specifications of First aid hose reel

IS : 15301 : Installation of firefighting pumps